KLE Technological University

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A Minor-Project report on,

“ Vision-Assisted Robotic Soldering Process Optimization For Electronic and Light Fabrication Applications ”

Automation and Robotics Department

2025-26

CERTIFICATE

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This is to certify that Shahista Khatib - 01FE23BAR006, Pratiksha Salecha - 01FE23BAR019, Shreyas Kubsad - 01FE23BAR021 & Rishita Kadam - 01FE23BAR033, Suchit Patil - 01FE24BAR401 have successfully submitted the project titled "Vision-Assisted Robotic Soldering Process Optimization For Electronic and Light Fabrication Applications" as the part Minor Project course for the academic year 2025-26 Even.

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Automation and Robotics Department

2025-26

Student Declaration

We the students of Automation and Robotics, 6th semester declare that all the sheets, CAD, report files and images submitted to the department as a part of Mini Project are solemnly modelled, captured and generated by us.

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Acknowledgment

We would like to thank the KLE Technological University for providing an opportunity and pedagogical improvement which helped us to get a hands-on experience in the 6th semester level.

We would thank our HOD, Dr. Vinayak Kulkarni for the competitive changes in curriculum and keeping the course content updated as per the industry requirement and standards.

We would thank our guide Mr. Girish Karikatti for his involvement and guidance throughout the course.

We would provide our heartly thanks to the coordinators, Mr. Girish Karikatti, & Mrs. Channavva Soori for their continuous involvement, assigning challenging tasks and quite an ease of delivery of content. The examples taken up in-session were easy to understand and practice, demonstrated all the tools required for the project fulfillment.

We thank all the staff members of A&R, our parents and teammates for the support and blissful guidance.

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INDEX

SL. No.	Content	Page No.
1	INTRODUCTION	1
2	THEME	2
2.1	Project Statement	2
2.2	Project Gantt	3
3	LITERATURE SURVEY	4-16
3.1	Patent Survey	17-18
3.2	Market Survey (Product Benchmarking)	19-20
3.3	VDI 2206 guidelines	20-21
4	PROBLEM DEFINITION	22
4.1	Requirements list	22
4.2	Constraints	23
4.3	Conceptual Design	24-28
4.4	Functional Architecture	28
4.5	Concept Evaluation	29
4.6	Screening	30
4.7	Finalized Concept	31
5	VIRTUAL PROTOTYPING	32
5.1	Freehand sketch	32-33

5.2	CAD Model and Drawings	33-38
5.3	Circuit Design	39
5.4	System Composer Architecture	40-41
5.5	Flowchart	42
5.6	Dataset & Processing	43
5.6.1	Image Training, Testing and Implementation	43-49
5.7	Simulation of Cartesian Robot in CoppeliaSim	50
6	Prototyping	51
6.1	Steps in Project Execution	51-54
7	CONCLUSION & LEARNINGS	55
8	FUTURE SCOPE	56
	REFERENCES	57-58
	Review Paper	
	Technical Paper	
	Portfolio	
	TEAM DETAILS	

List of Figures

Figure Number	Figure Name	Page No.
2.1	Project Gantt Chart delineating the weekly task breakdown and milestones.	**3**
3.3	VDI 2206 V-Model for Vision-Based Robotic Soldering System	**21**
4.3.1	Conceptual Design of SCARA-Based Vision-Assisted Robotic Soldering System	**24**
.4.3.2	Conceptual Design of 6-Axis Articulated Vision-Guided Robotic Soldering System	**25**
4.3.3	Conceptual Design of Cartesian (X-Y-Z) Vision-Assisted Robotic Soldering System	**26**
4.3.4	Conceptual Design of Modular Vision-Assisted PCB Robotic Soldering Workstation	**27**
4.3.5	Conceptual Design of Compact Vision-Guided Robotic Soldering System for PCB Assembly	**28**
4.4	Workflow of the proposed vision-assisted robotic soldering pipeline.	**28**
4.8	Finalized conceptual design of the proposed vision-based automated soldering system	**31**
5.1.1	Freehand sketch of the back plate	**32**
5.1.2	Freehand sketch of the front plate	**32**
5.1.3	Freehand sketch of the solder mount	**33**
5.2.1	3D CAD model of the proposed vision-based automated soldering system	**33**
5.2.2	SolidWorks CAD assembly model of the proposed vision-based automatic soldering system	**34**

5.2.3	Part drawings of the aluminum extrusion profiles used in the proposed vision-based automatic soldering system: (a) vertical aluminum profile and (b) horizontal aluminum profile.	**35**
5.2.4	Part drawings of the mechanical components used in the proposed vision-based automatic soldering system: (a) aluminum mounting bracket and (b) brass lead screw nut.	**35**
5.2.5	Part drawings of the structural mounting components used in the proposed vision-based automatic soldering system: (a) back plate and (b) NEMA 17 motor mounting clamp	**36**
5.2.6	Part drawings of the motion-guiding components used in the proposed vision-based automatic soldering system: (a) LMK8UU 8 mm linear bearing housing and (b) rod holder block	**36**
5.2.7	Part drawings of the supporting and guiding components used in the proposed vision-based automatic soldering system: (a) side plate and (b) smooth rod	**37**
5.2.8	Part drawings of the actuation components used in the proposed vision-based automatic soldering system: (a) soldering gun and (b) NEMA 17 stepper motor	**37**
5.2.9	Part drawings of the custom-designed mechanical components: (a) front plate and (b) solder mount.	**38**
5.2.10	Orthographic and isometric assembly views of the proposed vision-based automatic soldering system, including the top, front, side, and isometric views	**38**
5.3	Circuit diagram of the proposed vision-based automatic soldering system	**39**
5.4.1	Physical architecture of the proposed vision-based automatic soldering system developed using MATLAB System Composer	**40**
5.4.2	Logical architecture of the proposed vision-based automatic soldering system developed using MATLAB System Composer	**41**
5.5	Control program flow of the proposed vision-based automated soldering system	**42**
5.6.1	Detection of Front PCB Components Using the Proposed YOLOv8 Model	**44**

5.6.2	Confusion matrix for the front PCB component detection model	**45**
5.6.3	Performance evaluation of the front PCB component detection model using a normalized confusion matrix	**45**
5.6.4	Detection of Back PCB Holes Using the Proposed YOLOv8 Model	**47**
5.6.5	Confusion matrix for the back PCB hole detection model	**48**
5.6.6	Performance evaluation of the back PCB hole detection model using a normalized confusion matrix	**49**
5.7	CoppeliaSim simulation environment of the proposed vision-based automated soldering system	**50**
6.1	Structural Framework Fabrication & Mechanical Skeleton Assembly	**51**
6.2	Carriage , Tooling Mounts & Toolhead Fabrication	**52**
6.3	Automated Inversion (Flipping) Mechanism Generation	**53**
6.4	Electrical Subsystem Routing, Shield Coupling & Core Tuning	**54**
6.5	Unified System Integration & Complete Prototype Commissioning	**54**
	Team Photo	

List of Tables

Table Number	Table Name	Page No.
3.1	List of papers/articles	4-5
3.2	Summary of Recent Patents Related to Vision-Assisted PCB Inspection and Robotic Soldering	17
3.3	Emerging Technology Trends in AI-Based PCB Detection and Vision-Assisted Robotic Soldering	18
4.1.1	Functional Requirement List	22
4.1.2	Non-Functional Requirement List	23
4.2.1	Constraints List	24-25
4.6	Concept evaluation of the proposed robotic soldering system	29
4.7	Concept screening of the proposed robotic soldering system	30
5.6.1	Performance of YOLOv8 in Front PCB for solder hole detection	44
5.6.2	Performance of YOLOv8 in Back PCB for solder hole detection	48
	Team details	

Abbreviation List

Abbreviation	Definition
3-DOF / 3 DOF	Three Degrees of Freedom
6 DOF	Six Degrees of Freedom
AHP	Analytic Hierarchy Process
AMRWD	Automatic Multi-Wire Wire-Detection
AOI	Automated Optical Inspection
BGA	Ball Grid Array
CAD	Computer-Aided Design
CCD	Charge-Coupled Device
CMOS	Complementary Metal-Oxide-Semiconductor
CNC	Computer Numerical Control
CNIPA	China National Intellectual Property Administration
CNN	Convolutional Neural Network
CPC	Cooperative Patent Classification
DMAIC	Define, Measure, Analyze, Improve, Control
DOE	Design of Experiments

DRL	**Deep Reinforcement Learning**
EU-RoHS	**European Union Restriction of Hazardous Substances**
EV	**Electric Vehicle**
FCM	**Fuzzy C-Means**
FPCBs / FPCB	**Flexible Printed Circuit Boards**
HIL	**Hardware-in-the-Loop**
HLD	**High-Level Design**
HMI	**Human-Machine Interface**
HSV	**Hue, Saturation, Value**
IoT	**Internet of Things**
IP	**Internet Protocol**
LED / LEDs	**Light-Emitting Diode**
LSTM	**Long Short-Term Memory**
mAP	**Mean Average Precision**
MES	**Manufacturing Execution System**
ML/MB	**Machine Learning / Model-Based**
MQTT	**Message Queuing Telemetry Transport**

OFT	**OpenVLA-Optimized Fine-Tuning**
PCB / PCBs	**Printed Circuit Board**
PEFT	**Parameter-Efficient Fine-Tuning**
PFMEA	**Process Failure Mode and Effects Analysis**
PLC	**Programmable Logic Controller**
PPM	**Parts Per Million**
QA	**Quality Assurance**
ROI	**Region of Interest**
RPN	**Risk Priority Number**
SCARA	**Selective Compliance Assembly Robot Arm**
SEMP	**Systems Engineering Management Plan**
SIL	**Software-in-the-Loop**
SMPS	**Switched-Mode Power Supply**
SMT	**Surface Mount Technology**
SRN	**Student Registration Number**
THT	**Through-Hole Technology**
USART	**Universal Synchronous Asynchronous Receiver-Transmitter**

VDI	**Verein Deutscher Ingenieure**
VLA	**Vision-Language-Action**
YOLO / YOLOv3 / YOLOv8	**You Only Look Once**

1. INTRODUCTION

The modern engineering curriculum places a heavy emphasis on transforming theoretical principles into practical, industry-ready solutions. This minor project course serves as a critical bridge between concept design and physical product realization, requiring students to step out of structured textbook boundaries and manage an end-to-end engineering lifecycle. Modern industrial systems are no longer isolated within single engineering domains; rather, they demand a seamless fusion of mechanical structures, electronic hardware, computer vision algorithms, and embedded control software. Through this course, students experience the multi-layered responsibilities of a systems engineer navigating initial problem definition, conducting deep literature and patent surveys, mapping conceptual designs, sourcing industrial components, and executing custom physical fabrication.

By engaging with a real-world industrial theme like "Industrial Automation," this course exposes students to the realities of contemporary manufacturing, such as the strict standards of accuracy, resource optimization, and operational safety. Engineering design must always account for physical and economic constraints, including workspace footprints, component payload capacities, signal communication latency, and material constraints under high thermal loads. Following structured global paradigms, such as the VDI 2206 guidelines for mechatronic development, the course trains students to decompose a complex engineering challenge into manageable mechanical, electrical, and logical subsystems. Furthermore, the course cultivates crucial professional competencies including rigorous documentation, agile project management using Gantt schedules, risk mitigation, and cross-functional team collaboration, preparing entering engineers to drive value within highly automated smart factories and Industry 4.0 environments.

Course Objectives:

- **To Synthesize Multidisciplinary Knowledge:** To combine mechanical structural design, electronics, and software development into a unified mechatronic engineering workflow.
- **To Implement Structured Design Methodologies:** To apply formal industrial system lifecycles, such as the VDI 2206 standard and cross-domain concept screening, to transform technical requirements into viable product architectures.
- **To Develop Prototyping and Fabrication Skills:** To gain practical experience in engineering tasks, including custom circuit board layout etching, multi-axis hardware assembly, component integration, and system calibration.
- **To Master Intelligent System Control:** To integrate advanced sensors, microcontrollers, and real-time vision algorithms to develop automated, data-driven closed-loop machine controls.
- **To Cultivate Professional Project Management:** To build professional engineering competencies in structured technical writing, resource constraint analysis, timeline scheduling, and collaborative technical team execution.

2. THEME

Industrial Automation

Industrial automation refers to the integration of control systems, such as computers, microcontrollers, and robotics, with information technologies to handle various processes and machinery in an industry without requiring heavy human intervention. In modern manufacturing landscapes, automation has evolved from a tool for simple mass production into a sophisticated paradigm centered on flexibility, high precision, and digital connectivity. By replacing manual, repetitive tasks with automated sequences, industries can achieve unparalleled throughput, eliminate human error, and maintain strict quality standards that are impossible to duplicate manually.

At the core of industrial automation is the optimization of resource utilization and operational safety. In environments involving hazardous conditions, such as high-temperature soldering, chemical exposure, or repetitive strain, automated systems safeguard human workers by taking over high-risk operations. Furthermore, modern automation leverages closed-loop feedback systems where sensors continuously monitor the operational state, enabling real-time adjustments to maintain optimal performance. The integration of advanced sensory systems, like high-resolution machine vision, allows automated machinery to not only execute pre-programmed movements but also make intelligent, data-driven decisions on the factory floor, driving the transition toward smart factories and Industry 4.0.

2.1 Project Statement

In electronic and light fabrication industries, traditional soldering methods are manual or semi-automated, leading to inconsistent solder joints, human error, and inability to detect defects such as misplaced or missing components in real time.

The Aequs Koppal Toy Cluster, currently specializing in plastic and wooden toy manufacturing, does not have a production line for electronic toys. To enable this capability, a reliable and automated PCB soldering system with quality inspection is essential.

This project proposes a Vision-Assisted Robotic Soldering System that uses computer vision and machine learning to verify PCB component placement in real time and commands the robotic arm to solder only defect-free boards ensuring precision, consistency, and scalable quality in electronic toy production.

2.2 Project Gantt

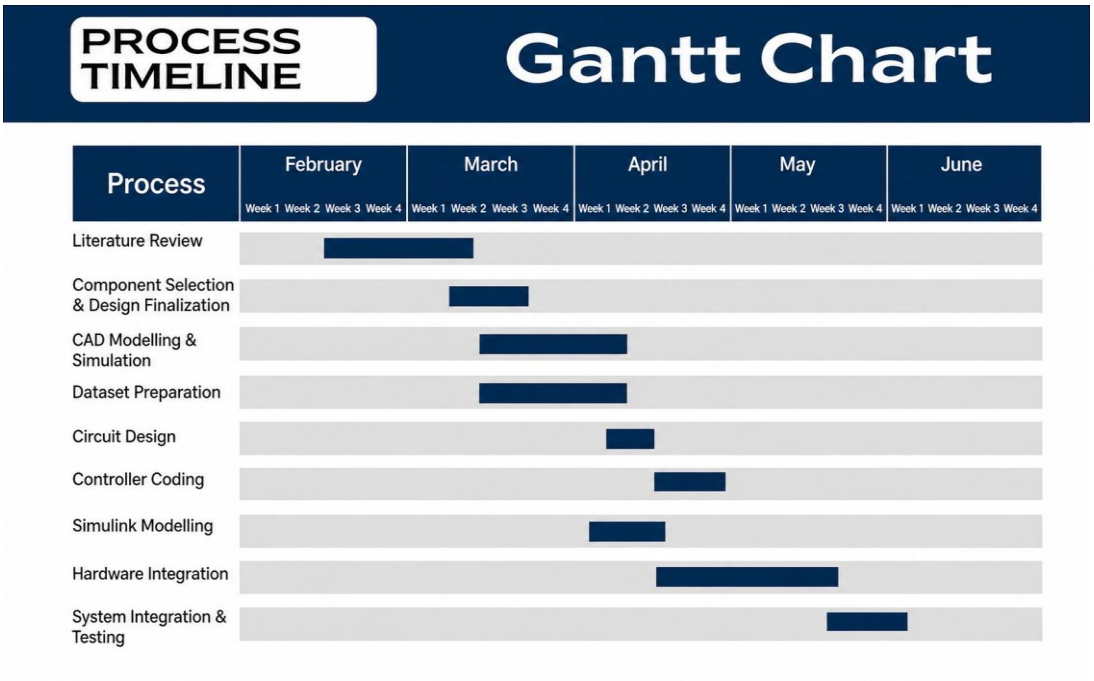


Fig.2.1: Project Gantt Chart delineating the weekly task breakdown and milestones.

3. LITERATURE SURVEY

A. Literature Survey

Table No. 3 1- List of papers/articles

Sl. No.	Article Name	Author & Publisher	Year
01	Deep Learning-Based Robotic Manipulation of Flexible PCBs	Gao et al.	2019
02	Application of Computer System in Control and Programming of a Robotized Soldering Station	Milecki et al.	2017
03	Development of an Automated System for Soldering and the Solder Inspection of an Assembly Line	Lee et al.	2018
04	Development of an automated system for the soldering of USB cables	Gao et al.	2023
05	Vision-Based Robotic Manipulation of Flexible PCBs	Li et al.	2018
06	Automatic Optical Inspection System for Solder Quality Classification of THT Solder Joints	Fonseka & Jayasinghe	2019
07	Deep Learning-Based PCB Defect Detection Using X-ray Images	Zhang et al.	2022
08	Deep Learning-Based Anomaly Detection for Solder Joint Inspection	Mendizabal-Arrieta et al.	2025
09	ChangeChip:Reference-Based Unsupervised Change Detection for PCB Defect Detection	Fridman et al.	2021
10	DemoTeach-Robot Programming with Motion Capture System for Foundry Applications	Li, Liu, Volk & Gunther	2025
11	Body-in-white Solder Joint Positioning Method Based on Machine Vision Technology	Yutong Yuan et al.	2024
12	Adaptive Template Matching for BGA Solder Joint Detection	Xueli Hao et al.	2018
13	PCB Soldering Defect Detection Using Template-Based Image Processing	S. Ozturk & B. Akdemir	2017

14	Deep Learning Based Solder Joint Defect Detection	Qianru Zhang et al.	2022
15	SMT Solder Joint Quality Inspection Using Surface Reconstruction	Nan Yu & Jing Zhang	2019
16	Vision-based soldering process parameters calculation for robotic soldering	Hao Ren et al.	2021
17	Experimental Path Tracking of a Cartesian Robot Using Image Processing	Allahyari & Fakhari	2023
18	Solderbot: Automated Soldering using Robot Arm	Jose Joseph Thandapral et al.	2021
19	Optimization of the soldering process by the DMAIC methodology	Michal Zasadzień	2016
20	Optimization of Process Parameters in Lead-free Wave Soldering Based on Orthogonal Experiment and Analytic Hierarchy Process	Yin et al.	2010
21	Development of a vision system integrated with industrial robots for online weld	Nguyen et al.	2024
22	RT-2: Vision-Language-Action Models Transfer Web Knowledge to Robotic Control	Brohan et al.	2023
23	OpenVLA: An Open-Source Vision-Language-Action Model	Xiao et al.	2024
24	π0 A Vision-Language-Action Flow Model for General Robot Control	Black et al.	2024
25	Fine-Tuning Vision-Language-Action Models	Kim et al.	2025

1. Deep Learning-Based Robotic Manipulation of Flexible PCBs

Gao et al. [1] introduces the positional uncertainty that classical rigid-body controllers cannot handle by the substrate with its deformable and dynamically shifting nature, addressing the problem of fully automatic robotic soldering of flexible PCBs. Their proposed system consists of a cooperative dual-arm configuration: an assistive arm which is equipped with a vacuum chunk responsible for the contacts and stabilizing of the upper flexible PCB, while a soldering end-effector performs the actual soldering operation.

The techniques used in the proposed research are the core vision component i.e. , the YOLOv3 based detection of solder pads. As feedback, detected bounding-box centroids are fed into the Jacobian-based dynamic controller. The dataset and tools used in this study comprise a custom dataset of 1,200 images captured in the real soldering environment across five scene categories-separated, adherent , overlapped, covered and under illuminated flexible PCBs with 1,000 images was used for training and validation. Hardware consisted of a Basler acA1300-60gc camera, an ADVANTECH PCI-1745 motion control card and an OptoSigma-GSC01 motion controller. Across all trials the study claims to have 100% contact success with position error reduced to less than 2 pixels in image space. The strength of the study is the YOLOv3 pad detection combined with a Jacobian dynamic controller which achieves sub-pixel accuracy without pre-calibration, generalizing across varied lighting conditions. The limitation faced was that the experiment was conducted without actual tin wire and validated only on flexible PCBs using custom dual-arm hardware which limits the applicability to small rigid PCBs.

2. Application of Computer System in Control and Programming of a Robotized Soldering Station

This study combines offline Gerber file parsing with an online vision system for pad coordinate recognition for THT soldering on rigid PCBs proposed by Milecki et al. [2]. The hardware components in this study consist of a station with a SCARA robot, a Siemens S7-1215 PLC, stepper motor-driven PCB transport and solder wire feed modules, a nitrogen supply system and a touch screen HMI. The technique used consists of 2 modes - a Gerber file parser for optimized trajectory generation which extracts pad coordinates and a classical image processing pipeline which includes a classical image processing pipeline. Using standard industrial hardware such as Siemens PLC and JVC stepper motors made it reproducible at moderate cost and it was evaluated on real PCB scans and camera images.

Against the Gerber reference the average pad detection error was found to be 0.06 mm sufficient for THT through-hole diameters of 0.8-1.5 mm. The highlight of this paper was the dual mode approach that offers flexibility, and off-the-shelf components ensure practical deployability without specialist hardware. The limitations of this study were that the camera based detection proved unreliable under variable lighting with results reliable only from a flatbed scanner and the system had no real-time process monitoring or adaptive parameter control.

3. Development of an Automated System for Soldering and the Solder Inspection of an Assembly Line

This study proposes an automatic robotic assembly cell for soldering and solder defect inspection in the electronic component manufacturing line by Lee et al. [3]. To perform operations like heat caulking, PCB soldering, and defect inspection, the system integrates offline robot programming, real-time visual feedback and automated process control. In this assembly line, vision sensors and industrial robotic manipulators are integrated. The redesigned automated cell enhanced the overall production dependability, improved soldering accuracy and fault detection capabilities.

When compared to previous manual or semi-automated setups, the experimental results showed reduced failure rates and improved operational efficiency. The limitation faced by this study was that this approach depends heavily on specialized industrial hardware and controlled factory environments which may limit easy deployment in smaller production setups. Adapting to other electronic configuration manufacturing processes may require redesign as the system only focuses on a specific camera assembly line. And real-time performance of the system can be affected by vision system accuracy and system calibration requirements.

4. Development of an automated system for the soldering of USB cables

A fully automated robotic system for the soldering of multi-wire USB cables is presented by Gao et al. [4]. The technique that the system uses is three sequential mechanical modules - a spring-loaded separating module, a rack-and-pinion sorting module that includes a camera-guided U-shaped gripper and a cylinder-driven pressing module. The transition takes place from one stage to another via the AMRWD deep vision network built on ResNet-50-FPN with Feature Pyramid Attention across varying illumination and cluttered backgrounds for robust wire detection. For the dataset, 2,000 images were captured covering all 24 colour permutations, two illumination bands and three background types in which 1,600 images were used for training and 400 for testing and for annotation Labelme was used alongside standard industrial hardware including stepper motor-driven modules, a PCI control card and a Basler camera.

The overall precision of 97.66% and recall of 97.48% was achieved by the AMRWD which outperformed PSPNet, DeepLabv3+, U-Net, and colour-based methods with an inference time of 420 ms versus 570 ms for U-Net and a 95% of module-level success rate for separation, >91% for sorting and ~100% for pressing. Mid-section wire deformation causes occasional sorting failures and the stepper motor based system yields a 32-second cycle time. There is no in-process solder quality monitoring integrated into the pipeline.

5. Vision-Based Robotic Manipulation of Flexible PCBs

Xiang Li et al. [5] propose a system that presents a vision-based robotic manipulation method that handles flexible printed circuit boards (FPCBs). The technique employed in this paper is

the Neural network estimators that are used for both PCB deformation behaviour and the camera mapping that enables adaptive control under visual and dynamic uncertainties. To observe the PCB during manipulation the experimental setup uses an industrial robot manipulator with a vision sensor and to guide the robot in real time image feedback is processed to estimate the deformation. For modeling uncertainties and to allow accurate shaping of PCBs for soldering tasks it is compensated by the adaptive neural network based approach.

System stability is evaluated and validated through Lyapunov-based analysis and the results are confirmed through practical robotic experiments. For conventional robotic systems it is difficult for automated manipulation of deformable electronic components but this approach makes it possible. The system avoids complex physical modeling and allows flexible PCB handling in robotic soldering applications by combining vision feedback with adaptive neural-network control. The limitation faced by this study is that this method depends heavily on vision feedback and neural network approximation which may be affected by lighting variations or occlusions. As the experiments are conducted in controlled environments, further work is required to improve robustness and make it effective and increase scalability for high-speed industrial production settings.

6. Automatic Optical Inspection System for Solder Quality Classification of THT Solder Joints

Fonseka and Jayasinghe [6] created an system optical inspection that checks the quality of the solder connections on printed circuit boards which uses through hole technology . For spot soldering flaws which occur during the electronic component production for this they have used machine vision technology. The system uses image processing methods, which contain taking photos next, editing them later on dividing them into few sections and identifying key features. This actually helps to find the problems occurring from solder like solder bridging, missing solder, too much solder. To take 300 pictures of the solder joint they have used a machine vision camera for analysis. Algorithms are used to examine these images taken from machine vision cameras to judge the quality of the solder connections. The results show that the system works well for speed up inspection and for detecting solder defects compared to the traditional methods.This approach allows for quick inspections, consistent and reducing errors in quality control. Although the system's performance depends on the lighting and image quality, it may need some changes when used with different printed circuit board designs.

7. Deep Learning-Based PCB Defect Detection Using X-ray Images

Zhang and his team [7] introduced a method that uses learning to find defects in printed circuit boards using X-ray images. This study aims to find solder defects that are hard to spot with regular optical inspection methods. The researchers proposed a deep learning framework that combines Convolutional Neural Networks with Long Short-term Memory networks. In this method the Convolutional Neural Networks model is used to find features from X-ray images. The Long Short-term Memory networks are used to find patterns in a sequence,

which improves the defect detection capabilities. The model was. Tested on a dataset of 3D X-ray images of printed circuit boards with various types of solder joint defects. The system's performance was checked using recall rate and false positive rate metrics. The results show that the deep learning model is better than inspection methods in finding defects. The good thing about this approach is that it can find defects in solder joints. However this method needs a lot of labeled data and powerful computers, which can make training and implementation more complicated.

8. Deep Learning-Based Anomaly Detection for Solder Joint Inspection

Mendizabel-Arrieta and his team [8] introduced a system to find anomalies in solder joints using learning. This system is mainly built for use in manufacturing environments. The study aims to improve quality control in industries by using computer vision and artificial intelligence. The authors developed a model that can find solder patterns without relying on predefined defect criteria. To acquire images of solder joints the system utilizes a camera for formations from various angles. These images are then processed to detect possible defects. A set of collected images representing solder joint formations, got from real-world manufacturing scenarios, was used for training and testing models. The benefit of the proposed system was tested by using F1 score and recall matrics. An F1 score close to 0.90 was achieved, which shows a high degree of accuracy. The major advantage of this method is its ability to detect unusual patterns on its own. Nonetheless, the system necessitates controlled lighting and high quality camera setups for dependable results, and the model might require retraining if production conditions alter.

9. ChangeChip:Reference-Based Unsupervised Change Detection for PCB Defect Detection

Fridman et al. have presented an unsupervised defect detection technique named as Change chip which aims at inspecting the printed circuit boards by referencing. This study focuses on the use of computer vision for automated PCB defect detection without use of labeled datasets. And the method identifies the detect by comparing a reference PCB image with a test image. To simplify this principle component analysis is employed for decreasing the dimensionality while K-means clustering is used to defect the areas. Moreover the scale invariant feature transform is applied to align the reference and test images accurately for comparison. They have developed a dataset called CD-PCB which has 20 pairs of PCB images with labeled defect areas for assessment. The overall effectiveness of the algorithm was calculated by using the precision and recall metrics, and achieved precision of 0.80 and recall of 0.87. The main gain point of this method is that it detects various PCB defects without having a large amount of labeled training data. However the approach is open to image misalignment which can lead to detection of noise or minor variations as defects.

10. DemoTeach-Robot Programming with Motion Capture System for Foundry Applications

Li et al.[10] introduced a robot programming system named DemoTeach which is based on motion capture and designed for industrial manufacturing purpose.This study aims to enhance human-robot interactions by enabling workers to instruct robots through physical demonstrations rather than intricate programming. In this approach workers execute tasks using a dummy tool fitted with reflective markers and their movements are recorded using an OptiTrack motion capture system. Then the robotic arm can follow the robot trajectories which are created from the recorded motion data. To perform deburning operations on aluminum casting components a KUKU iiwa robotic arm and an OptiTrack camera setup were used in this system's implementation. The outcome of the experiments show that the robot could successfully finish the given task and precisely mimic the human movement displayed. The main advantage of this system is that it makes robot programming easier, making it possible for operators without programming experience to easily train robots. Although, minor trajectory errors may arise from variations in hum hand movements,requiring extra correction mechanisms.

11. Machine Vision Based Solder Joint Positioning

Yutong Yuan et al.[11] proposed system which is on machine vision for the positioning of solder joints in body in white welding inspection in the automotive industry.The main goal of the study is to precisely locate the solder joints, even in harsh environments where lighting, oil stains, and electrode wear are present.The authors categorized the issue under machine vision-based techniques for solder joint localization. Rather than relying on conventional template matching methods, which are prone to nose and lighting fluctuations, the study employs texture feature analysis in conjunction with regional statistical analysis to pinpoint solder joint locations.The proposed methodology begins with the extraction of the region of interest from the images, followed by the conversion of the images into grayscale and HSV modes.The next step in the proposed algorithm is the application of texture filtering,which enhances the edge features of the solder joints. The algorithm then uses a circular sliding window scan method to locate the region with the maximum feature density, thus indicating the solder joint. The image acquisition and processing are carried out with the integration of cameras,lenses,lighting, and an industrial computer.The algorithm was evaluated in both standard and complex industrial settings.Experimental findings show that our method works in accurately positioning solder joints and resists lighting interference. Most solder joints are detected within errors. However the system still needs image quality and predefined limits to work properly. This might be a problem in noisy environments. Overall, the study illustrated that integrating texture features with statistical analysis can greatly enhance the accuracy of solder joint positioning in industrial apps.

12. Adaptive Template Matching for BGA Solder Joint Detection

Hao et al.[12] Introduction a method for adaptively matching techniques for detecting Ball Grid Array (BGA) solder joints using x-ray images of printed circular boards.The research

aims to enhance detection accuracy ,particularly when solder joints are obscured by chip packages. This study is categorized under image processing and template matching techniques for identifying solder joints .Since BGA solder joints are not easily visible with standard light, the authors employ X-ray imaging to examine the internal solder structures.This process starts with the extraction of the region of interest from the PCB image after taking an X-ray image. Then edge detection and threshold segmentation are performed to obtain an edge template. The template matching process is supported by the fact that the proposed system utilizes direction vectors based on edge features as prior knowledge. To enhance robustness, the algorithm conducts a global traversal search using an image pyramid structure,which aids in identifying potential solder joints at various scales .Finally ,least optimization is applied to fine -tune the matching results. Experimental findings indicate that the proposed method achieves rapid and precise solder joint detection, even in a complex environment with noise and varying illumination .However, the method may encounter difficulties when solder joint exhibits significant deformation or overlapping structures ,which can impact the accuracy of template matching.

13. PCB Soldering Defect Detection Using Template-Based Image Processing

S. Ozturk and B. Akdemir [13] introduced a computer vision-based system designed to inspect and identify soldering defects on printed circuit boards(PCBs). This Research falls under the category of automated optical inspection(AOI) systems that utilize image processing methods. The main aim is to spot soldering problems like short circuits between solder points, which can mess up how a circuit works.We need to find these issues because they can cause problems.Manual checking is not good because it is slow and people can make mistakes. It does not work well for making a lot of electronics.So we need a system that can automatically check for these problems.The method we suggest uses a way of looking at images along with a type of computer clustering called Fuzzy C-Means or FCM.First we change the picture of the printed circuit board or PCB from color to shades of gray.Then the FCM algorithm groups the image, into parts that show the background of the PCB the paths of the circuit and the solder joints.This helps us to see the different parts of the PCB.We use this technique to make it easier to spot any soldering issues.Once the joints are identified, a joint template is created based on the PCB design to establish the relationship between adjacent solder joints. In the course of inspection, each solder joint is compared with the other solder joints near to it, as indicated by the template. This template-based inspection process reduces computation time and increases the inspection speed. The template also helps identify the solder joints that are deliberately connected, thus enabling the inspection system to identify only unwanted sort circuits. The results obtained from the experiments on a PCB with 50 solder joints indicated that the system can identify the soldering defects. The inspection process was repeated 85 times with 100% accuracy for the PCB model. The proposed method for inspection is effective and fast; however, it requires a template to be created manually and the PCB to be positioned correctly for inspection. The study proves that template-based computer vision can be highly effective for the inspection of solder defects on a PCB.

14. Deep Learning Based Solder Joint Defect Detection

Zhang et al. [14] introduced a system for detecting solder joint defects using deep learning techniques applied to X-ray images of industrial PCBs. The aim of the research is to minimize the workload of inspecting the solder joints manually.Their research falls under the deep learning techniques used in the analysis of the X-rays. In the research, the authors use different AI techniques,including CNN,3D CNN, and LSTM. In the research, the authors use the following steps:the authors pre-process the images.In the pre-processing, the authors use techniques that involve the extraction of the region of interest and stacking the slices. Different pre-processing techniques were used to handle the varying number of slices and the presence of noise. Four AI models were assessed to identify the most effective architecture for detecting defects. The experimental findings indicate that the deep learning models considerably enhanced detection accuracy compared to conventional inspection techniques.The new system also makes things easier for specialists. It filters out a lot of solder joints, which is about 73.29%. This means that experts can focus on the solder joints that are actually defective.The problem with using learning is that it needs a lot of labeled datasets and strong computers, which can make it hard to use in real life.A lot of work is needed to prepare the data. The computers need to be very powerful which might limit how deep learning approaches can be used.

15. SMT Solder Joint Quality Inspection Using Surface Reconstruction

Yu and Zhang[15] introduced a system using surface reconstruction techniques for inspecting 3D solder joints. By using 3D analysis instead of the traditional 2D image inspection their study looks for improvement of solder joint quality inspection. They classify their method for solder joint analysis under 3D image reconstruction techniques.Initially, the system applies image denoising through wavelet packet decomposition, Wiener filtering, and median filtering to eliminate noise from the image of solder joints.Following the denoising process the system rebuilds the surface of the solder joint using Shape-from-Shading technology.This method uses a lighting model that includes both diffuse and specular reflection parts to accurately rebuild the solder joint surface.The reconstructed 3D model of the solder joint is then examined.From this we can find out parameters like the shape, height and curvature of the solder joint.These parameters help in finding defects and checking the quality of the solder joint accurately than traditional 2D methods.The solder joint parameters give a picture of its quality.The 3D model of the solder joint is very helpful in this process.It provides information about the solder joint.The Shape-from-Shading technology plays a role here.It helps in creating a 3D model of the solder joint. Experimental findings indicate that the proposed system effectively extracts 3D quality information of solder joints and enhances inspection reliability. Nonetheless, the method involves intricate computations and relies significantly on precise illumination modeling, which may lead to increased processing time.

16. Vision-based soldering process parameters calculation for robotic soldering

Hao Ren, Xinyu, Wu, and Wanfeng Shang [16] focuses on Through Hole Technology (THT) assembly automation as it is highly required in aerospace and military to form more robust connections because of its high reliability rather than simple surface mount alternatives. The paper presents the current industrial need for building vision assisted fully autonomous frameworks rather than teaching playbacks, which require manual training and debugging. The method used here is mathematical modeling rather than the traditional trial and error method, as the mathematical approach calculates the required solder volume precisely using high order curves and also approximately gives the longitudinal profile of a standard solder joint. Directly from the PCB, the pad radius, pin cross sectional area and through hole radius data are extracted using Hough transforms, HSV color space ,asking and Canny edge detection. These data are then combined with a soldering database to provide consistency among different types of components like indicators and LEDs, it also prevents duplicate computations.During the performance analysis the system shows 99% of similarity when empirical data was optimized against existing literature data. The average computational time for vision is 18.99 ms and for calculation it requires 12.40 ms which means to solder a new joint the total average processing time is 31.39 ms which is very much faster and more efficient compared to the system requiring human intervention.

17. Experimental Path Tracking of a Cartesian Robot Using Image Processing

Allahyari and Fakhari [17] focuses on an essential framework that has high flexibility path tracking and is also cost effective using a 3 DOF cartesian robot. For simulation they have used matlab software which includes IP camera software, Machine vision toolbox, and stepper motor driven linear actuators as they provide a viable entry point for small scale soldering automation. To define non linear paths like sinusoidal, circular courses they have used passive optical sensing. Based on image processing they have captured images to create a dataset which is exported to matlab to extract the coordinates of image, these coordinates are then sent to an Arduino UNO microcontroller using a USART protocol at 38,400 baud rate. For processing they have used an HSV color model where they set a threshold at >0.7 saturation level in order to obtain better robustness in industrial lighting conditions without the need of using an external light source. The system shows a high level image processing precision with up to 1 mm of maximum image processing error and roughly 0.48 pixels at mean reprojection error but the mechanical tracking performance it depends on the path complexity, for linear paths it shows acceptable deviations while for circular path tracking it exhibits error up to 3mm. The major benefit of this approach is that it is accessible, reduces multi pixel pathways to single pixel trajectory but on the flip side the memory constraint of the Arduino UNO is major setback as thousands of paths are to be reduced to a maximum of 128 co-ordinate, due to the reduction in this data the discretization error are evident during high curvature paths as the robot does not travel in smooth true curve instead it follows points between short linear segments. In this paper the vision pipeline is very accurate for soldering trajectories but the

hardware mechanical planning lacks accuracy in real time due to the memory limitation of the microcontroller.

18. Solderbot: Automated Soldering using Robot Arm

This paper presents various robotic arm applications and automated manufacturing techniques that includes deep learning based robot controls, vision based soldering, point cloud based motion planning and visual programming for robots. In this paper they have used OpenCV for image processing, RoboDK for offline motion planning and Gazebo for 6 DOF robotic simulation.The reason they chose OpenCV instead of yolo or other deep learning methods is because calculation of soldering targets does not require the complex neural network computation as they are geometrically straightforward.

The advantage of this bot is that it is cost effective and does not use complex AI models or expensive hardware but it has significant limitations. The results are only validated through simulation while for real time hardware the marker based recognition is not suitable for small metal contacts of common PCBs, the amount of solder to be dispensed is supposed to be calculated mathematically rather than by any closed loop feedback and this method does not generalize to densely populated boards. This method is only suitable for educational grade rather than industrial grade systems.

19. Optimization of the soldering process by the DMAIC methodology

This paper follows DMAIC (Define - Measure - Analyze - Improve - control) methodology. It focuses on automotive PCB soldering applications, mainly soldering of pins that connect a controller board and a wire harness plug. This process has three observable modes of failure i.e solder alloy breakage of spool, operator correction time and part scrapping. The research shows that DMAIC effectively reduces the variability of soldering defects and also reduces its magnitude with a drop from 4000 - 16000 PPM range to 1000-2700 PPM which is below 3000 PPM target using PFMEA analysis, sustained control plan, and targeted corrective actions. The study specifically addresses seven highest RPN root causes which includes oxidised solder alloy, operator skill and soldering pin condition but it also has some limitations such as the comparative analysis against different approaches such as DOE based parameter optimization is not performed, it does not have any statistical, simulation or modelling data to validate the corrective actions before the implementation and mainly depends on human trainings and compliance with new procedures and thus the system stability is not guaranteed.

20. Optimization of Process Parameters in Lead-free Wave Soldering Based on Orthogonal Experiment and Analytic Hierarchy Process

Yin et al.[20] presents the research on the transition to lead free soldering driven by EU-RoHS directive, it was observed that though much research is done on lead free solder alloy formation using Sn- Zn, SnAgCu etc. To examine and figure out the four parameters,

flux quantity, preheat temperature,track speed, and soldering temperature at three levels they have used the L9(3^4) orthogonal array as it is a structured and effective substitute for full factorial experiments. The analytical method to quantify and rank the influence of each parameter level the AHP (analytical hierarchy process) is introduced using a pairwise comparison matrix derived from observed PPM defect rates. The dataset consist of 100 PCBs per experimental group which consist of 446 solder joints each. The primary result of this paper gives the important process parameter that is track speed of 44.5% total AHP weight, followed by 26.6% of flux quantity, 16.1% soldering temperature, and 12.8% of preheat temperature with ideal settings of 40 mL/min flux, 110 degree celsius preheat, 1.6 m/min track speed and 265 degree celsius soldering temperature. Another important finding is that if we reverse the pcb direction it reduces defects from 10,291 PPM to 4,574 PPM at the same parameter values.

21. Development of a vision system integrated with industrial robots for online weld

Seam tracking Representing the stage of highly specialized industrial vision systems, this study conducted by Nguyen et al. develops a precision-focused robotic tracking platform that employs a line laser projector and a CMOS camera to acquire 3D object data via triangulation. To handle complex demands like tracking square-groove butt joints, the researchers integrated a discriminated convolution tracker refined by deep reinforcement learning (DRL), which continuously outputs corrective pixel-movement actions based on visual states. Validated in controlled lab environments on custom datasets, this methodology achieved an absolute mean welding error of merely 0.31 mm. While its extreme precision and reliability in visually noisy industrial environments represent a significant strength, the system's reliance on rigid calibration and narrow applicability highlight the inherent limitations of task-specific AI compared to modern generalist approaches.

22. RT-2: Vision-Language-Action Models Transfer Web Knowledge to Robotic Control

Marking a huge breakthrough toward generalist autoregressive policies, Brohan et al. RT-2 is a foundational Vision-Language-Action (VLA) model that directly combines internet-scale knowledge into robotic control systems. Rt-2 trains itself to produce robotic trajectories by representing physical action in the form of test token within standard language model vocabulary, which is achieved by employing massive pre-training on a set of visual question and image captioning datasets. Which enables Rt-2 for unprecedented emergent semantic generalization, this gives the robot a perspective about the properties of object or recognizes improvised tool zero short in real world performance hence allowing reasoning. One the other hand, the innovative ability of mapping visual and warble linguistic inputs to produce an action comes at a cost due to its massive parameter count of up to 55B making it difficult for local development and making it slow for high-frequency control application as it works on autoregressive generation mechanism.

23. OpenVLA: An Open-Source Vision-Language-Action Model

To modify and improve upon generalist autoregressive VLA architectures, Xiao et al. present OpenVLA is a 7B-parameter fully open-source foundation model. Built by combining advanced Vision Transformers DINOv2 and SigLIP with a LLaMA 2 backbone, the model was trained on the Open X-Embodiment dataset, encompassing over 970,000 robot episodes of robot morphologies. By capturing spatial-semantic features, OpenVLA exhibits a significant improvement in generalization capabilities, outperforming RT-2-X model by 16.5% in task success rates across unseen robot embodiments. Its open-source availability and support for parameter-efficient fine-tuning (PEFT) make it a highly versatile tool for researchers; nevertheless, because it relies on the same discrete token-by-token decoding as RT-2, it still faces limitations regarding inference latency in highly dynamic, real-time physical environments.

24. π_0 A Vision-Language-Action Flow Model for General Robot Control

To address the inference and skill challenges of autoregressive VLA models, Black et al. propose **π_0,** a versatile policy that moves towards continuous control methods. Rather than breaking down physical actions into text tokens, **π_0** pairs a pre-trained Vision-Language Model backbone with an independent "action expert". This expert creates continuous actions through flow matching, which is a type of diffusion. The model was trained on a large cross-embodiment dataset with over 10,000 hours of robot manipulation data. It was tested on complex tasks that take time, like bussing tables and folding laundry in sequence. The flow matching design effectively connects high-level semantic thinking with fast (up to 50 Hz) continuous motor control. However, it still faces challenges due to the significant computing resources needed for pre-training and the lack of high-quality demonstration data for completely new robot designs.

25. Fine-Tuning Vision-Language-Action Models

Optimizing Speed and Success Focusing on the optimal deployment and adaptation of existing foundation models, Kim et al. developed the OpenVLA-Optimized Fine-Tuning (OFT) recipe to overcome the speed limitations of standard autoregressive generation. By modifying the base OpenVLA architecture to utilize parallel decoding, continuous action representations, and a simplified L1 regression learning objective, the model can predict entire chunks of future actions simultaneously in a single forward pass. Evaluated heavily on the LIBERO simulation benchmark and real-world ALOHA bimanual teleoperation setups, OpenVLA-OFT achieved a staggering 26x to 43x increase in action generation throughput, which directly correlated with boosting the average success rate from 76.5% to a state-of-the-art 97.1%. This methodology provides a vital strength by enabling high-speed, reliable fine-tuning for new environments, although fully capitalizing on these optimizations still requires starting from a heavily resource-intensive pretrained base model.

3.1 Patent Survey

Patent Landscape Overview

Global patent activity in robotic soldering and vision-guided electronics assembly has been intensifying. China (CNIPA) accounts for approximately 46% of all global patent filings and dominates soldering automation volume. Collaborative robot patent filings peaked at around 797 applications in 2022 under CPC class B25J9. The most active filers in vision-assisted robotic soldering are Cognex, KEYENCE, Apollo Seiko, Japan Unix, and BAE Systems.

Main patent classification areas for this domain:

Table No. 3.2 - Summary of Recent Patents Related to Vision-Assisted PCB Inspection and Robotic Soldering

Patent	Title	Assignee	Link	Year
US 20240428399	Hybrid ML/MB for PCB Defect Detection via autoencoder	BAE Systems	View	2024
US 11961218 B2	Auto-generate vision inspection jobs from PCB ROI using AI ensemble	—	View	2024
US 20240062549	Standalone vision system verifying solder joint formation + MES output	—	View	2024
CN 118493414 A	Robot vision path generation from 3D PCB model — replaces teach-playback	Shuzhi Huineng	View	2024
Apollo Seiko	Position calibration camera + Remoca-FF 4K vision + Gerber integration	Apollo Seiko	View	Ongoing

Key Patents (2020–2025)

- **US 20240428399 : Hybrid ML/MB for Automated PCB Defect Detection** Assignee: BAE Systems Information and Electronic Systems Integration Filed: June 2023 | Published: December 2024 [26]
  Uses a hybrid autoencoder combining model-based and machine learning features at the latent space. Detects solder defects missing components, cold joints, and misplacements via image reconstruction error. Targets defense-grade THT PCB inspection for aerospace and military applications.

- **US 11961218 B2 : Auto-Generating Vision Inspection Jobs from PCB ROI** Filed: 2022 | Granted: April 2024 [27]

Trains a vision learning model on region-of-interest patches from PCB images using an ensemble of AI models. Outputs a machine vision job deployable to an imaging device that detects success or failure of solder joints, missing components, or blown capacitors without requiring expert vision engineers to hand-program each new product.

- **US 20240062549 : Standalone Vision System for Verifying Soldering Operations** Filed: August 2022 | Published: February 2024 [28]
  A vision program that verifies and reports that a selected soldering operation has been completed measuring the length and location of components, confirming correct solder joint formation, and sending output signals to manufacturing execution systems.

- **CN 118493414 A : Intelligent Robot Vision Image Acquisition Device and Method** Assignee: Shuzhi Huineng Dalian Technology Co., Ltd. Filed: July 2024 | Published: August 2024 [29]
  Generates an optimal image acquisition track from a 3D entity model of the PCB and sequences robot motion accordingly. Directly solves vision-guided robot path planning from CAD/3D model input replacing manual teach-playback for soldering trajectory generation.

- **Apollo Seiko : Position Calibration Camera + Remoca-FF Vision Integration** Ongoing patent portfolio covering 55+ years of continuous filings in selective soldering [30]. Includes patented position calibration camera (exclusive to their robots), Remoca-FF 4K vision system for teaching and process monitoring, and Gerber data integration for automatic path generation.

Identified Gaps & Research Opportunities

The patent survey reveals several areas that are currently underpatented, representing open space for novel research and IP development:

THT-specific vision-guided soldering : Most existing vision patents target surface mount technology (SMT). Through-Hole Technology specific vision guidance for aerospace and military applications is sparse in the patent landscape despite its industrial importance.

Mathematical solder volume modeling from PCB geometry : Computing the required solder volume directly from vision-extracted PCB parameters (pad radius, hole radius, pin cross-section) has no direct patent coverage despite strong practical value.

Real-time defect-adaptive closed-loop soldering : Systems that detect a defect during the soldering process and adjust parameters in real time rather than performing post-hoc inspection are essentially absent from the current patent landscape.

Low-cost vision pipelines for small-scale soldering : Budget-accessible vision systems using tools like OpenCV and MATLAB have no industrial-grade patent coverage, leaving the space open for research contributions.

3.2. Market Survey

Market Size & Growth

The global automated soldering machine market was valued at **USD 3.46 billion in 2024**, projected to reach **USD 5.5 billion by 2035** at a CAGR of 4.3%. The soldering robotics sub-segment alone is expected to reach **USD 0.37 billion by 2033** at 5.6% CAGR. Asia-Pacific leads with approximately 50% of global market share, driven by electronics manufacturing hubs in Japan, China, and South Korea.

Key growth drivers include:

- EV production (projected 40 million units/year by 2025) and 5G rollout driving high-density PCB demand
- EU-RoHS lead-free soldering mandates increasing process complexity
- Labor cost pressures accelerating full factory automation
- Industry 4.0 requirements — MES integration, traceability, IoT connectivity

Key Players in the Market

- **Japan Unix — Contact + Laser + Vision Soldering** [31]: One of the global leaders in robotic soldering. Their systems support both contact and laser soldering with an AOI vision sensor add-on for real-time position correction, fiducial correction camera, MES integration, and full board traceability.

- **Apollo Seiko — Selective Soldering with 4K Camera** [32,33]: A pioneer in selective soldering since 1969. Their Remoca-FF 4K camera system enables vision-based teaching and process monitoring. Gerber data integration allows automatic path generation without manual teach-playback.

- **Kurtz Ersa — Wave + Selective + Reflow** [34]: A leading European manufacturer covering all major soldering technologies. Strong presence in automotive and industrial electronics.

- **HAKKO — Iron Tip Robotic Soldering** [35,36]: Their HU-200 all-in-one soldering robot integrates solder feed control and programming software via tablet. Supports joystick-based teaching down to 0.01 mm precision. Also offers a Universal Robot integration kit for flexible automation.

- **QUICK — Desktop Vision Soldering Robots** [37,38]: The QUICK 9433 and 9434 Vision series feature closed-loop temperature sensors and a vision system that makes

real-time tip position adjustments on the fly — providing process forgiveness for positional variation.

- **Han's Laser — Laser Soldering for High-Density PCBs** [39,40]: Offers fully automated laser soldering systems with integrated visual positioning for PCB and IC soldering. Non-contact processing with local heating, small heat-affected zone, and precision down to 250 μm tin ball diameter. CCD real-time monitoring improves speed and accuracy.

Technology Trends (2023–2025)

Table.No 3.3 - Emerging Technology Trends in AI-Based PCB Detection and Vision-Assisted Robotic Soldering

Trend	What's Happening
AI replacing classical computer vision	Deep learning models replacing HSV thresholding and Hough transforms in commercial systems
Gerber/CAD auto-teaching	PCB design files auto-generate soldering paths, eliminating manual teach-playback
Laser + closed-loop vision feedback	Thermal and optical cameras regulate heat delivery in real time during soldering
Lead-free process adoption	Transition to eco-friendly alloys (SnAgCu, Sn-Zn) increasing process complexity
EV and 5G demand surge	High-density, heat-sensitive PCBs driving precision soldering requirements
Cobot integration	Vision-safe collaborative robots entering mixed human-robot soldering environments

3.3 VDI 2206

VDI 2206 Guidelines Used for the Project Flow

The proposed **Vision-Based Automated Cartesian Robotic Soldering System** follows the **VDI 2206 V-Model** shown in fig 3.3, a standardized design methodology for mechatronic systems. The development begins with defining system requirements such as automated PCB inspection, precise soldering, and safety interlocks. These requirements are decomposed into mechanical, electrical, electronic, and software subsystems, including the Cartesian robot, vision system, motion controller, soldering mechanism, and control logic.

After implementation, each module undergoes unit testing and is progressively integrated to verify communication, motion accuracy, and vision-guided operation. Finally, complete system validation ensures reliable PCB inspection, accurate coordinate-based soldering, safe operation through homing and emergency stop mechanisms, and successful completion of the automated soldering cycle. This structured approach minimizes design errors, improves traceability, and ensures seamless integration of all mechatronic components.

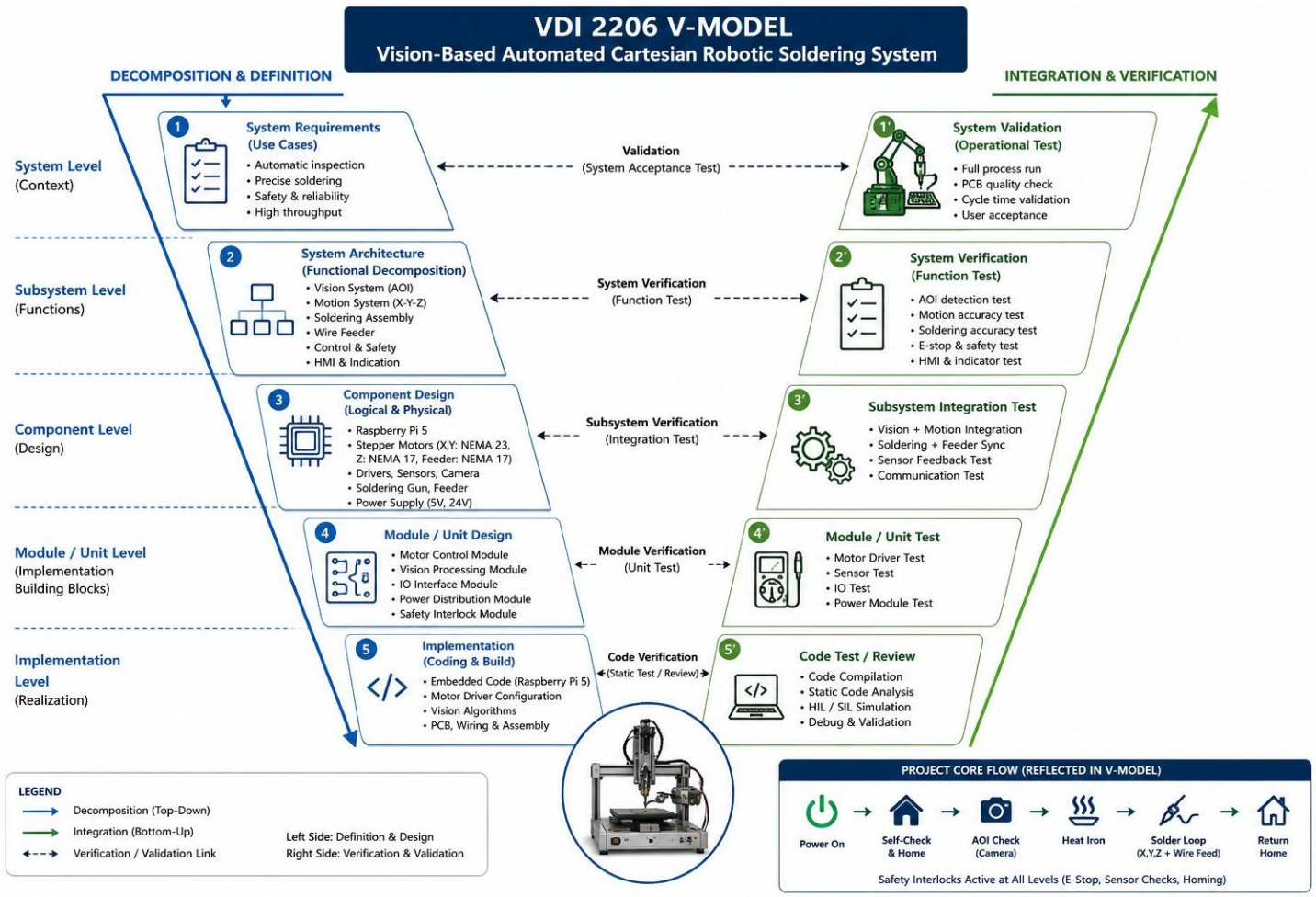


Fig.3.3 - VDI 2206 V-Model for Vision-Based Robotic Soldering System

4. PROBLEM DEFINITION

In electronic and light fabrication industries, traditional soldering methods are manual or semi-automated, leading to inconsistent solder joints, human error, and inability to detect defects such as misplaced or missing components in real time.

The **Aequs Koppal Toy Cluster**, currently specializing in plastic and wooden toy manufacturing, does not have a production line for electronic toys. To enable this capability, a reliable and automated PCB soldering system with quality inspection is essential.

This project proposes a **Vision-Assisted Robotic Soldering System** that uses computer vision and machine learning to verify PCB component placement in real time and commands the robotic arm to solder only defect-free boards ensuring precision, consistency, and scalable quality in electronic toy production.

4.1 Requirements List

4.1.1 Functional Requirement

Table.No 4.1.1 - Functional Requirement List

SI. No.	Functional Requirement
1	The system shall capture PCB images using a vision system for solder joint localization.
2	The system shall detect PCB position and orientation using vision-based algorithms.
3	The system shall identify solder pads, tracks, and component leads from PCB images.
4	The system shall extract coordinate points of solder joints for trajectory generation.
5	The system shall control robotic end-effector movement to reach solder joints.
6	The system shall perform motion along X, Y, and Z axes.
7	The system shall feed solder wire during the soldering operation.
8	The system shall control contact between the soldering tip and PCB joints.
9	The system shall retract the soldering tool after completing each joint.
10	The system shall perform system initialization before soldering operations.
11	The system shall restrict robotic motion within defined workspace limits.
12	The system shall avoid collision with PCB components during movement.
13	The system shall apply flux on the PCB surface before soldering.
14	The system shall inspect solder joints after completion using a vision system.
15	The system shall provide real-time visual feedback during soldering.

4.1.2 Non-Functional Requirement

Table.No 4.1.2 - Non-Functional Requirement List

Sl. No.	Non-Functional Requirement
1	The system shall operate in a stable manner during soldering operations.
2	The system shall maintain controlled thermal behavior at the soldering tip.
3	The system shall process vision data in real-time during operation.
4	The system shall ensure smooth and controlled robotic motion.
5	The system shall maintain coordination between subsystems.
6	The system shall operate without excessive vibration.
7	The system shall maintain consistent performance under working conditions.
8	The system shall initialize properly before execution.
9	The system shall prevent overheating during operation.
10	The system shall maintain uninterrupted subsystem communication.
11	The system shall operate within defined workspace boundaries.
12	The system shall provide clear operational status indication.

4.2 Constraints

Table.No 4.2.1 - Constraints List

Sl. No.	Constraint
1	The system shall operate only within the defined X, Y, and Z workspace limits of the Cartesian robot.
2	The PCB size shall not exceed the maximum working area of the robotic soldering setup.
3	The PCB shall be fixed firmly in the fixture and shall not move during image capture or soldering.
4	The camera shall be properly calibrated with the robot coordinate system.
5	Proper and uniform lighting shall be provided for accurate PCB image detection.
6	The soldering tip temperature shall remain within the safe temperature range for the PCB and components.
7	The robot speed and acceleration shall be limited to avoid damage to PCB tracks and components.
8	The robotic arm shall avoid collision with PCB components, fixtures, and other obstacles.

9	The soldering tip shall not apply excessive force on the PCB surface.
10	The solder wire feed rate shall be controlled to avoid excess or insufficient solder.
11	The system shall require a stable power supply for the robot, camera, controller, and soldering unit.
12	The system shall stop operation if PCB detection, coordinate extraction, or robot communication fails.

4.3 Conceptual Designs

4.3.1 Design 1

This design in fig 4.3.1 focuses on high-speed, precision performance in a controlled environment. The SCARA (Selective Compliance Assembly Robot Arm) architecture is uniquely rigid in the horizontal plane, enabling rapid, jerky movements between solder points without loss of accuracy. Functionally, it uses a parallel-mounted camera to provide real-time vision correction for flexible PCBs, all housed within a safety cabinet to manage fumes and lighting. The design philosophy is centred on industrial reliability and high-throughput production of standardised electronics.

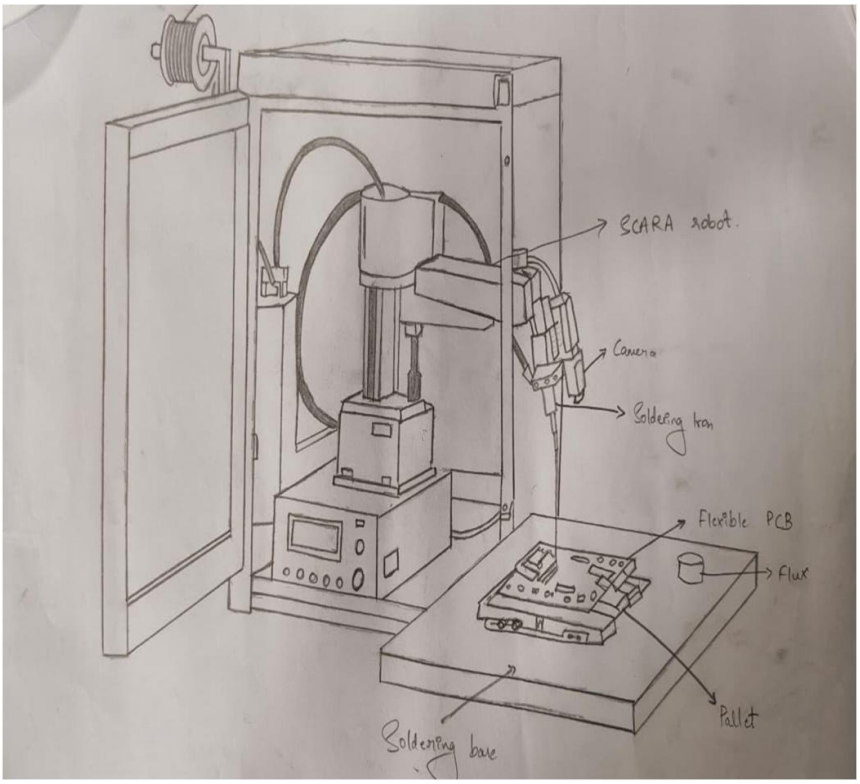


Fig.4.3.1 Conceptual Design of SCARA-Based Vision-Assisted Robotic Soldering System

4.3.2 Design 2

The 6-Axis Articulated Arm design in fig 4.3.2 represents the most versatile configuration for complex fabrication, utilizing a high-precision linear transport slide to position the PCB. The uniqueness of this system is defined by the arm's "human-like" degrees of freedom, which allow the soldering head to approach the workpiece from virtually any angle, enabling the soldering of 3D components or vertical pins that top-down systems cannot reach. Functionally, the lead-screw-driven transport line serves to precisely index the PCB back and forth under the camera's focal point, ensuring the vision system can provide the most accurate coordinates to the robot's controller.

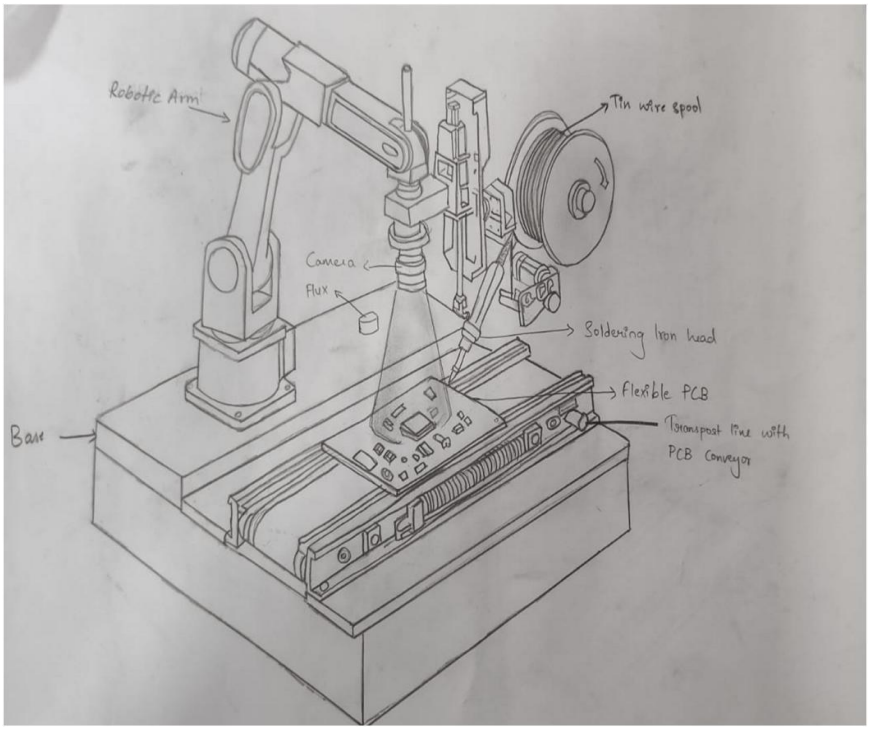


Fig.4.3.2 Conceptual Design of 6-Axis Articulated Vision-Guided Robotic Soldering System

4.3.3 Design 3

This configuration uses a linear coordinate system (X-Y-Z) to provide maximum structural stability and scalability as shown in fig 4.3.3. Its uniqueness lies in its simple, rigid frame and lead-screw driven movement, which offers the highest level of positional accuracy for dense component layouts. Functionally, it features an onboard controller for localised processing and an open-bed design that can be easily scaled for larger workpieces. The design philosophy emphasises versatility and ease of maintenance, making it ideal for light fabrication and prototyping.

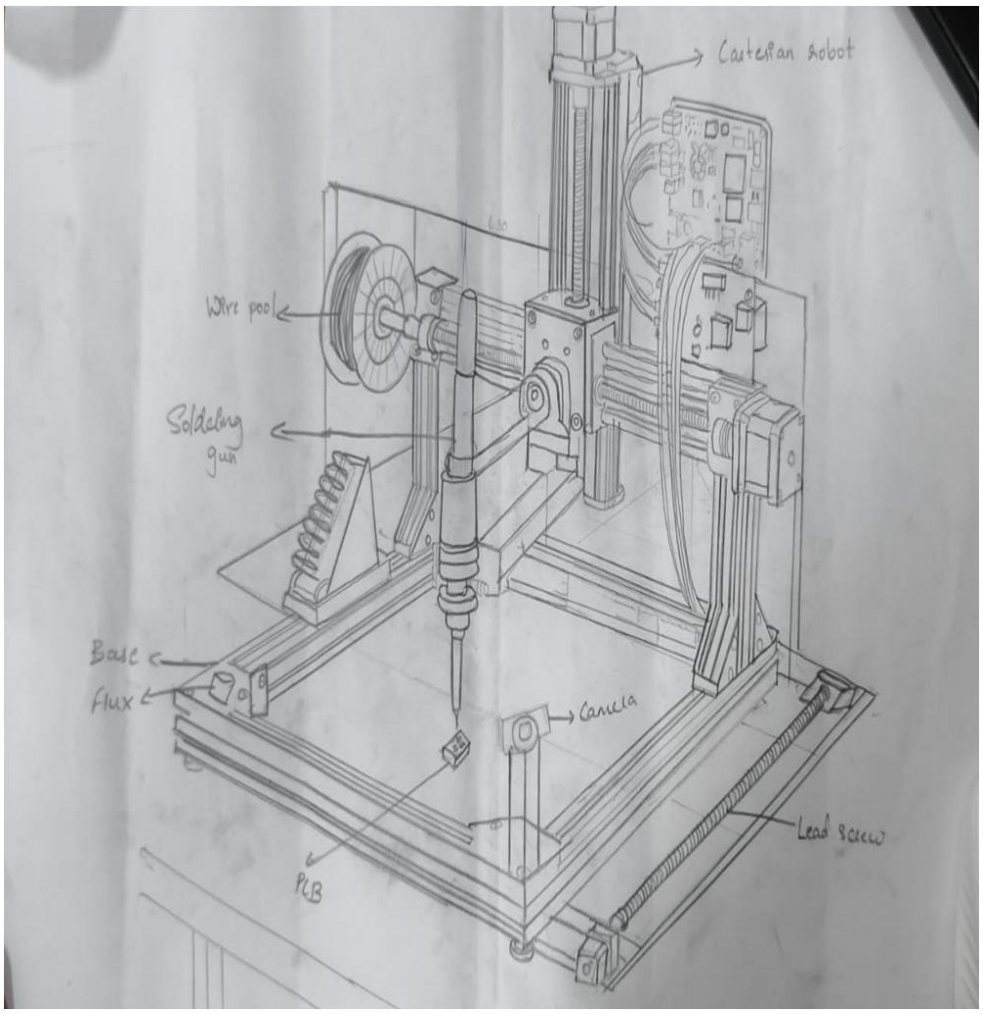


Fig.4.3.3 Conceptual Design of Cartesian (X-Y-Z) Vision-Assisted Robotic Soldering System

4.3.4 Design 4

This design in fig 4.3.4 represents a vision-assisted robotic soldering station with a structured and modular setup. It includes an HD vision camera mounted above the work area to capture real-time images of the PCB during soldering. The robotic arm performs precise soldering operations on a PCB fixed on an XY worktable, enabling accurate positioning and movement. A monitoring system displays the process, allowing observation and control of the operation. The PCB fixture ensures stability and proper alignment during soldering. The system focuses on improving soldering accuracy and reducing manual errors through vision guidance. Overall, the design emphasizes precision, stability, and efficient automation for electronic assembly tasks

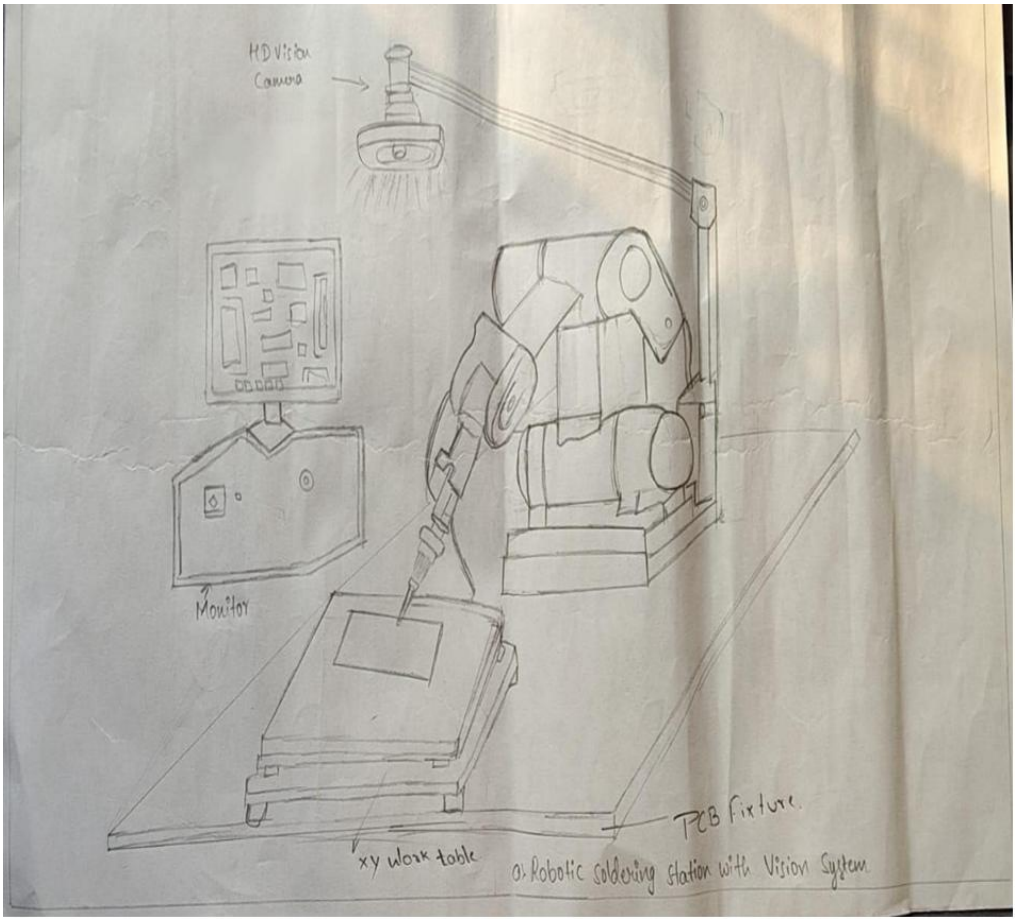


Fig.4.3.4 Conceptual Design of Modular Vision-Assisted PCB Robotic Soldering Workstation

4.3.5 Design 5

This design in fig 4.3.5 shows a compact vision-guided robotic soldering system suitable for small-scale or lab environments. An industrial camera with LED lighting is mounted near the robotic arm to capture clear images of the PCB for accurate solder joint detection. The robotic arm performs precise soldering operations using an automated soldering iron as the end effector. A PCB fixture securely holds the board in position to ensure stability during the process. The system is controlled through a touchscreen control panel, allowing easy operation and monitoring. A separate monitor displays real-time vision feedback and system status. Overall, the setup focuses on compactness, automation, and improved soldering accuracy with minimal human intervention.

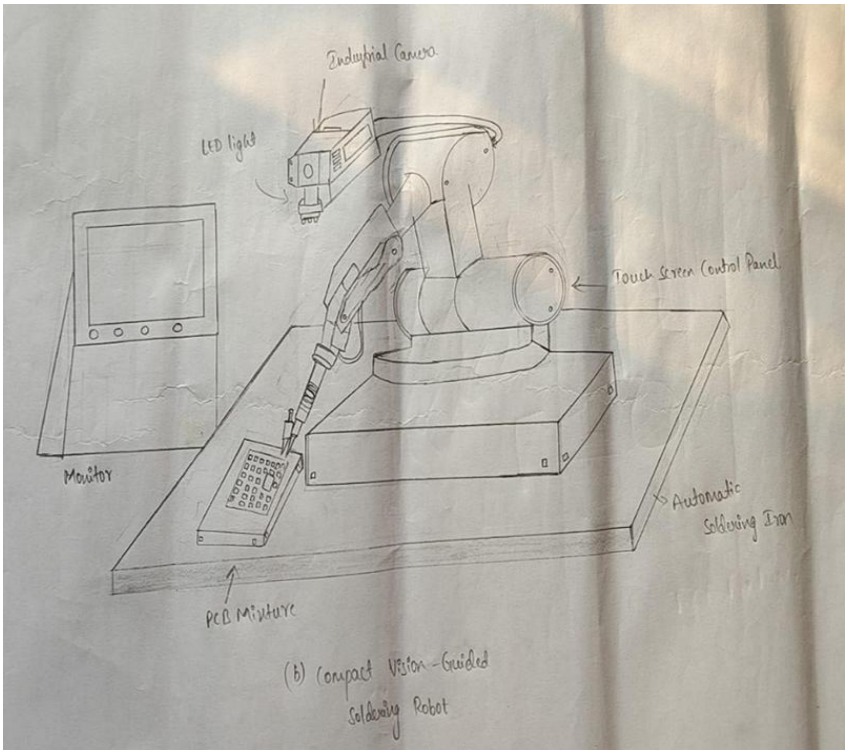


Fig.4.3.5 Conceptual Design of Compact Vision-Guided Robotic Soldering System for PCB Assembly

4.4 Functional Architecture

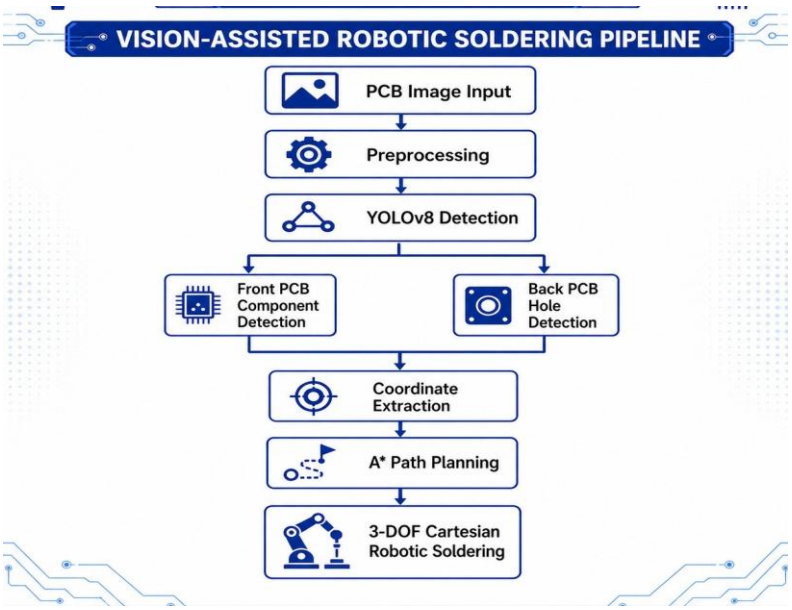


Fig.4.4 Workflow of the proposed vision-assisted robotic soldering pipeline.

The fig 4.4 shows the process flow of a **Vision-Assisted Robotic Soldering Pipeline**. First, an image of the PCB is captured and preprocessed to improve its quality. Then, the YOLOv8 AI model detects the components on the front side of the PCB and the holes on the back side. After detection, the exact coordinates of the soldering points are extracted. Using the A* path-planning algorithm, the system finds the shortest and safest path for robot movement.

Finally, a 3-DOF Cartesian robotic arm moves in X, Y, and Z directions and performs accurate soldering at the required PCB locations.

4.6 Concept Evaluation

Table 4.6. Concept evaluation of the proposed robotic soldering system.

Sl. No.	Selection Criteria	Weight	D1 Rating	D1 Score	D2 Rating	D2 Score	D3 Rating	D3 Score	D4 Rating	D4 Score	D5 Rating	D5 Score
1	Work Envelope / Reach	8	3	24	4	32	3	24	4	32	4	32
2	Power Consumption Efficiency	7	3	21	3	21	5	35	2	14	2	14
3	System Processing Speed (Real-Time Capability)	8	3	24	2	16	2	16	5	40	5	40
4	Robotic Arm Precision (X, Y, Z Movement)	9	3	27	2	18	5	45	5	45	5	45
5	Reliability Under Continuous Operation	10	3	30	2	20	5	50	3	30	3	30
6	Ease of Maintenance and Servicing	7	3	21	4	28	5	35	3	21	3	21
7	Cost of Implementation	5	3	15	2	10	5	25	3	15	3	15
8	Lead Time	6	3	18	3	18	5	30	2	12	5	30
9	Mechanical Stability and Rigidity	9	3	27	3	27	5	45	2	18	2	18
10	Safety	10	3	30	2	20	5	50	2	20	2	20
	Total Weighted Score			237		210		355		247		265
	Rank			4		5		1		3		2

4.7 Screening

Table 4.7.Concept screening of the proposed robotic soldering system

SI. No.	Selection Criteria	Weight	Design 1 (Datum)	Design 2	Design 3	Design 4	Design 5
1	Work Envelope / Reach	8	0	+	0	+	+
2	Power Consumption Efficiency	7	0	0	+	-	-
3	System Processing Speed (Real-Time Capability)	8	0	-	-	+	+
4	Robotic Arm Precision (X, Y, Z Movement)	9	0	-	+	+	+
5	Reliability Under Continuous Operation	10	0	-	+	0	0
6	Ease of Maintenance and Servicing	7	0	+	+	0	0
7	Cost of Implementation	5	0	-	+	0	0
8	Lead Time	6	0	0	+	-	+
9	Mechanical Stability and Rigidity	9	0	0	+	-	-
10	Safety	10	0	-	+	-	-
PLUSES			-	2	8	3	4
SAME			-	3	0	3	3
MINUS			-	5	1	4	3
NET			-	-3	7	-1	1
RANK			-	4	1	3	2
CONTINUE?			No	No	Yes	No	No

4.8 Finalized Concept

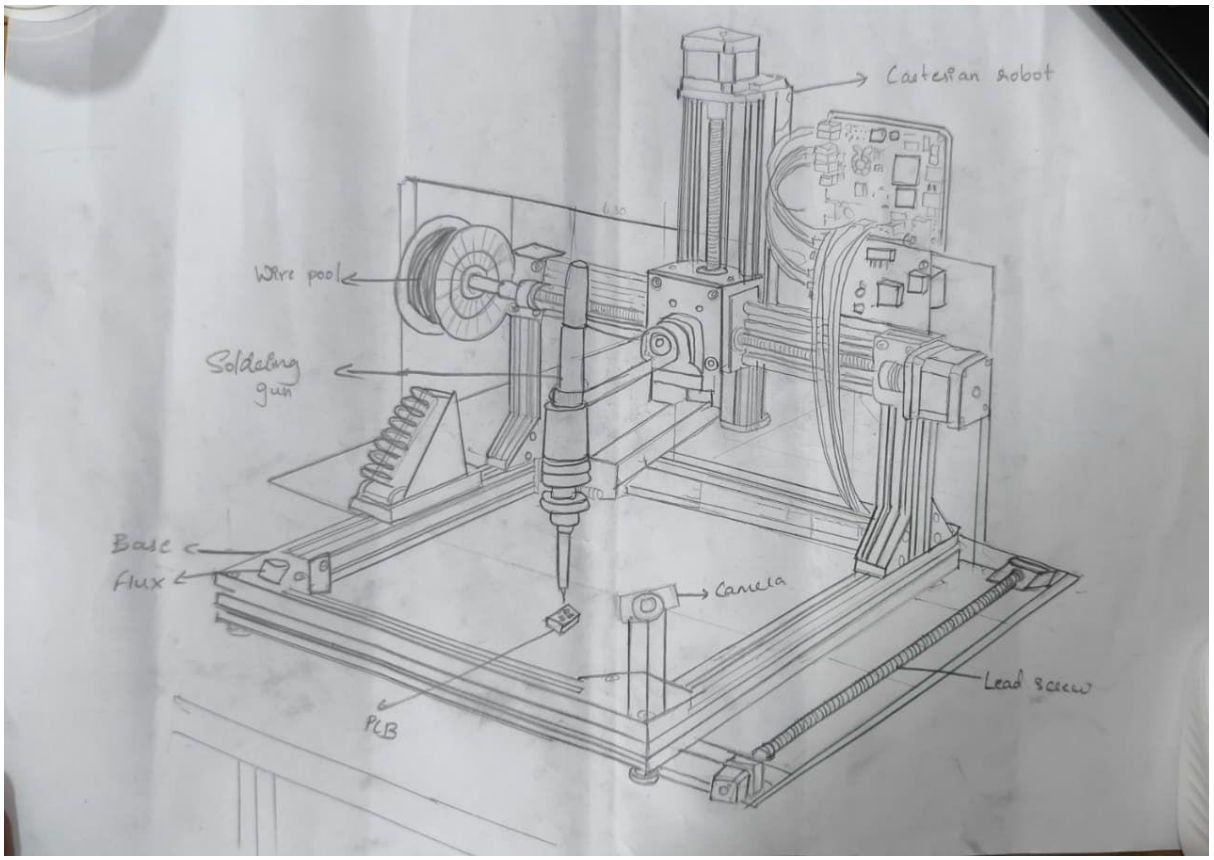


Fig.4.8. Finalized conceptual design of the proposed vision-based automated soldering system.

5. VIRTUAL PROTOTYPING

5.1 Freehand Sketch

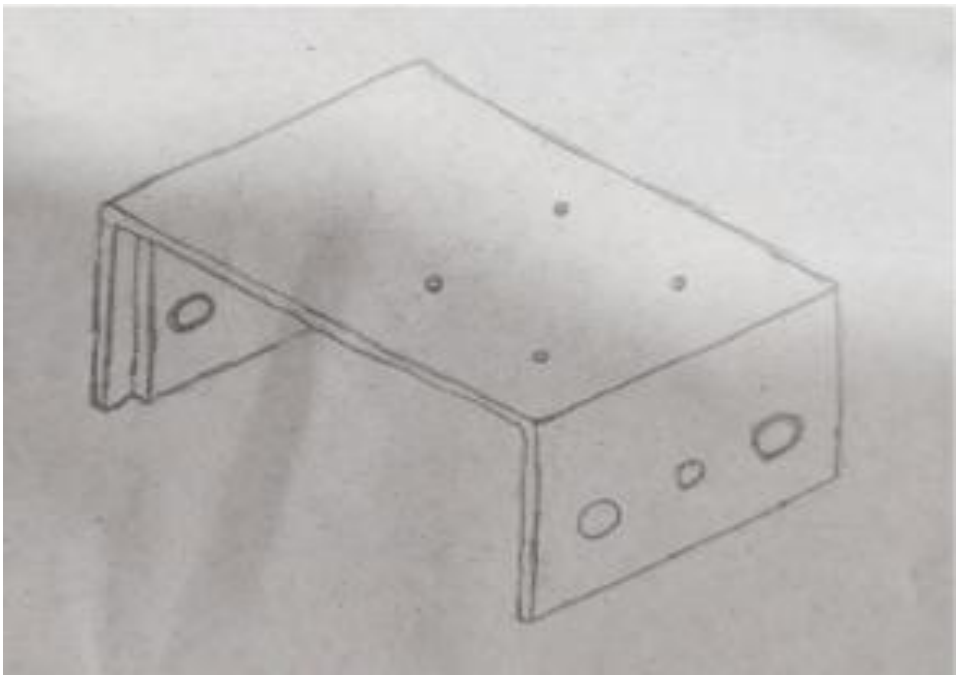


Fig.5.1.1. Freehand sketch of the back plate.

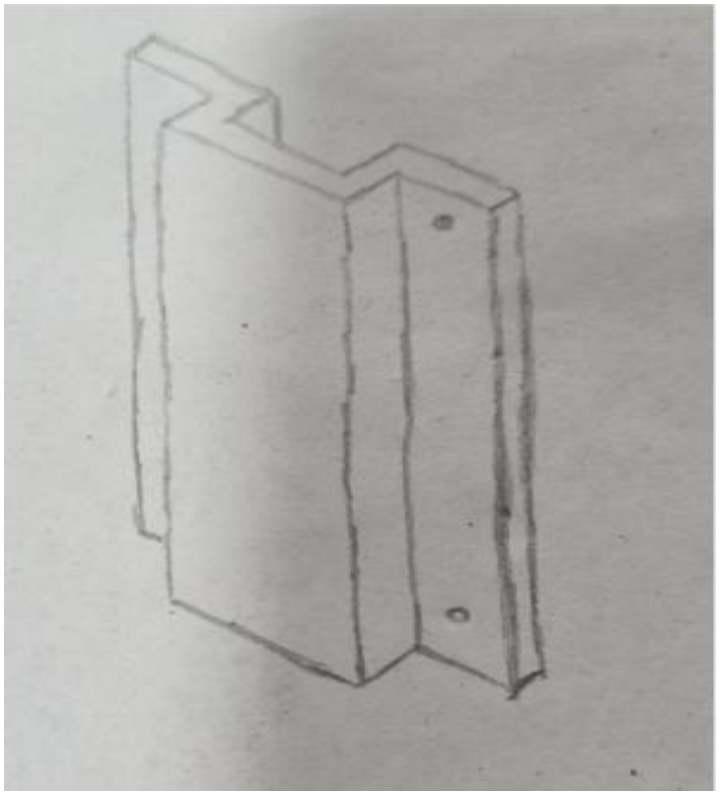


Fig.5.1.2. Freehand sketch of the front plate.

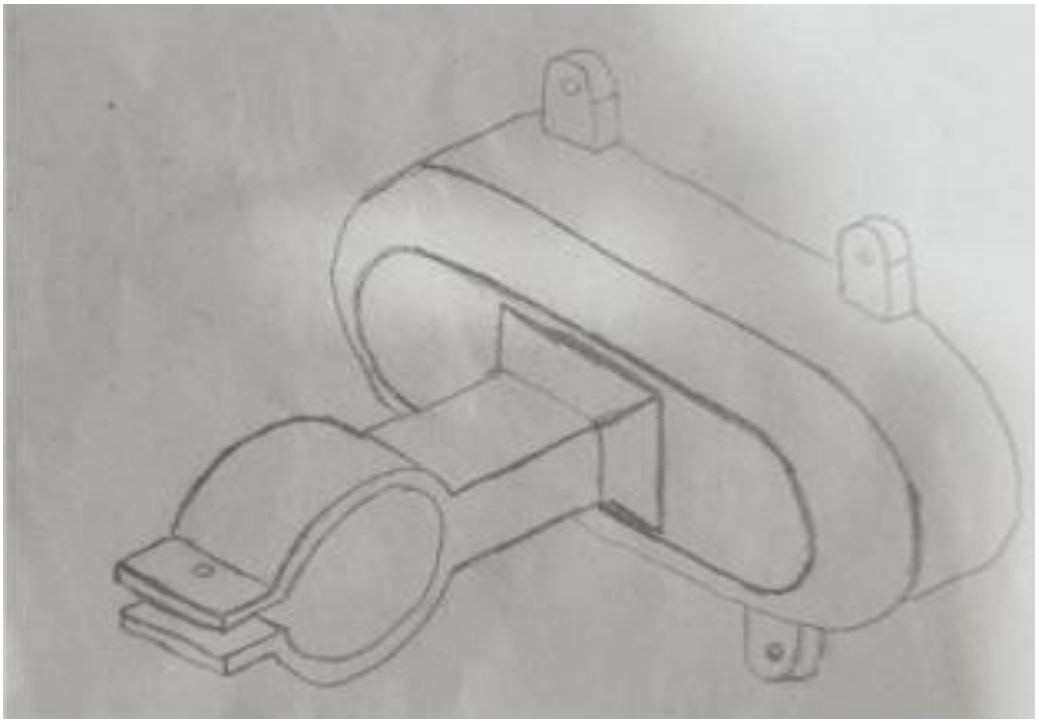


Fig.5.1.3 Freehand sketch of the solder mount.

5.2 CAD Model and Drawings

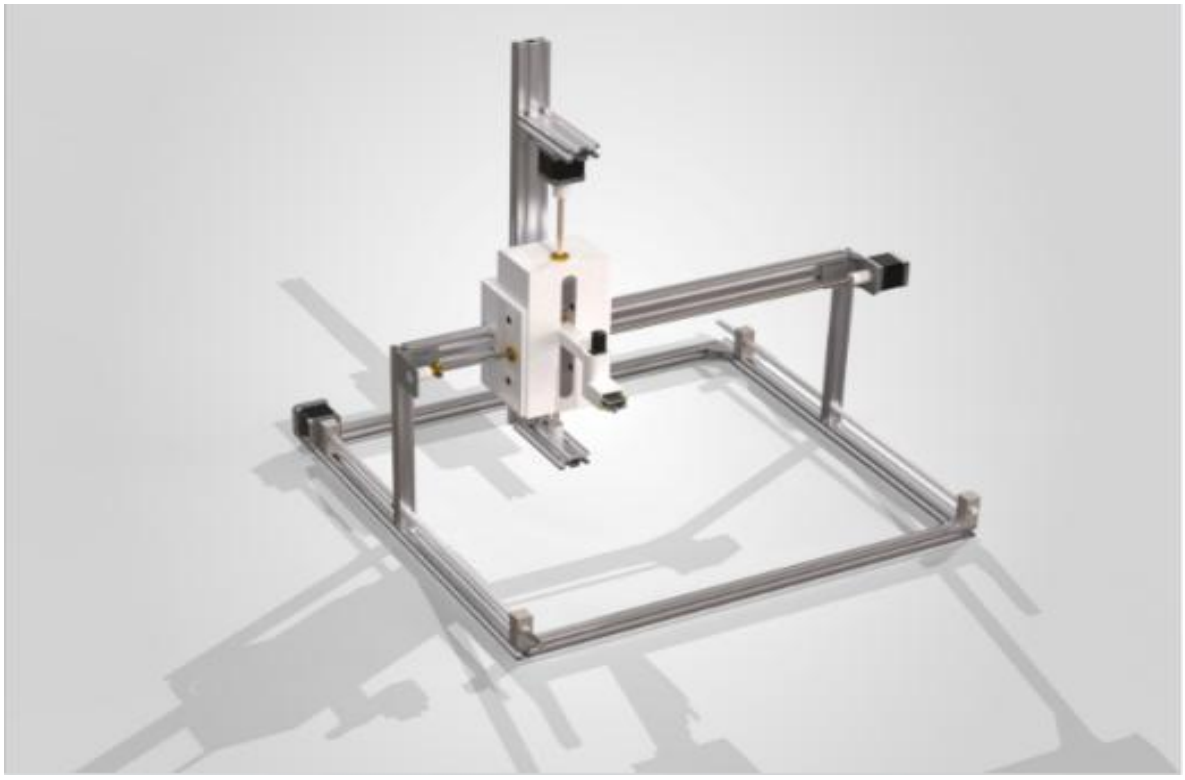


Fig.5.2.1. 3D CAD model of the proposed vision-based automated soldering system.

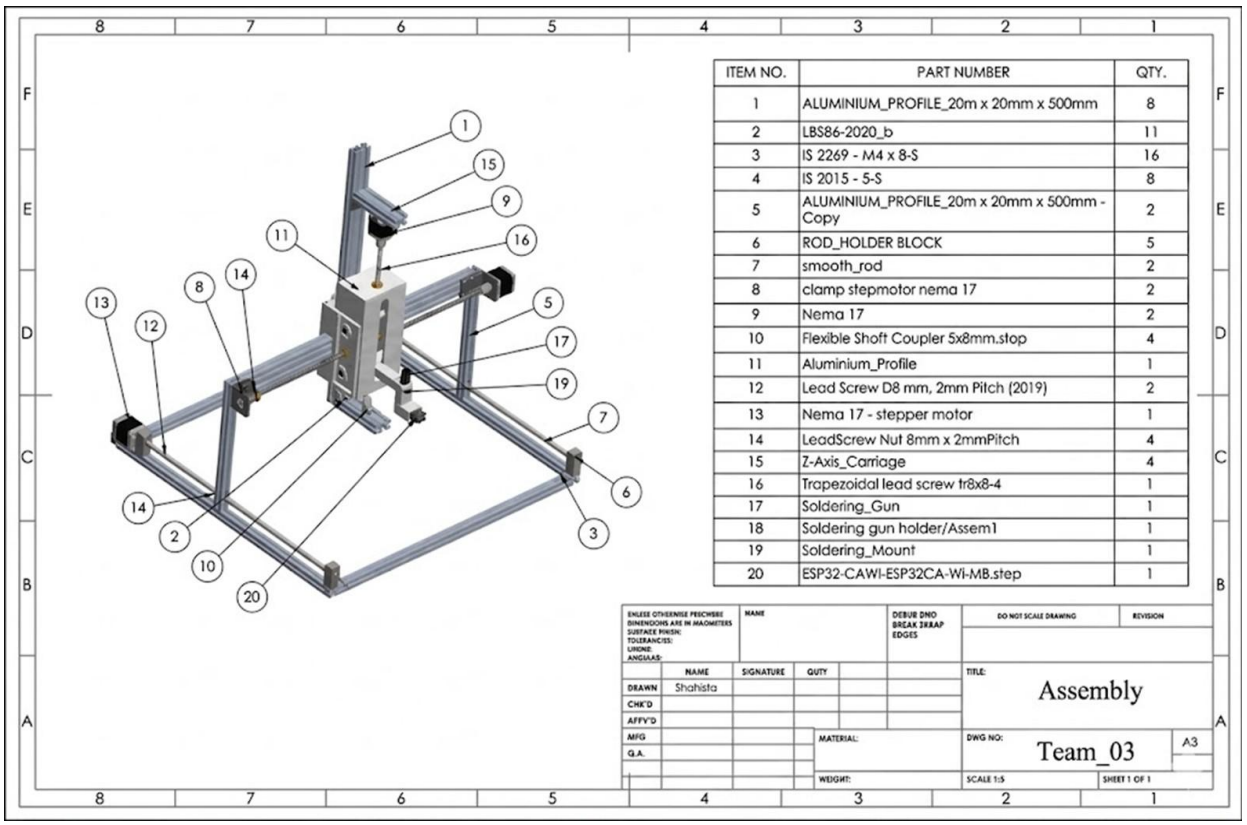


Fig.5.2.2 SolidWorks CAD assembly model of the proposed vision-based automatic soldering system

This Cartesian robot was designed and assembled in SolidWorks using an aluminum profile frame structure to achieve precise linear motion along the X, Y, and Z axes. The robot consists of stepper motors, lead screws, smooth rods, couplers, and linear guides that work together to provide accurate positioning and controlled movement in three-dimensional space. The rigid frame improves stability while the modular design allows easy assembly and maintenance. Components such as motor holders, bearing supports, and sliding mechanisms are integrated to ensure smooth motion transmission and reduce vibration during operation. This type of Cartesian robot is suitable for applications such as pick-and-place operations, CNC systems, 3D printing, and automated material handling due to its high accuracy, simple mechanical structure, and reliable motion control.

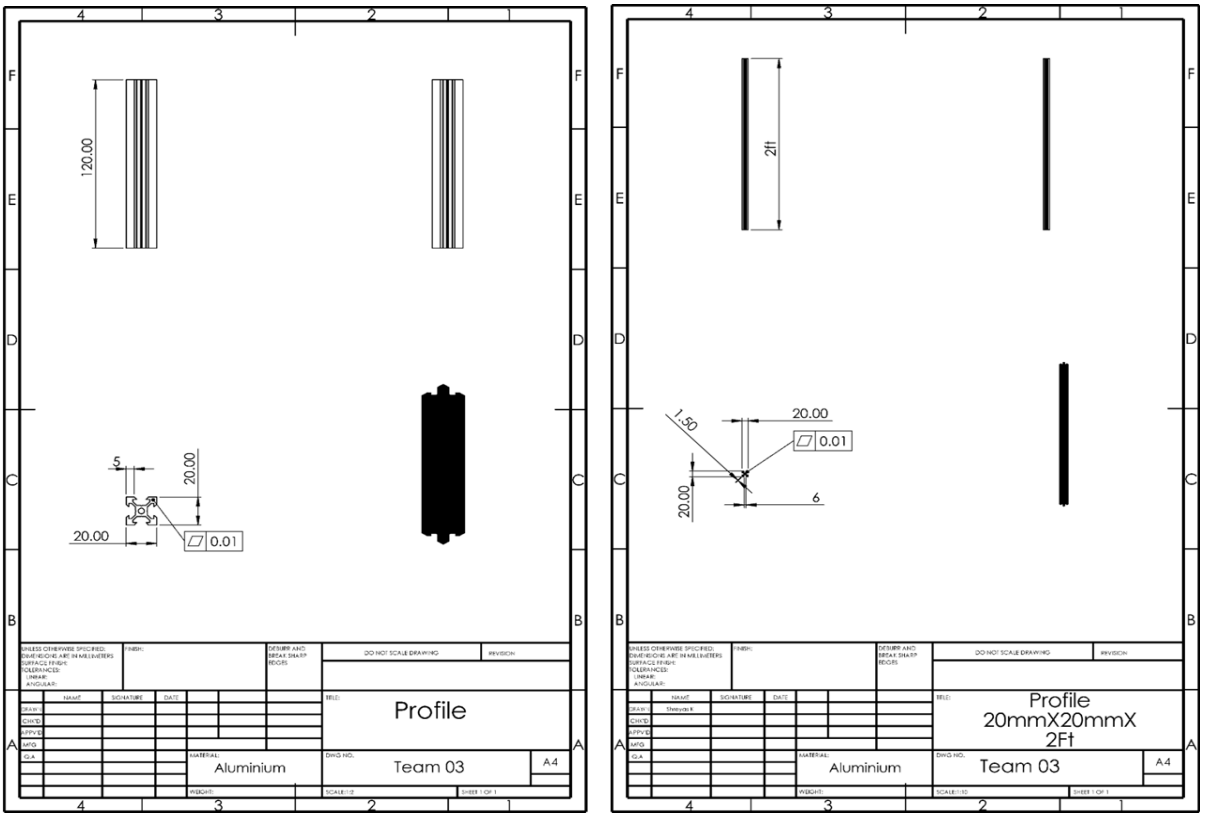


Fig.5.2.3. Part drawings of the aluminum extrusion profiles used in the proposed vision-based automatic soldering system: (a) vertical aluminum profile and (b) horizontal aluminum profile.

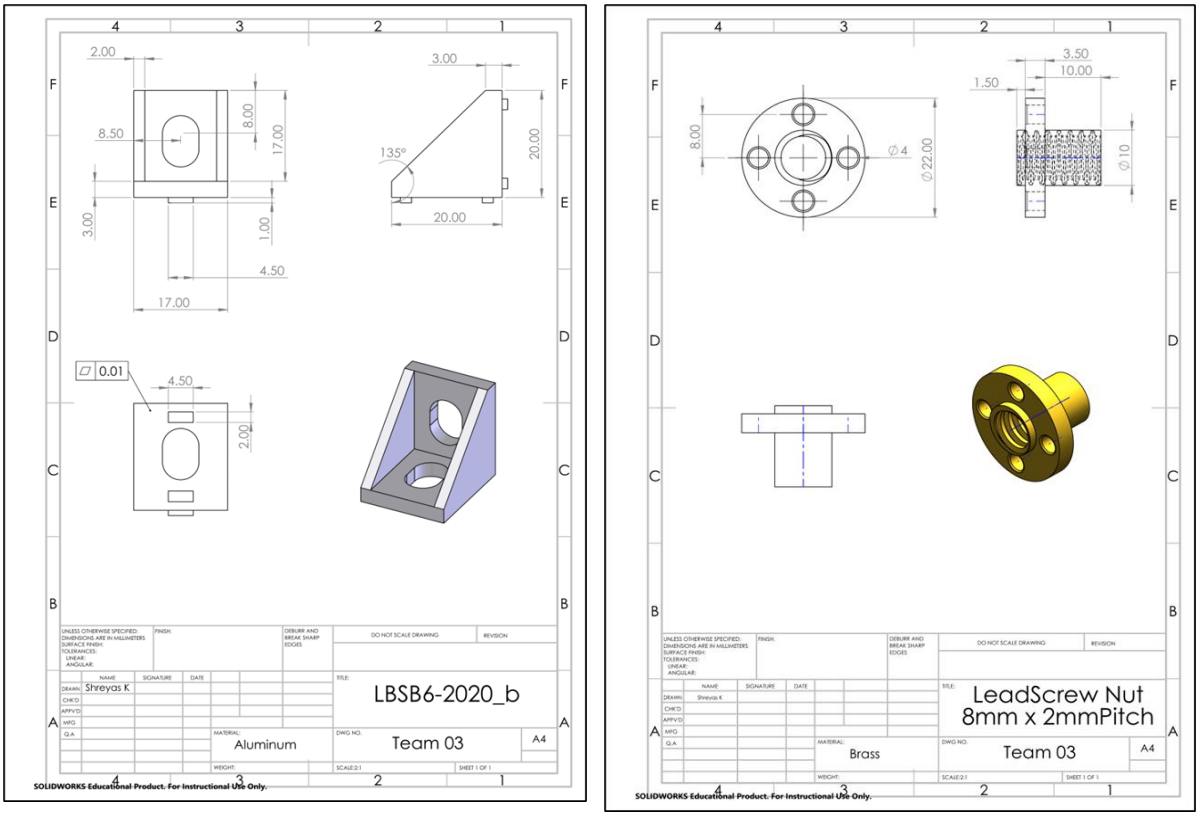


Fig.5.2.4. Part drawings of the mechanical components used in the proposed vision-based automatic soldering system: (a) aluminum mounting bracket and (b) brass lead screw nut.

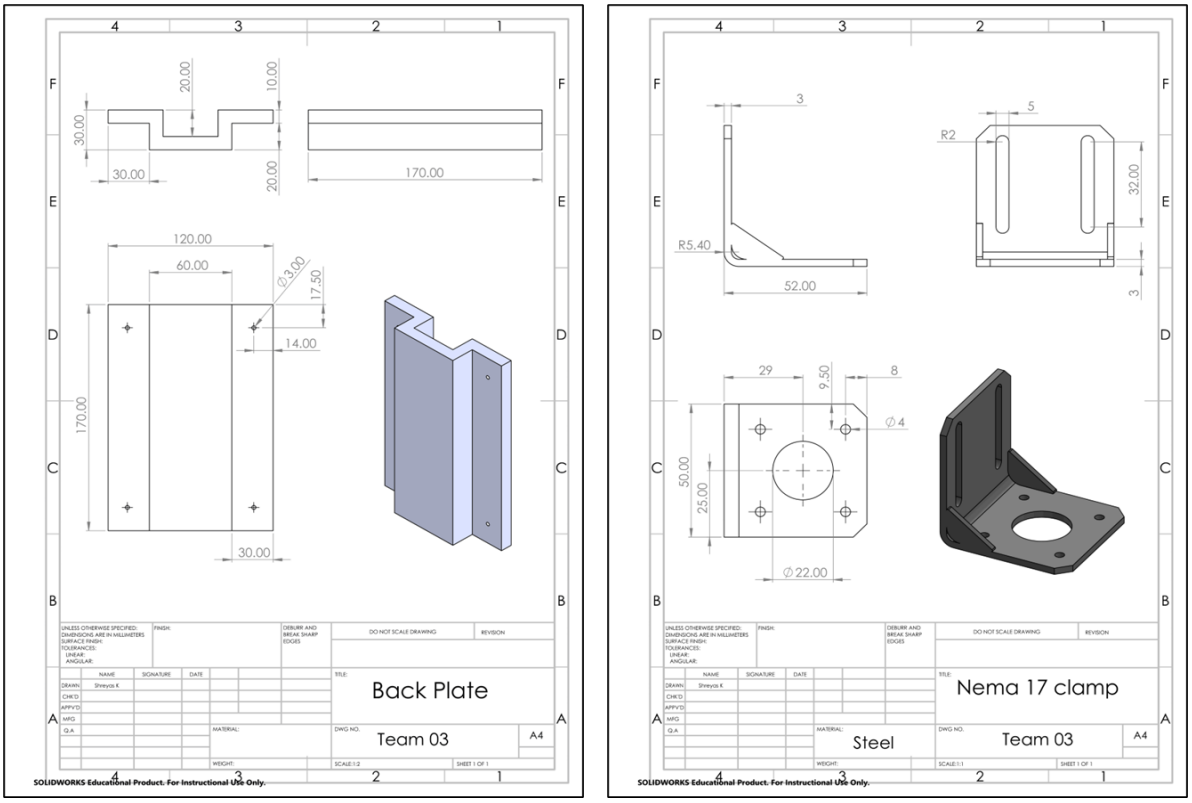


Fig.5.2.5. Part drawings of the structural mounting components used in the proposed vision-based automatic soldering system: (a) back plate and (b) NEMA 17 motor mounting clamp.

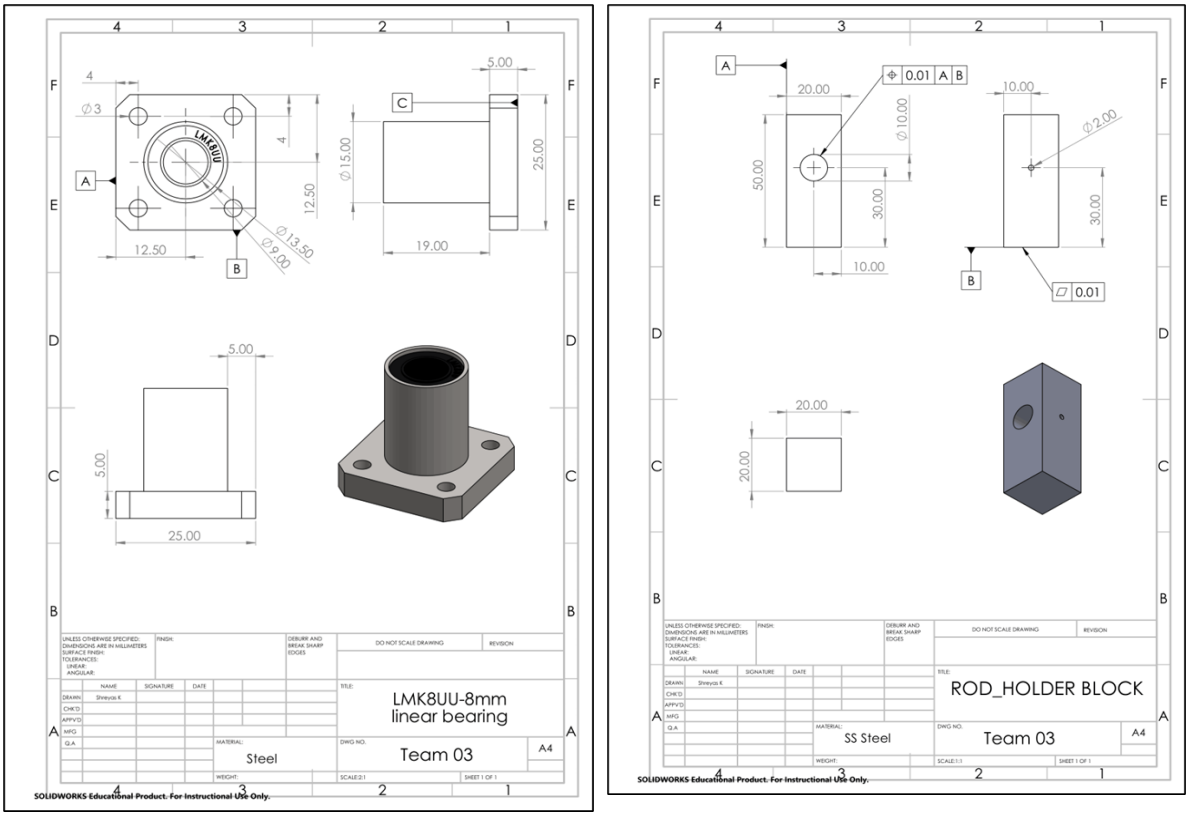


Fig.5.2.6. Part drawings of the motion-guiding components used in the proposed vision-based automatic soldering system: (a) LMK8UU 8 mm linear bearing housing and (b) rod holder block.

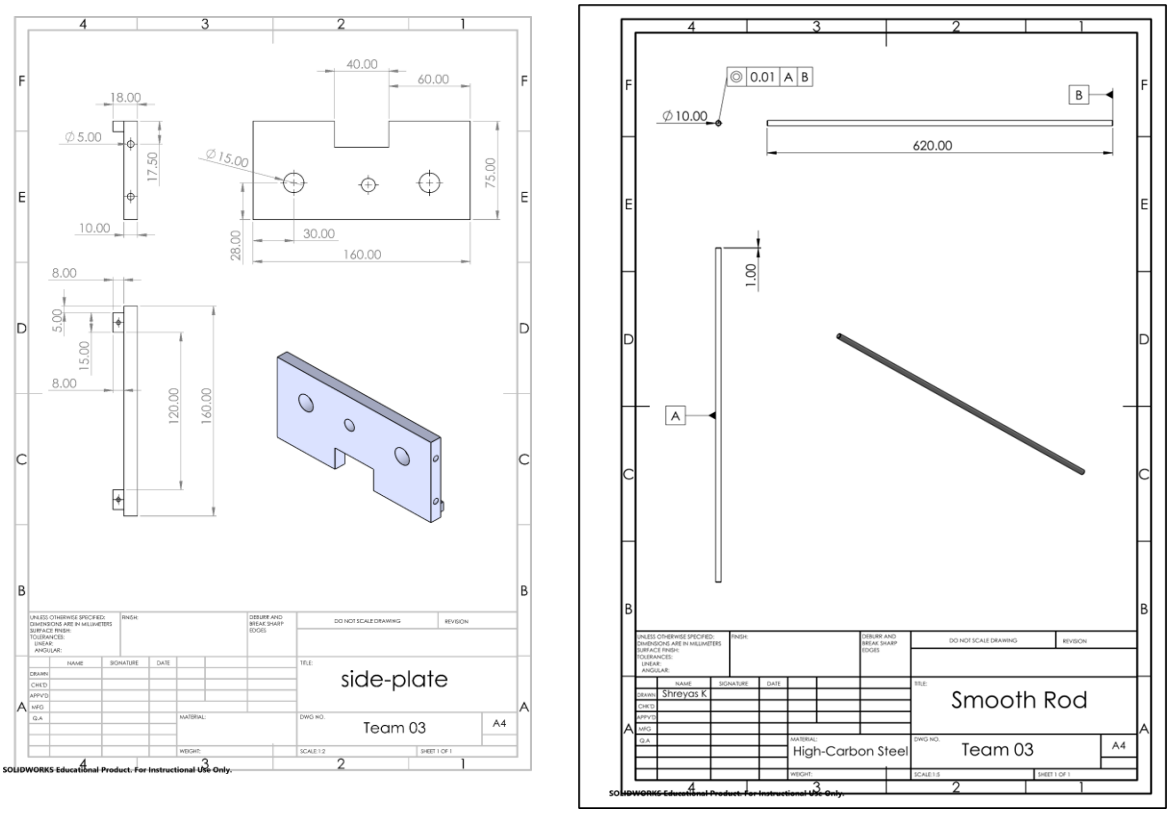


Fig.5.2.7. Part drawings of the supporting and guiding components used in the proposed vision-based automatic soldering system: (a) side plate and (b) smooth rod.

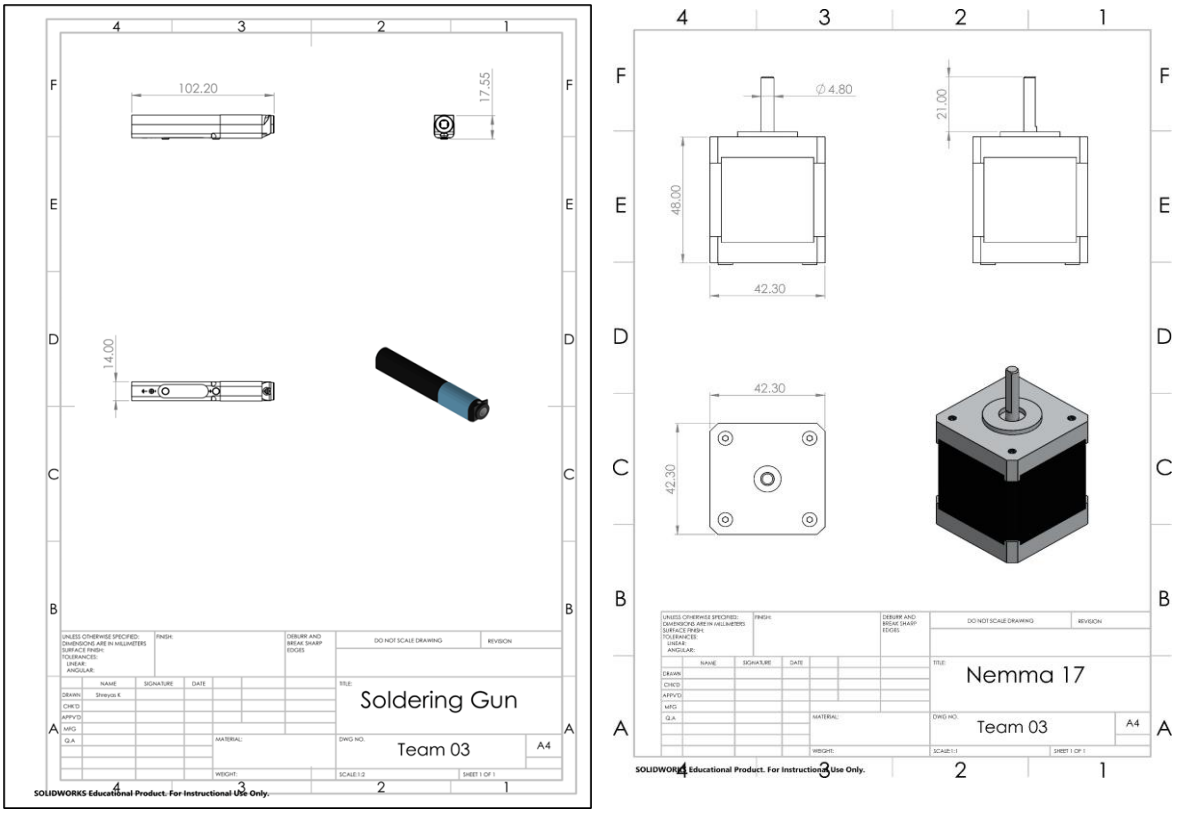


Fig.5.2.8. Part drawings of the actuation components used in the proposed vision-based automatic soldering system: (a) soldering gun and (b) NEMA 17 stepper motor.

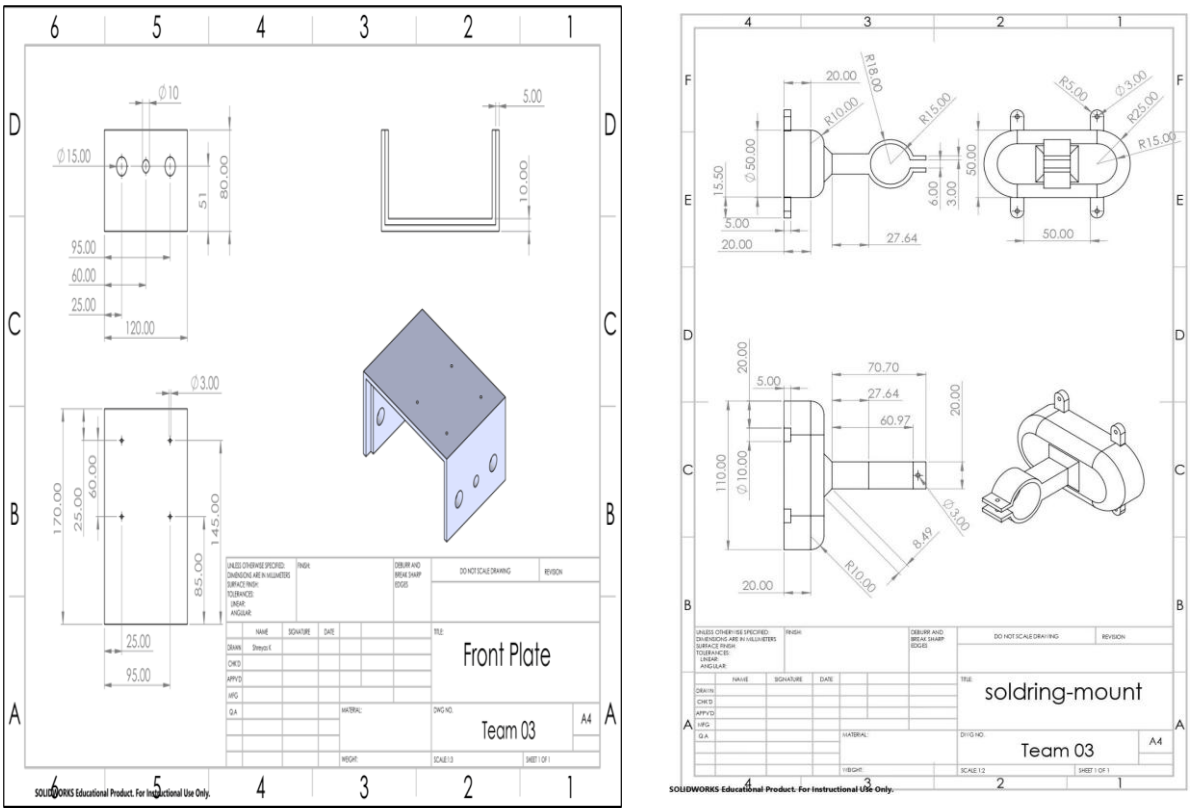


Fig.5.2.9. Part drawings of the custom-designed mechanical components: (a) front plate and (b) solder mount.

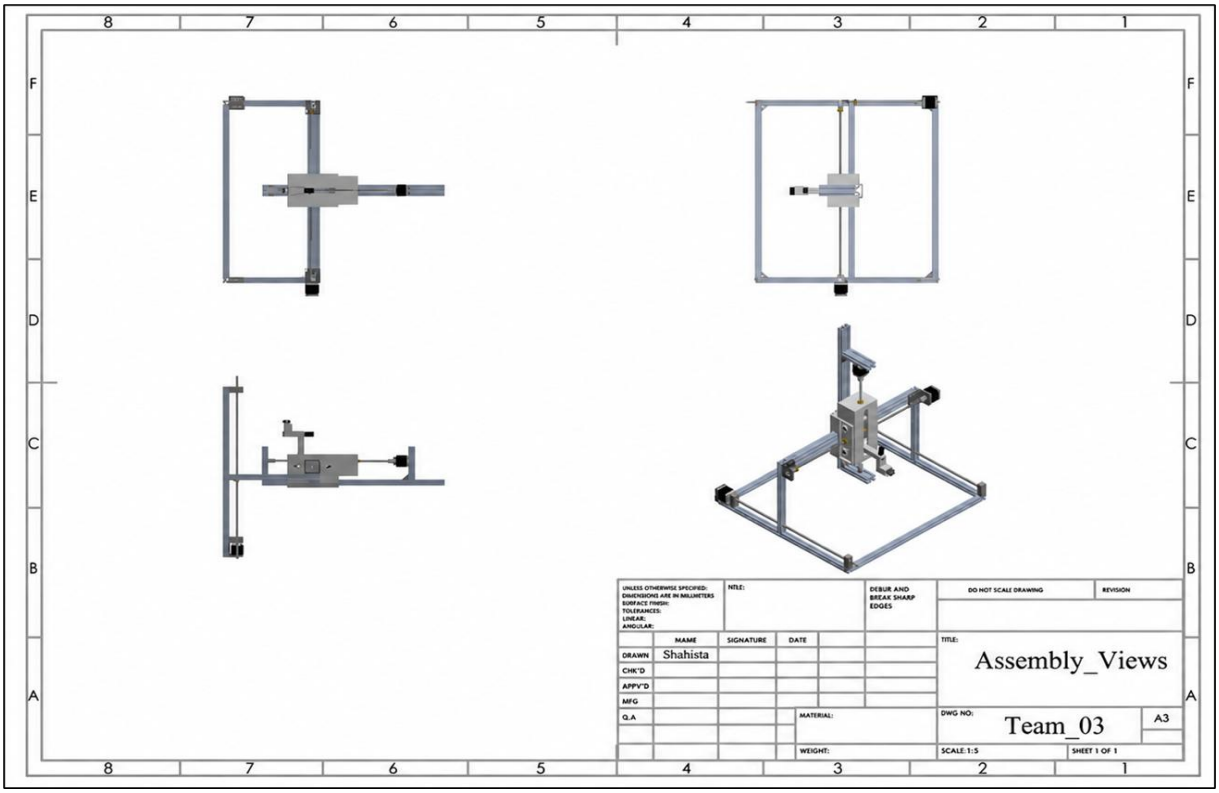


Fig.5.2.10. Orthographic and isometric assembly views of the proposed vision-based automatic soldering system, including the top, front, side, and isometric views.

5.3 Circuit Design

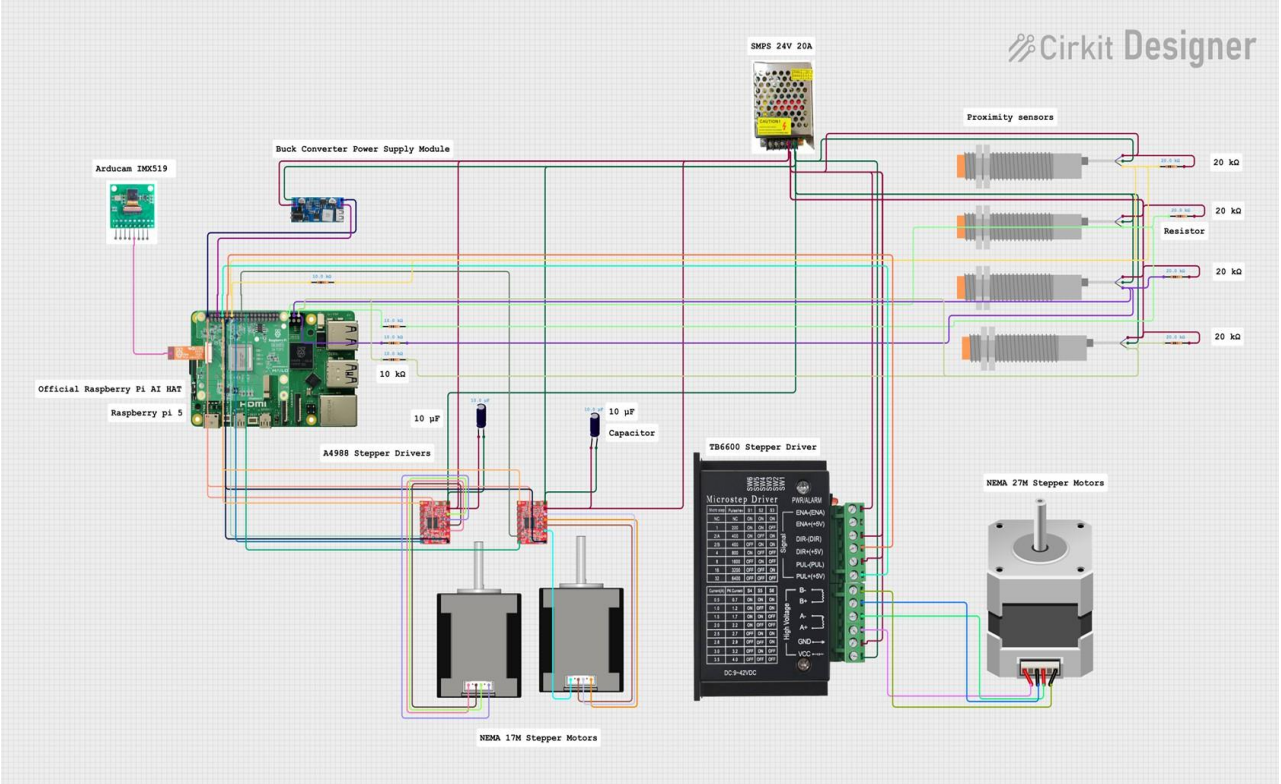


Fig.5.3. Circuit diagram of the proposed vision-based automatic soldering system.

The hardware circuit of the proposed vision-based automatic soldering system is centered around a Raspberry Pi 5 and an Arduino UNO, where the Raspberry Pi performs image acquisition and processing while the Arduino executes motion control. An Arducam Camera Module 2 is interfaced with the Raspberry Pi to capture PCB images for component detection and localization. The system is powered by a 24 V SMPS, with a buck converter used to step down the supply voltage to provide the required operating voltages for the Raspberry Pi, Arduino, sensors, and other low-voltage peripherals. Four proximity sensors are incorporated as limit switches to establish home positions and prevent overtravel along the motion axes. The X- and Y-axis movements are driven by two NEMA 17 stepper motors through A4988 stepper motor drivers, whereas the Z-axis is actuated by a high-torque NEMA 23 stepper motor controlled using a TB6600 stepper motor driver. All motion commands generated from the vision system are transmitted to the Arduino via serial communication, enabling synchronized control of the three-axis positioning mechanism. This integrated hardware architecture provides stable power distribution, precise motion control, reliable position feedback, and accurate soldering operations, ensuring repeatable and automated PCB soldering.

5.4 System Composer Architecture

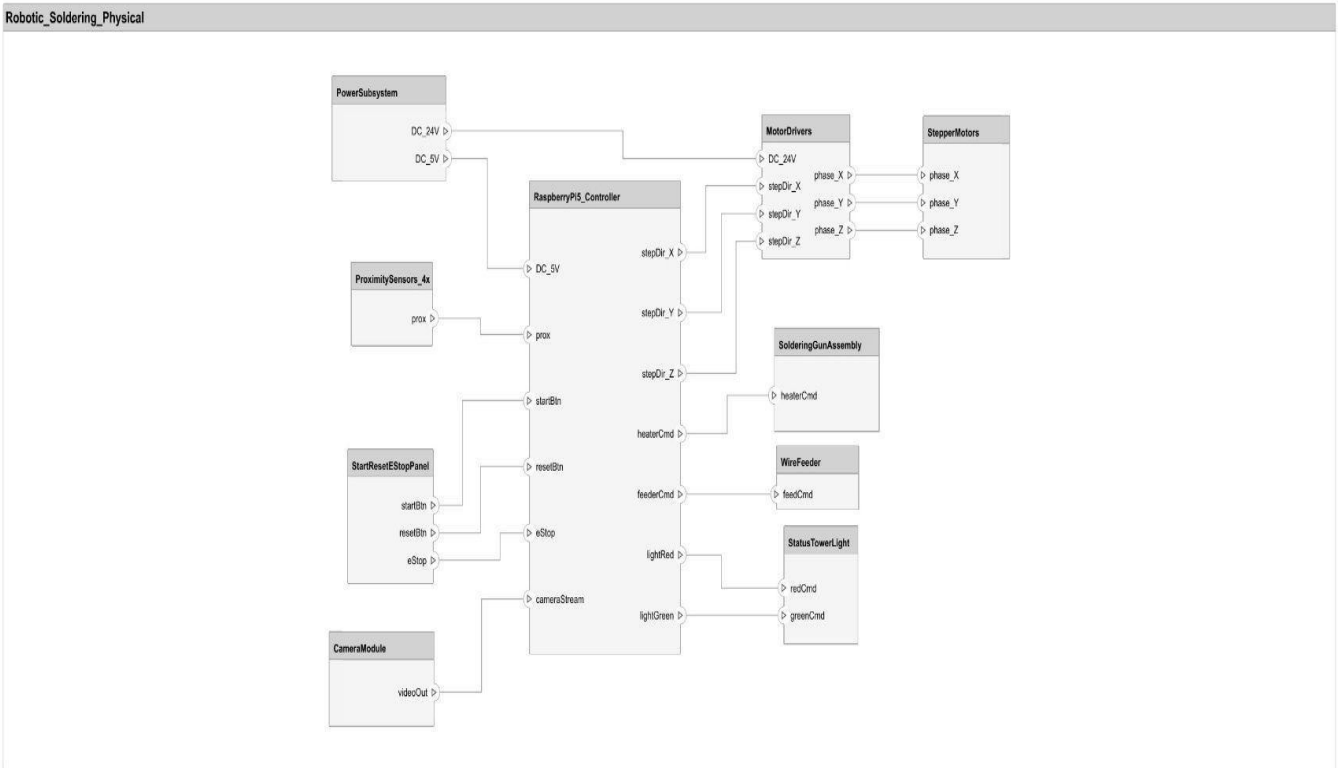


Fig.5.4.1 Physical architecture of the proposed vision-based automatic soldering system developed using MATLAB System Composer.

The physical architecture of the proposed vision-based robotic soldering system shown in fig 5.4.1 illustrates the hardware organization and interconnection of all major components. The system consists of a Raspberry Pi 5, Arduino UNO, Arducam Camera Module 2, three-axis motion mechanism, stepper motor drivers, power supply unit, and soldering module. The Raspberry Pi acquires PCB images through the camera and performs vision-based processing to identify soldering locations. Motion control is executed by the Arduino UNO, which interfaces with two A4988 drivers for the X- and Y-axis NEMA 17 stepper motors and a TB6600 driver for the Z-axis NEMA 23 stepper motor. A 24 V SMPS supplies power to the system, while a buck converter provides regulated voltages for low-power devices. Four proximity sensors are installed to establish home positions and ensure safe operation by preventing axis overtravel. The physical architecture demonstrates the integration of sensing, processing, power distribution, and actuation hardware required for autonomous PCB soldering.

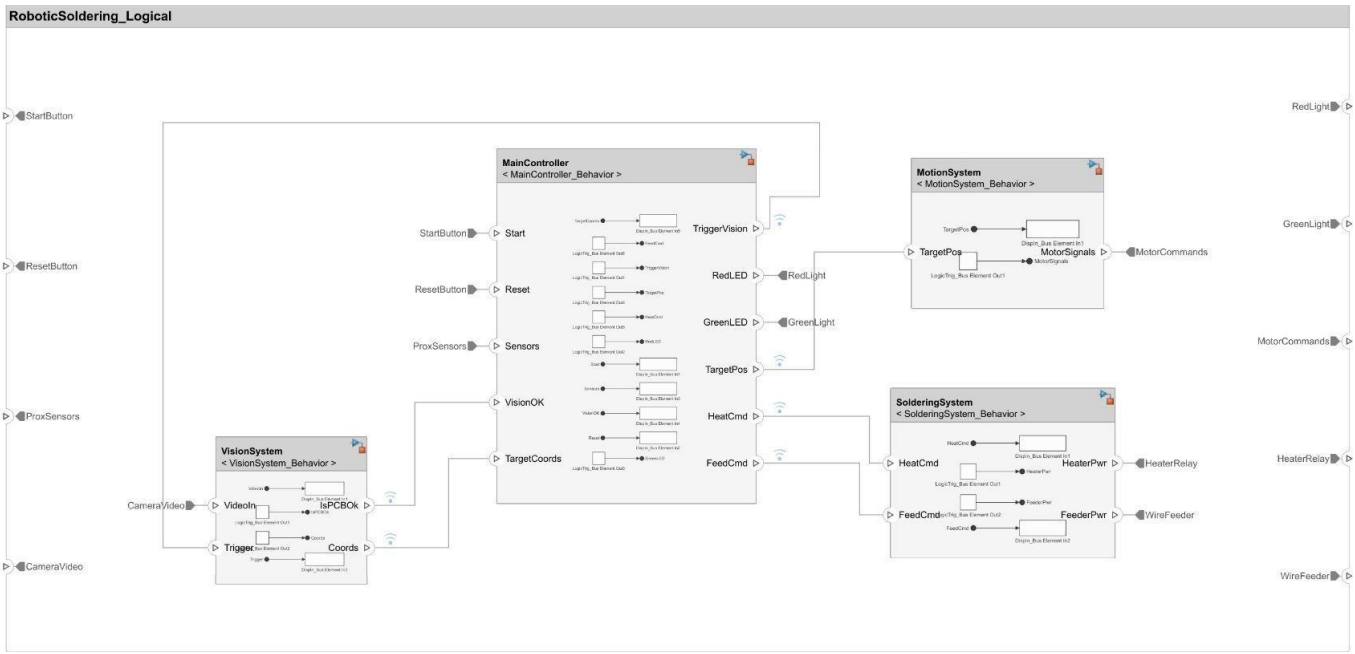


Fig.5.4.2 Logical architecture of the proposed vision-based automatic soldering system developed using MATLAB System Composer.

The logical architecture in fig 5.4.2 represents the functional workflow and interaction between the software and hardware modules of the proposed robotic soldering system. The process begins with image acquisition using the Arducam Camera Module 2, followed by image preprocessing and PCB feature detection on the Raspberry Pi 5. Based on the detected soldering points, coordinate data are generated and transmitted to the Arduino UNO through serial communication. The Arduino interprets these commands and controls the A4988 and TB6600 motor drivers to position the X-, Y-, and Z-axis motors accurately. Feedback from the proximity sensors is continuously monitored for homing, position verification, and safety. Once the soldering tool reaches the target location, the soldering operation is executed before the system proceeds to the next coordinate. The logical architecture highlights the sequential flow of data, control commands, and feedback required to achieve precise, synchronized, and fully automated soldering operations.

5.5 Flowchart

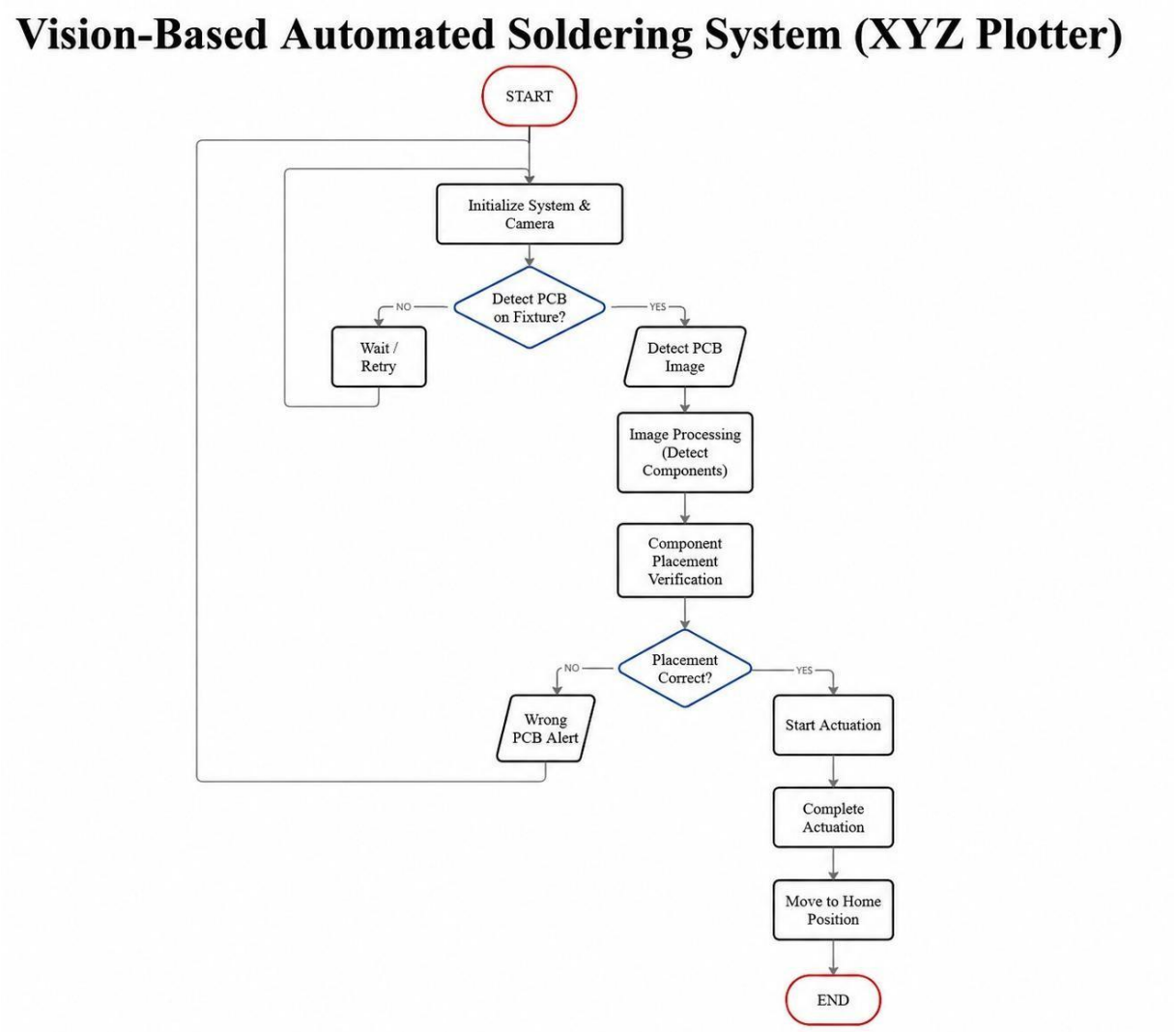


Fig.5.5. Control program flow of the proposed vision-based automated soldering system.

The workflow of the proposed vision-based automated soldering system in fig 5.5 begins with the initialization of the system and camera. Once initialized, the system checks whether a PCB is placed on the fixture. If no PCB is detected, the system enters a wait/retry state until a PCB is available. Upon successful detection, the camera captures an image of the PCB, which is then processed to detect the electronic components and their locations. The detected component placement is subsequently verified to ensure that all components are correctly positioned. If an incorrect placement is identified, the system generates a wrong PCB alert and returns to the PCB detection stage for corrective action. If the placement is verified successfully, the system initiates the soldering actuation process. After completing the soldering operation, the motion system returns to its home position, and the workflow terminates, indicating the successful completion of the automated soldering cycle.

5.6 Dataset & Processing

The performance of the vision-guided robotic soldering system depends on the quality of the image dataset and the preprocessing pipeline. A custom dataset of Printed Circuit Board (PCB) images was created by capturing images using an Arducam camera under different lighting conditions, viewing angles, and PCB orientations. The dataset contains images of solder pads, electronic components, and PCB layouts that represent real industrial scenarios. To improve model robustness, images were collected with varying backgrounds and illumination levels. Data augmentation techniques such as image rotation, horizontal flipping, scaling, brightness adjustment, and contrast enhancement were applied to increase dataset diversity and reduce overfitting during training.

Before training, all images were resized to the required input resolution of the detection model and manually annotated using bounding boxes around the target components. The annotated dataset was divided into training, validation, and testing subsets to ensure unbiased model evaluation. The trained model was then integrated with the Raspberry Pi-based vision system, where the detected component coordinates were transferred to the robotic controller for accurate soldering path generation.

5.6.1 Image Training, Testing and Implementation

The image processing module was developed in three sequential stages: front-side PCB detection and component localization, back-side solder point detection, and integration of both detection modules with the robotic soldering system. This modular approach ensures accurate identification of component locations on the front surface of the PCB and corresponding solder pads on the rear surface before initiating the soldering process.

A. Front PCB Image Training and Detection

The first stage focuses on detecting electronic components mounted on the front side of the PCB. A custom image dataset was created by capturing high-resolution PCB images using the Arducam camera under different lighting conditions, camera orientations, and PCB positions. The dataset contained images of resistors, capacitors, integrated circuits, connectors, LEDs, and other commonly used electronic components. To improve the robustness of the model, images were collected with varying illumination levels, slight rotations, and different backgrounds.

Each image was manually annotated using bounding boxes, where every component was assigned a corresponding class label. Data augmentation techniques including random rotation, horizontal and vertical flipping, brightness adjustment, contrast variation, scaling, and translation were applied to increase dataset diversity and improve generalization.

The YOLO object detection model was trained using the annotated dataset until convergence was achieved. During training, the model continuously optimized localization and classification losses while validation images were used to monitor learning performance and prevent overfitting. After training, the model was evaluated on previously unseen testing images as shown in fig 5.6.1, where it successfully detected PCB components with high confidence defined in table 5.6.1. The detected component coordinates were then stored as reference locations for subsequent alignment with the backside solder pads.

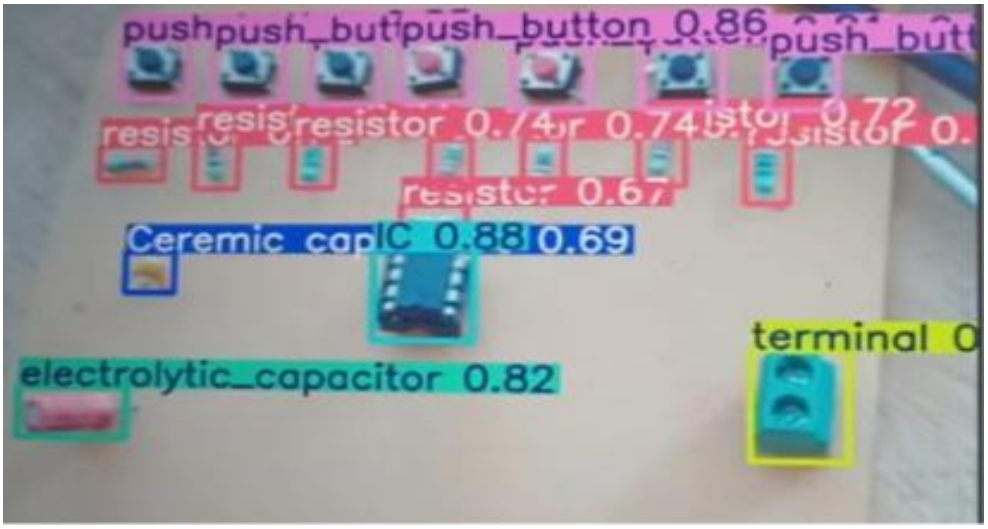


Fig.5.6.1. Detection of Front PCB Components Using the Proposed YOLOv8 Model

TABLE 5.6.1: Performance of YOLOv8 in Front PCB component detection

Class	Images	Instances	P	R	mAP50	mAP50-95
all	1646	3879	0.95	0.97	0.92	0.81
Ceramic Capacitors	121	121	0.98	0.98	0.98	0.66
Electrolytic Capacitor	122	122	0.97	1.0	0.99	0.83
IC	122	122	0.98	0.99	0.99	0.84
Push Button	122	852	0.99	0.99	0.98	0.81
Resistor	121	969	0.90	0.85	0.93	0.52
Terminal	120	120	0.96	0.99	0.98	0.88

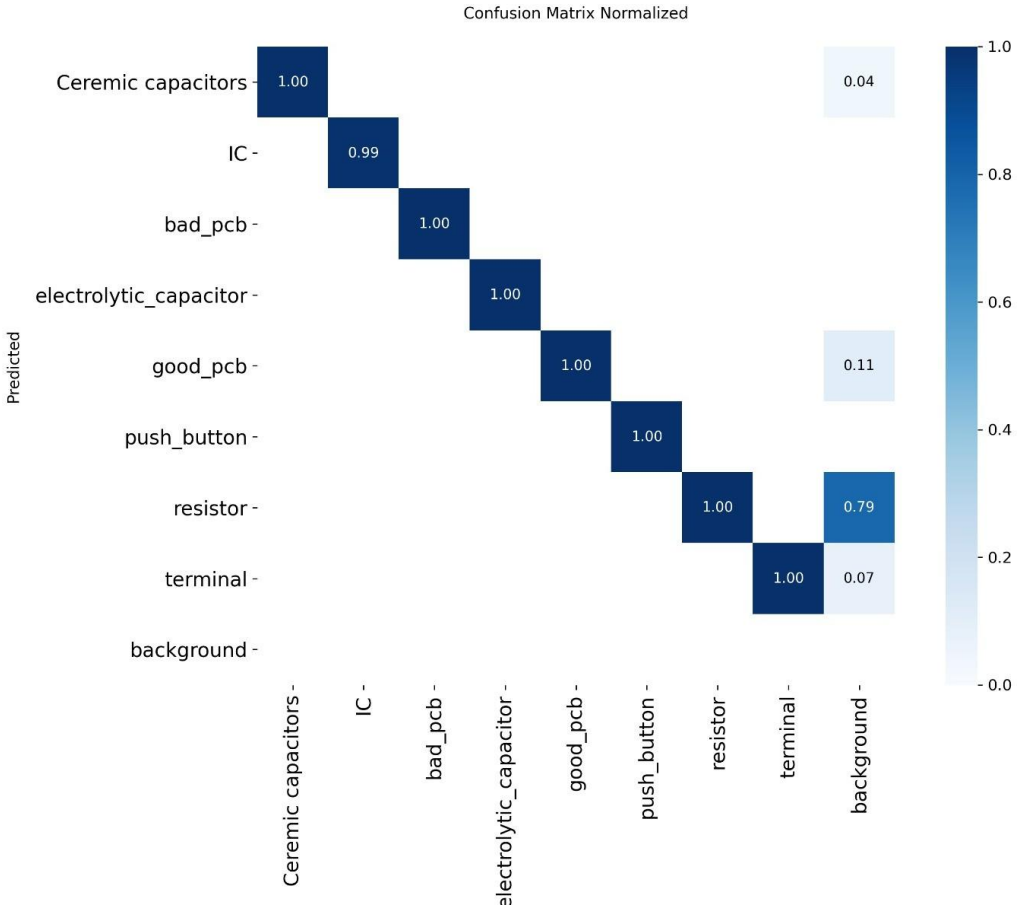


Fig.5.6.2. Confusion matrix for the front PCB component detection model.

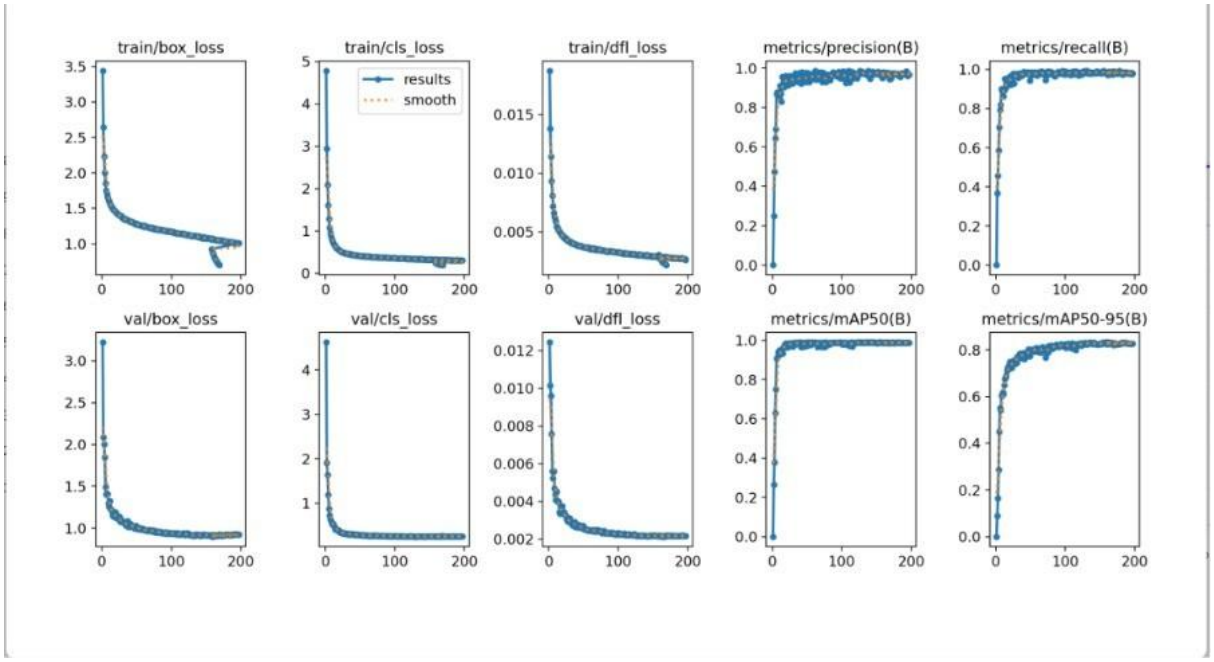


Fig.5.6.3. Performance evaluation of the front PCB component detection model using a normalized confusion matrix.

- **Normalized Confusion Matrix**

  Figure 5.6.2 presents the normalized confusion matrix of the YOLO-based front PCB component detection model. The matrix demonstrates excellent classification performance, with nearly all component classes, including ceramic capacitors, ICs, electrolytic capacitors, push buttons, resistors, terminals, and good/bad PCB classes, achieving prediction accuracies close to 100%. The strong diagonal values indicate that the model correctly distinguishes between different PCB components with minimal misclassification. A small amount of confusion is observed with the background class, which is expected due to variations in image backgrounds and object boundaries. Overall, the confusion matrix confirms the robustness and reliability of the trained model for accurate front PCB component detection and classification.

- **Training and Validation Performance Curves**

  Figure 5.6.3 illustrates the training and validation performance curves of the YOLO-based front PCB component detection model. The box loss, classification loss, and distribution focal loss decrease steadily throughout the training process, indicating stable convergence of the model. Simultaneously, the precision, recall, mAP@0.5, and mAP@0.5:0.95 metrics increase rapidly and stabilize at high values, demonstrating the model's ability to accurately detect and classify PCB components. The close agreement between the training and validation curves suggests good generalization with minimal overfitting. These results confirm that the trained model achieves high detection accuracy and is well suited for reliable front PCB inspection in the proposed vision-based automated soldering system.

B. Back PCB Solder Pad Training and Detection

After successful front-side detection, a second dataset consisting of backside PCB images was prepared for solder pad detection. Unlike the front-side dataset, this dataset primarily contained solder joints, solder pads, through-hole pins, and PCB tracks. Images were captured under different illumination conditions to account for reflections produced by metallic solder surfaces.

Each solder pad was manually annotated using bounding boxes before training. Similar preprocessing operations including image resizing, normalization, brightness correction, and augmentation were performed to improve detection performance under practical industrial conditions. The trained detection model accurately identified individual solder pads as shown in fig 5.6.4 and generated their pixel coordinates within the captured image.

To improve positional accuracy, camera calibration was performed to compensate for lens distortion and perspective errors. The detected solder pad coordinates were converted from image coordinates into real-world coordinates using calibration parameters. This coordinate transformation enabled precise localization of soldering points required for robotic motion planning.

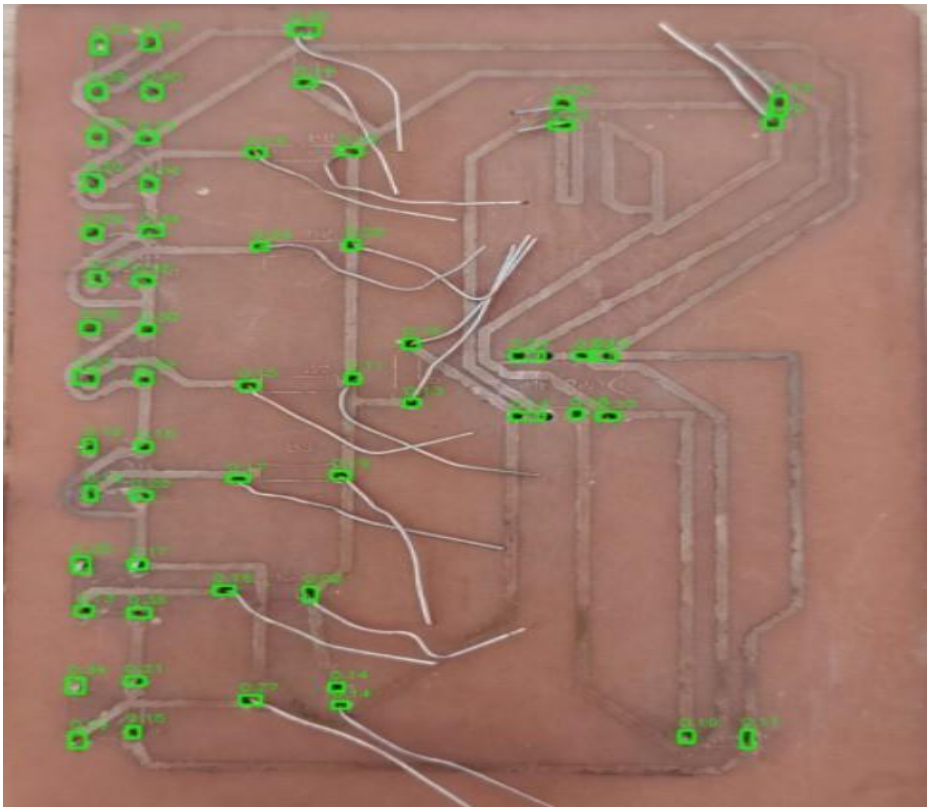


Fig.5.6.4. Detection of Back PCB Holes Using the Proposed YOLOv8 Model

- **Confusion Matrix**

  Figure 5.6.5. presents the confusion matrix of the YOLO-based back PCB hole detection model. The matrix illustrates the classification performance for the hole and background classes. The model successfully identifies the majority of hole instances while maintaining effective discrimination from the background. Although a small number of misclassifications are observed due to similarities between the hole features and background regions, the overall performance demonstrates the model's capability to accurately detect PCB mounting holes. These results validate the effectiveness of the trained model for reliable hole localization, which is essential for accurate PCB alignment and positioning in the proposed vision-based automated soldering system.

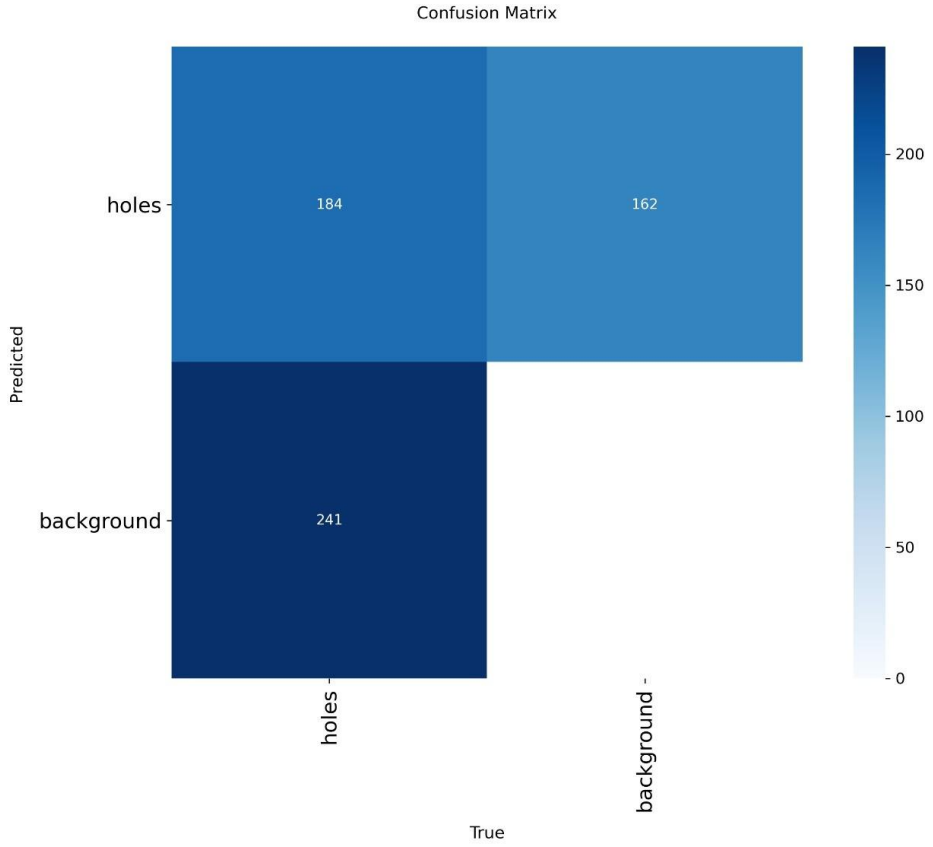


Fig.5.6.5. Confusion matrix for the back PCB hole detection model.

- **Training and Validation Performance Curves**

Figure 5.6.6 illustrates the training and validation performance curves of the YOLO-based back PCB hole detection model. The box loss, classification loss, and distribution focal loss decrease progressively throughout the training process, indicating stable learning and convergence of the model. At the same time, the precision, recall, mAP@0.5, and mAP@0.5:0.95 metrics show a consistent improvement with increasing training epochs, demonstrating the model's ability to accurately detect PCB holes. The close agreement between the training and validation losses indicates good generalization with minimal overfitting. These results confirm that the trained model provides reliable and consistent hole detection performance, making it suitable for PCB alignment and automated soldering applications.

TABLE 5.6.2: Performance of YOLOv8 in Back PCB for solder hole detection

Class	Images	Instances	P	R	mAP50	mAP50-95
all	1152	6734	0.54	0.78	0.60	0.45

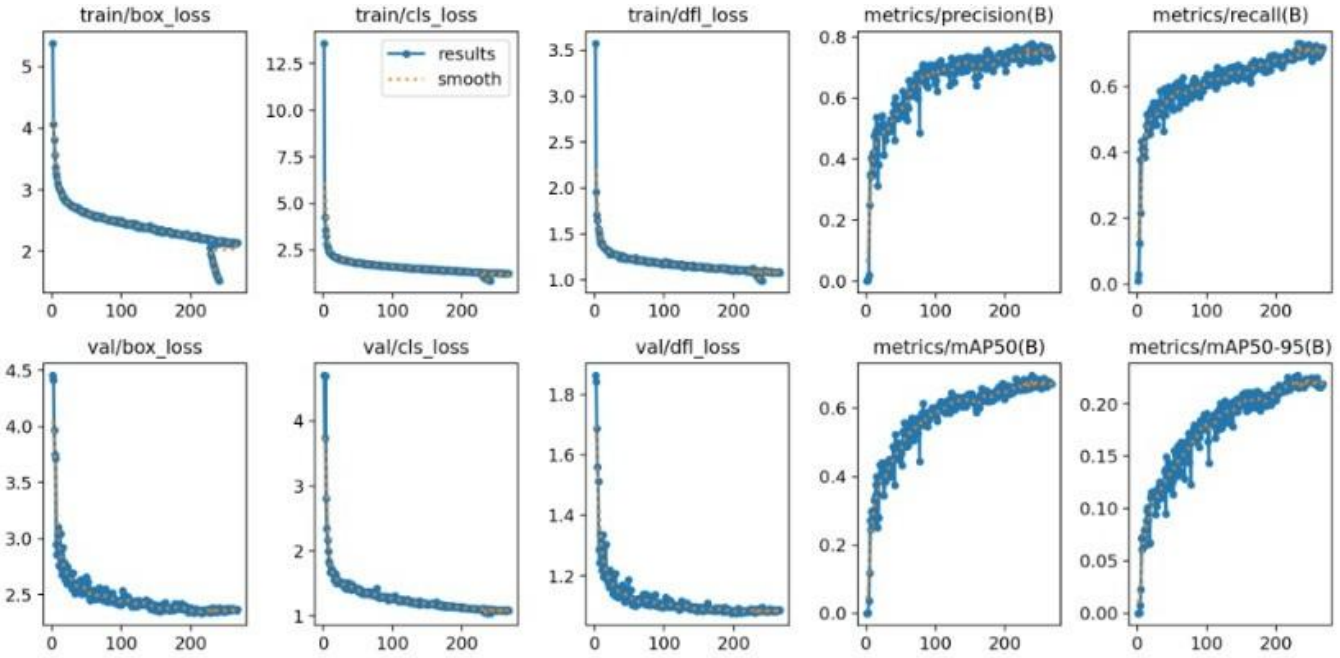


Fig.5.6.6. Performance evaluation of the back PCB hole detection model using a normalized confusion matrix.

C. Integration of Front and Back PCB Detection

Once both detection models were individually validated, they were integrated into a unified robotic soldering framework. Initially, the front-side PCB image is captured to verify the placement and orientation of all electronic components. The detected component positions serve as reference points for PCB alignment. After verification, the PCB is flipped automatically, and the backside image is acquired for solder pad detection.

The vision system establishes correspondence between the front-side component locations and the backside solder pad positions through PCB coordinate mapping. After successful alignment, the detected solder pad coordinates are transmitted from the Raspberry Pi to the robotic controller. These coordinates are converted into robotic workspace coordinates through hand-eye calibration and inverse kinematic calculations.

Finally, the robotic arm follows the generated trajectory to position the soldering iron accurately over each solder joint. Throughout the implementation process, the vision module continuously verifies the detected positions before soldering, thereby minimizing positioning errors and improving soldering accuracy. The complete integration of front-side inspection, backside solder pad detection, and robotic motion control enables an automated vision-guided soldering system capable of performing reliable PCB soldering with minimal human intervention.

5.7 Simulation of Cartesian Robot in CoppeliaSim

A vision based automatic soldering system was developed and tested using the CoppeliaSim simulation environment in order to overcome the limitations of manual soldering such as inconsistent solder quality, positioning errors and reduced productivity. To carry out the automated soldering process of the PCB, a three-axis Cartesian robotic manipulator with X, Y, and Z axes was designed. A vision sensor was installed next to the soldering tip to continuously monitor the soldering environment and take real-time images of the PCB while it was in use. The PCB with several soldering holes was placed inside the robot's workspace and the manipulator moved the soldering tip in the direction of the intended soldering spots by following predetermined motion trajectories.Additional virtual cameras were integrated into the simulation environment to improve the visualization and monitoring of the robotic operation. A top view camera was used to observe the overall PCB workspace and robot movement while a side view camera monitored the vertical motion of the soldering tool and tip positioning accuracy. To validate the appropriate workspace coverage and movement execution the vision sensor and virtual cameras continuously monitored the robot's interaction with the PCB during the simulation. The modeled simulation demonstrates that combining both robotic automation and machine vision is feasible for soldering applications. It also provides a framework for future research on intelligent trajectory planning, automatic pad detection and real-time soldering process monitoring.

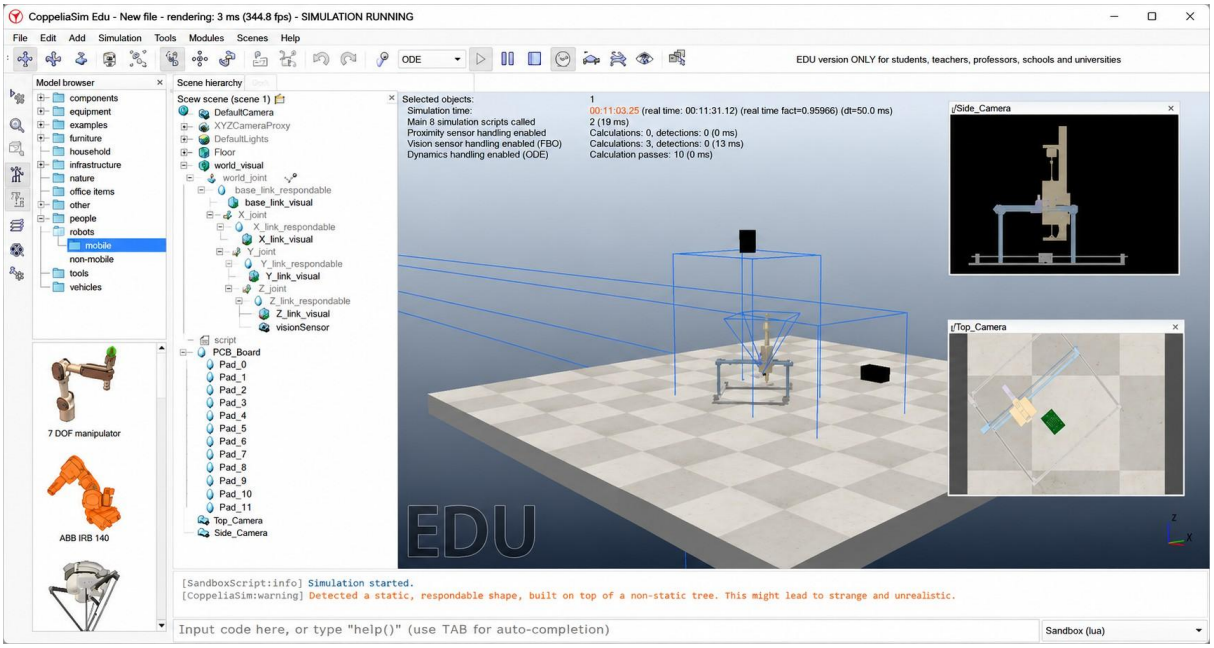


Fig.5.7. CoppeliaSim simulation environment of the proposed vision-based automated soldering system.

6. PROTOTYPING

6.1 Steps in Project Execution

Phase 1: Structural Framework Fabrication & Mechanical Skeleton Assembly

High-strength 2020 aluminum extrusion profiles were precisely measured and cut according to the dimensions derived from the SolidWorks 3D model. Appropriate-sized T-slot nuts, M5 fasteners, and 2020 L-brackets were utilized to securely interface the structural components, establishing the primary 2 m x 2 m x 2 m Cartesian workspace envelope. This framework forms the basic skeleton of the entire robotic workstation, ensuring excellent mechanical stability and structural integrity. It serves as the primary load-bearing foundation that completely supports the static load of the multi-axis moving assemblies while dampening operational vibrations.

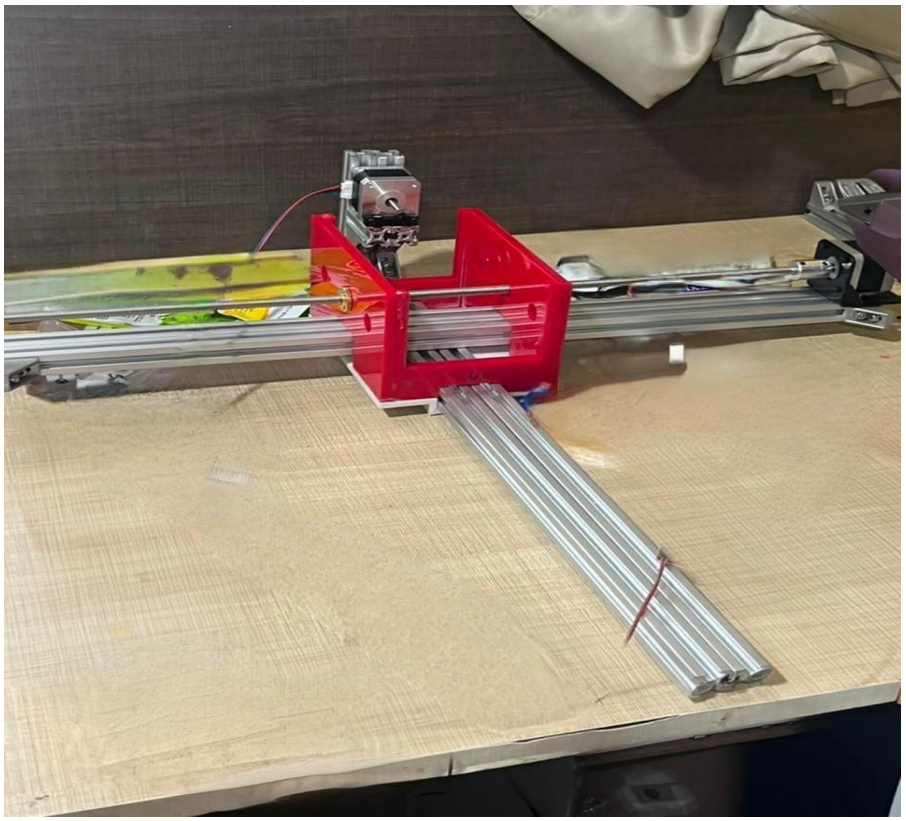


Fig.6.1 Structural Framework Fabrication & Mechanical Skeleton Assembly

Phase 2: Carriage Sizing, Tooling Mounts & Toolhead Fabrication

A custom multi-axis carriage was engineered based on the dimensions of the rigid outer frame to establish a central holding mount for the lead screws of the Y and Z travel vectors. Dedicated placement spaces were integrated into the toolhead layout to support the secure alignment of the 12 MP Arducam autofocus camera and the automated selective soldering gun. The structural housing elements were fabricated via 3D printing and custom laser-cut acrylic pieces, which were subsequently fused together using a high-strength adhesive to ensure superior structural strength. Finally, the printed holder and the outer casing box were combined to form a complete toolhead assembly, and a partial load-bearing test was

conducted to verify that the integrated carriage could easily withstand the mechanical weight of the soldering gun during rapid direction changes.



Fig.6.2 Carriage , Tooling Mounts & Toolhead Fabrication

Phase 3: Automated Inversion (Flipping) Mechanism Generation

A subsequent substrate flipping mechanism was engineered to automate the process transition immediately after the top-down vision camera completes its quality assurance scan. The primary purpose of this mechanism is to physically rotate the PCB by 180°c clockwise once the component counts are verified as correct and all manufacturing standards are successfully met. The mechanism was built by mounting a high-torque NEMA 23 stepper motor directly to the frame's corner using an L-shaped structural extrusion. An internal mounting sub-frame was attached to the motor shaft to hold the PCB securely, creating a highly rigid, reliable rotation assembly that remains perfectly aligned under continuous flipping operations.

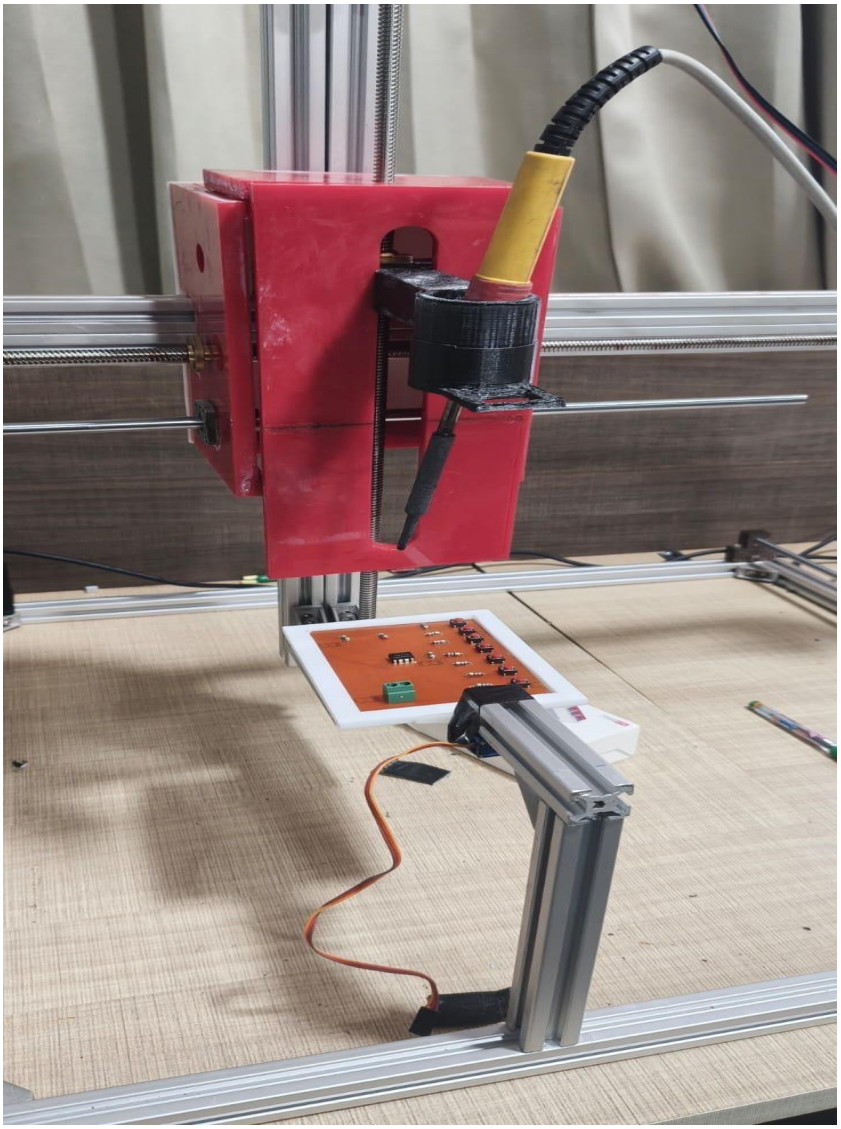


Fig.6.3 Automated Inversion (Flipping) Mechanism Generation

Phase 4: Electrical Subsystem Routing, Shield Coupling & Core Tuning

The electrical infrastructure was completed by mounting the industrial NEMA stepper motors to their designated spots along the linear X, Y, and Z coordinate axes. A NEMA 23 stepper motor was installed on the X-axis for high-torque base movement, while independent NEMA 17 motors were integrated to drive the linear motions of the Y and Z axes respectively. Electrical routing was systematically executed across the frame, linking a stable 24V 20A Switched-Mode Power Supply (SMPS) through a buck converter to handle low-voltage peripheral logic levels. The automated soldering gun was connected directly to a heavy-duty hardware relay module governed by an ESP32 microcontroller, ensuring that the master program could directly cycle the tool's power state. The motors were coupled to a CNC shield populated with micro-stepping drivers, which was wired back to the main control board. Uploading a primitive test script demonstrated the successful, synchronous actuation of all three axes simultaneously, allowing us to calibrate motor steps, minimize harmonic vibrations, and establish smooth gantry routing.

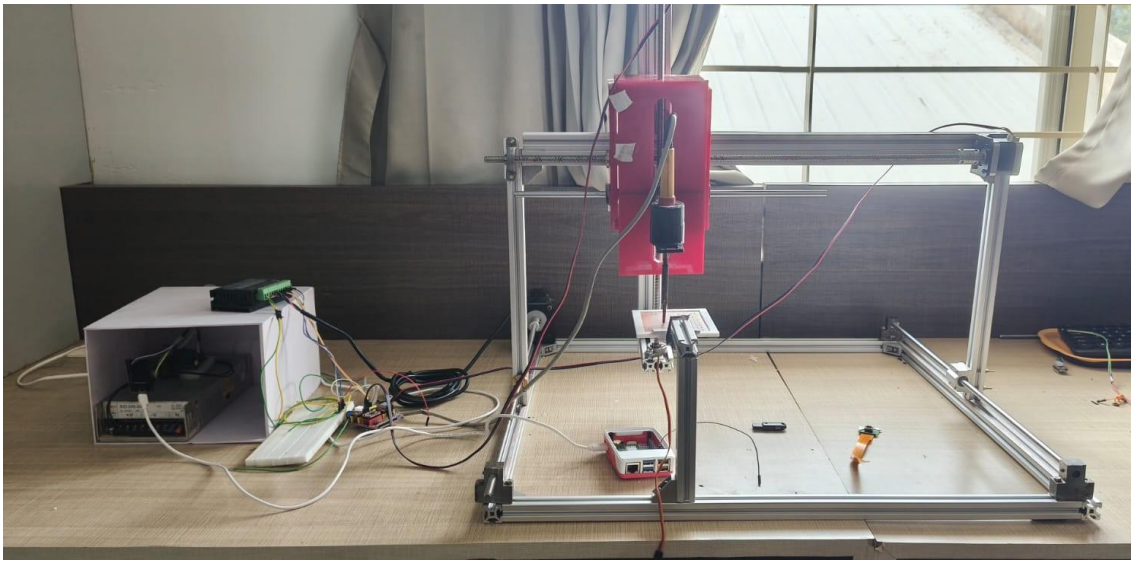


Fig 6.4 Electrical Subsystem Routing, Shield Coupling & Core Tuning

Phase 5: Unified System Integration & Complete Prototype Commissioning

The final integration phase successfully combined all three independent subsystems: the mechanical gantry framework, the localized electronic hardware routing, and the computer vision automation software into a singular operational prototype. By establishing stable serial communication between the central processor and the underlying motion shield, Inductive proximity limit sensors were securely mounted at the travel bounds of each linear axis to indicate maximum displacement and stop the carriage safely before physical over-travel collisions could occur. The 12 MP Arducam camera was mounted at its dedicated terminal and integrated with the image processing backend code. This setup allows the camera system to analyze top-down PCB images, extract precise pixel data, count electronic components, and check for defects. If the component count matches manufacturing reference metrics, the board is successfully passed, triggering the ESP32 logic to flip the PCB via the NEMA 23 motor and execute the automated selective soldering toolpath cleanly.

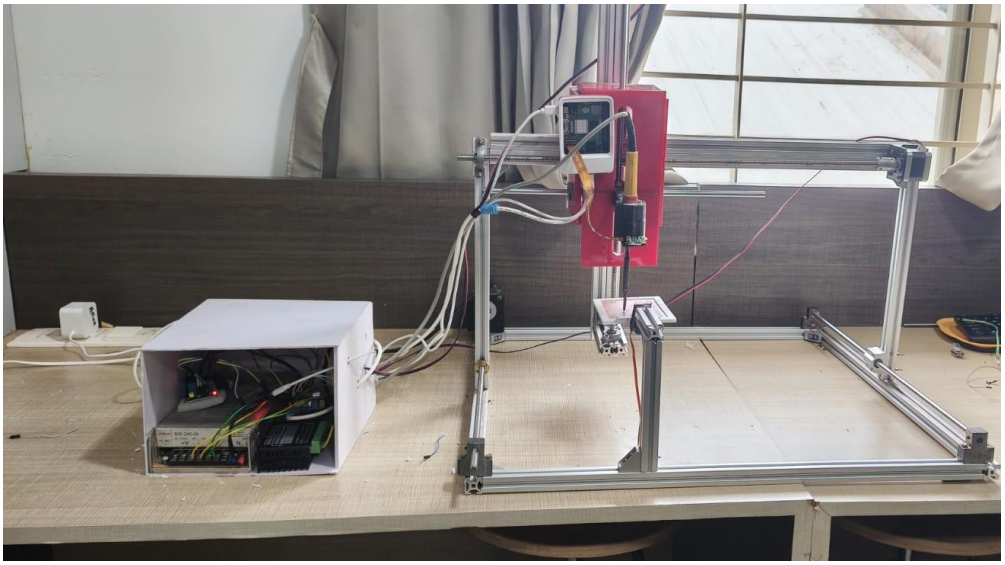


Fig 6.5 Unified System Integration & Complete Prototype Commissioning

7. CONCLUSION & LEARNINGS

The successful execution of this minor project demonstrates the design, development, and validation of a fully functional, vision-assisted automated selective soldering workstation. By deploying a robust 2 m x 2 m x 2 m Cartesian gantry driven by NEMA 17 stepper motors, the system accomplishes stable, highly repeatable, multi-axis linear positioning within a target accuracy threshold of ±1 mm. The successful integration of a high-resolution 12 MP Arducam module forms a real-time, data-driven computer vision pipeline. This feedback loop accurately detects component profiles, automates topological feature extraction, and verifies real-time component counts against an optimized reference layout using bounding-box confidence scores. This ensures that soldering sequences are triggered exclusively on non-defective, correctly populated circuit boards.

Furthermore, the introduction of a custom-designed, high-torque flipping mechanism powered by a NEMA 23 stepper motor resolves a critical production line challenge. It safely indexes and rotates the fixed PCB by 180° This allows a single, unified workstation to handle top-side optical quality assurance and bottom-side selective soldering within a compact footprint, eliminating the need for separate industrial machine cells. The system's electrical infrastructure is fully anchored by a custom-etched control PCB, which safely isolates high-current inductive motor spikes from sensitive microcontroller signaling. Hard-wired end-stop limit switches define safe axis boundaries and provide automated homing calibration. Throughput calculations indicate a reliable processing cycle capable of handling up to 20 boards per hour.

PROJECT LEARNINGS

The development lifecycle of this mechatronic system provided essential technical competencies, broken down across the following core implementation steps:

1. **VDI 2206 System Lifecycle Application:** Mastered the structured mechatronic design approach by systematically decomposing the project into mechanical, electrical, and control domains during specification, and using horizontal verification loops to validate parts systematically during integration.

2. **Computer Vision & Coordinate Mapping:** Gained practical expertise in image processing (grayscale conversion, Gaussian smoothing, and thresholding) and learned to mathematically map flat pixel coordinates into absolute millimeter movements (X_real, Y_real) for robot navigation.

3. **Kinematics & Multi-Axis Motion Control:** Acquired deep insights into managing stepper motor torque curves, step loss limits, acceleration profiles, and hardware boundaries using limit switches to ensure smooth path execution.

4. **Custom Electronics Layout & Board Etching:** Developed hands-on skills in chemical PCB path etching, trace-width safety calculations for high-load motor driver lines, and industrial electromagnetic isolation to safeguard control signals.

8. FUTURE SCOPE

To expand this selective soldering and optical inspection platform into a fully ruggedized, high-throughput manufacturing cell, the following advancements are proposed:

- **Transition to Closed-Loop Servo Control:** Replacing the current open-loop stepper motors with industrial-grade brushless AC or DC servo motors equipped with absolute rotary encoders. This will completely eliminate step-loss risk and track drift, dramatically increasing positioning speeds while improving axis tolerances to sub-micron accuracy levels.

- **Integration of Machine Learning for Solder Joint Quality (Pre & Post-Solder Tracking):** Upgrading the image processing backend from traditional edge thresholding to deep CNN classifiers. This allows the Arducam module to perform complex pre-solder component orientation monitoring, as well as post-solder joint analysis to automatically detect and log microscopic thermal defects like bridging, cold joints, or voiding.

- **Dual-Head Tooling & Multifunctional End-Effectors:** Redesigning the Z-axis carriage into a modular dual-head configuration. This setup can house an automatic liquid flux dispensing nozzle alongside the soldering gun tip, enabling inline automated fluxing right before thermal deposition to significantly improve joint wetting and cycle-time efficiency.

- **Automated Fume Extraction & Safety Enclosure:** Building an integrated industrial safety hood featuring UV-filtering acrylic paneling and a dedicated HEPA-carbon smoke filtration unit. This would safely trap toxic rosin-core soldering gases and regulate ambient workcell lighting to keep external light variations from affecting camera threshold performance.

- **Industrial IoT & MES Network Connectivity:** Interfacing the central microcontroller platform with cloud-based analytics backends using standard communication protocols like MQTT or OPC-UA. This enables real-time throughput tracking, component defect reporting, and predictive machine maintenance scheduling directly integrated into factory-level Manufacturing Execution Systems (MES).

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A Review of Intelligent Vision-Assisted Robotic Soldering Systems for Electronics Manufacturing

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Abstract—The necessity for precise, accurate and dependable soldering techniques for flexible and compact printed circuit boards (PCBs) has greatly increased due to the rapid development of electronics manufacturing. Traditional manual soldering methods often have drawbacks including inconsistent joint quality, high dependence on skilled labour and low productivity. As a result, automated soldering devices have become increasingly important in modern production environments. This study focuses on the advancements in vision-guided robotic soldering with the primary focus on increasing efficiency, accuracy and flexibility. Recent studies show a transition from conventional based detection methods towards a hybrid perception framework that combines deep learning models with traditional image processing methods. Algorithms based on deep neural networks including YOLO based detectors have shown excellent performance in identifying solder pads and junctions under a variety of lighting conditions and complex backgrounds. This research also looks into various robotics designs used in automated soldering systems such as SCARA, Delta and 6DOF manipulators as well as calibration techniques that enable accurate mapping between robotic coordinates and visual inputs. To improve motion efficiency and soldering accuracy trajectory planning and optimisation techniques including B-spline interpolation and evolutionary algorithms are also covered. The issues that were noticed were flexible PCB deformation, fine-pitch pad alignment and multi-wire soldering in small scale assemblies. alignment and flexible PCB deformation are also analyzed.

Key words: Robotic Soldering, Computer Vision, Deep Learning, Printed Circuit Boards (PCB), Vision-Guided Robotics, Trajectory Optimization, Industrial Robotics, Motion Planning, Adaptive Control.

I. INTRODUCTION

Robotic manipulators are at the forefront of modern manufacturing, surgery and logistics because of the persistent advancement in Industry 4.0. As these systems advance from simple repetitive activities to intelligent agents with complex kinematic duties the risk of their use has significantly increased. At present, in the industrial automation field, the "Build and Pray" approach of directly executing code on hardware is no longer an option. Safety, reliability, and validation must come first. One singularity in the trajectory plan, an unstable logic, or a microcontroller timing error can lead to catastrophic hardware failure, costly downtime, or potential harm to human operators. As a result, effective validation tools are no longer a nicety but an absolute requirement for robotic system development. This calls for an immediate transition to Model-Based Design (MBD) with a focus on Software-in-the-Loop (SIL) and Hardware-in-the-Loop (HIL) simulation. Software-in-the-Loop (SIL) is the first line of defense in validating algorithms. In this approach, the control logic and the physical plant (the robotic arm) are both modeled mathematically in a virtual environment such as MATLAB/Simulink. SIL is extremely helpful in validating the mathematical correctness of Inverse Kinematics (IK) solvers and trajectory planners. It operates in an ideal world where processing speed is infinite, and communication is instantaneous. But this is also its biggest drawback. SIL cannot forecast how an algorithm will perform under the constraints of real-world clock cycles, memory, and communication delays of a microcontroller. It validates the logic, not the implementation.

To move from the world of abstract mathematics to the real world, Hardware-in-the-Loop (HIL) simulation is used. In HIL simulation, the virtual controller is replaced by the actual embedded hardware that will be used in the final product. This hardware controller is then used to interact with the simulated robot in real-time, as if it were controlling real motors. This method uncovers system dynamics that are hidden from SIL simulation, such as UART communication delays, noise from analog-to-digital converters, and the time taken for floating-point operations on the processor.

II. LITERATURE SURVEY

1.1 Deep Learning-Based Robotic Manipulation of Flexible PCBs

Gao et al. [1] introduces the positional uncertainty that classical rigid-body controllers cannot handle by the substrate with its deformable and dynamically shifting nature, addressing the problem of fully automatic robotic soldering of flexible PCBs. Their proposed system consists of a cooperative dual-arm configuration: an assistive arm which is equipped with a vacuum chunk responsible for the contacts and stabilizing of the upper flexible PCB, while a soldering end-effector performs the actual soldering operation.

The techniques used in the proposed research are the core vision component i.e., the YOLOv3 based detection of solder pads. As feedback, detected bounding-box centroids are fed into the Jacobian-based dynamic controller. The dataset and tools used in this study comprise a custom dataset of 1,200

images captured in the real soldering environment across five scene categories-separated, adherent, overlapped, covered and under illuminated flexible PCBs with 1,000 images was used for training and validation. Hardware consisted of a Basler acA1300-60gc camera, an ADVANTECH PCI-1745 motion control card and an OptoSigma-GSC01 motion controller. Across all trials the study claims to have 100% contact success with position error reduced to less than 2 pixels in image space. The strength of the study is the YOLOv3 pad detection combined with a Jacobian dynamic controller which achieves sub-pixel accuracy without pre-calibration, generalizing across varied lighting conditions. The limitation faced was that the experiment was conducted without actual tin wire and validated only on flexible PCBs using custom dual-arm hardware which limits the applicability to small rigid PCBs.

1.2 Application of Computer System in Control and Programming of a Robotized Soldering Station

This study combines offline Gerber file parsing with an online vision system for pad coordinate recognition for THT soldering on rigid PCBs proposed by Milecki et al. [2]. The hardware components in this study consist of a station with a SCARA robot, a Siemens S7-1215 PLC, stepper motor-driven PCB transport and solder wire feed modules, a nitrogen supply system and a touch screen HMI. The technique used consists of 2 modes - a Gerber file parser for optimized trajectory generation which extracts pad coordinates and a classical image processing pipeline which includes a classical image processing pipeline. Using standard industrial hardware such as Siemens PLC and JVC stepper motors made it reproducible at moderate cost and it was evaluated on real PCB scans and camera images.

Against the Gerber reference the average pad detection error was found to be 0.06 mm sufficient for THT through-hole diameters of 0.8-1.5 mm. The highlight of this paper was the dual mode approach that offers flexibility, and off-the-shelf components ensure practical deploy ability without specialist hardware. The limitations of this study were that the camera-based detection proved unreliable under variable lighting with results reliable only from a flatbed scanner and the system had no real-time process monitoring or adaptive parameter control.

1.3 Development of an Automated System for Soldering and the Solder Inspection of an Assembly Line

This study proposes an automatic robotic assembly cell for soldering and solder defect inspection in the electronic component manufacturing line by Lee et al. [3]. To perform operations like heat caulking, PCB soldering, and defect inspection, the system integrates offline robot programming, real-time visual feedback and automated process control. In this assembly line, vision sensors and industrial robotic manipulators are integrated. The redesigned automated cell enhanced the overall production dependability, improved soldering accuracy and fault detection capabilities.

When compared to previous manual or semi-automated setups, the experimental results showed reduced failure rates and improved operational efficiency. The limitation faced by this study was that this approach depends heavily on specialized industrial hardware and controlled factory environments which may limit easy deployment in smaller production setups. Adapting to other electronic configuration manufacturing processes may require redesign as the system only focuses on a specific camera assembly line. And real-time performance of the system can be affected by vision system accuracy and system calibration requirements.

1.4 Development of an automated system for the soldering of USB cables

A fully automated robotic system for the soldering of multi-wire USB cables is presented by Gao et al. [4]. The technique that the system uses is three sequential mechanical modules - a spring-loaded separating module, a rack-and-pinion sorting module that includes a camera-guided U-shaped gripper and a cylinder-driven pressing module. The transition takes place from one stage to another via the AMRWD deep vision network built on ResNet-50-FPN with Feature Pyramid Attention across varying illumination and cluttered backgrounds for robust wire detection. For the dataset, 2,000 images were captured covering all 24 colour permutations, two illumination bands and three background types in which 1,600 images were used for training and 400 for testing and for annotation Labelme was used alongside standard industrial hardware including stepper motor-driven modules, a PCI control card and a Basler camera.

The overall precision of 97.66% and recall of 97.48% was achieved by the AMRWD which outperformed PSPNet, DeepLabv3+, U-Net, and colour-based methods with an inference time of 420 ms versus 570 ms for U-Net and a 95% of module-level success rate for separation, >91% for sorting and ~100% for pressing. Mid-section wire deformation causes occasional sorting failures and the stepper motor-based system yields a 32-second cycle time. There is no in-process solder quality monitoring integrated into the pipeline.

1.5 Vision-Based Robotic Manipulation of Flexible PCBs

Xiang Li et al. [5] propose a system that presents a vision-based robotic manipulation method that handles flexible printed circuit boards (FPCBs). The technique employed in this paper is the Neural network estimators that are used for both PCB deformation behaviour and the camera mapping that enables adaptive control under visual and dynamic uncertainties. To observe the PCB during manipulation the experimental setup uses an industrial robot manipulator with a vision sensor and to guide the robot in real time image feedback is processed to estimate the deformation. For modelling uncertainties and to allow accurate shaping of PCBs for soldering tasks it is compensated by the adaptive neural network-based approach.

System stability is evaluated and validated through Lyapunov-based analysis and the results are confirmed through practical robotic experiments. For conventional robotic systems it is difficult for automated manipulation of deformable electronic components but this approach makes it possible. The system avoids complex physical modelling and allows flexible PCB handling in robotic soldering applications by combining vision feedback with adaptive neural-network control. The limitation faced by this study is that this method depends heavily on vision feedback and neural network approximation which may be affected by lighting variations or occlusions. As the experiments are conducted in controlled environments, further work is required to improve robustness and make it effective and increase scalability for high-speed industrial production settings.

1.6 Automatic Optical Inspection System for Solder Quality Classification of THT Solder Joints

Fonseka and Jayasinghe [6] created a system optical inspection that checks the quality of the solder connections

on printed circuit boards which uses through hole technology. For spot soldering flaws which occur during the electronic component production for this they have used machine vision technology. The system uses image processing methods, which contain taking photos next, editing them later on dividing them into few sections and identifying key features. This actually helps to find the problems occurring from solder like solder bridging, missing solder, too much solder. To take 300 pictures of the solder joint they have used a machine vision camera for analysis. Algorithms are used to examine these images taken from machine vision cameras to judge the quality of the solder connections. The results are pretty clear, like the whole system makes inspections run quicker and it catches solder defects more efficiently than the old-school approach. With this kind of setup, quality control feels faster and more uniform, and, yes, mistakes drop. But, lighting and image quality matter a lot, so performance can swing pretty noticeably. If you're dealing with different PCB designs, you will probably need to tune some settings, or at least re-calibrate it a bit, because it doesn't magically stay perfect everywhere.

1.7 Deep Learning-Based PCB Defect Detection Using X-ray Images

Zhang and his group [7] tested a defect detection technique on printed circuit boards, using X-ray images. They fed their model a dataset of 3D X-ray images, where every sample contains different varieties of solder joint defects. Then they evaluated results using recall and false positive rates. This study aims to find solder defects that are hard to spot with regular optical inspection methods. The researchers proposed a deep learning framework that combines Convolutional Neural Networks with Long Short-term Memory networks. In this method the Convolutional Neural Networks model is used to find features from X-ray images. The Long Short-term Memory networks are used to find patterns in a sequence, which improves the defect detection capabilities. The model was. It turned out that their deep learning model detects defects better than older inspection methods. Honestly, what stands out is how reliably it identifies solder joint issues. Even so, this style of method still demands a lot of labeled data and serious computing power, so both training and rolling it into real use is not exactly effortless.

1.8 Deep Learning-Based Anomaly Detection for Solder Joint Inspection

Mendizabel-Arrieta and his team [8] proposed a machine learning system for finding solder joint anomalies, designed specifically for manufacturing settings. The aim is improvement in quality control across the industry, via computer vision and AI. Unlike some other models, theirs can detect solder patterns directly, without leaning on fixed defect rules or preset criteria. They capture solder joint images from several angles with a camera, then push those images through their pipeline to flag potential defects. For evaluation, they used a real-world dataset from manufacturing situations. The system reached close to 0.90 on the F1 score, so it looks pretty accurate overall. The main advantage is that it can recognize odd or atypical patterns automatically. Still, you really need steady, controlled lighting and high quality cameras for stable results. Also, if production conditions shift, retraining the model becomes unavoidable in practice.

1.9 ChangeChip: Reference-Based Unsupervised Change Detection for PCB Defect Detection

Fridman et al. have presented an unsupervised defect detection technique named as Change chip which aims at inspecting the printed circuit boards by referencing. This study focuses on the use of computer vision for automated PCB defect detection without use of labelled datasets. And the method identifies the detect by comparing a reference PCB image with a test image. To simplify this principal component analysis is employed for decreasing the dimensionality while K-means clustering is used to defect the areas. Moreover, the scale invariant feature transform is applied to align the reference and test images accurately for comparison. They have developed a dataset called CD-PCB which has 20 pairs of PCB images with labelled defect areas for assessment. The overall effectiveness of the algorithm was calculated by using the precision and recall metrics, and achieved precision of 0.80 and recall of 0.87. The main gain point of this method is that it detects various PCB defects without having a large amount of labelled training data. However, the approach is open to image misalignment which can lead to detection of noise or minor variations as defects.

1.10 DemoTeach-Robot Programming with Motion Capture System for Foundry Applications

Li et al.[10] introduced a robot programming system named DemoTeach which is based on motion capture and designed for industrial manufacturing purpose. This study aims to enhance human-robot interactions by enabling workers to instruct robots through physical demonstrations rather than intricate programming. In this approach workers execute tasks using a dummy tool fitted with reflective markers and their movements are recorded using an OptiTrack motion capture system. Then the robotic arm can follow the robot trajectories which are created from the recorded motion data. To perform deburring operations on aluminium casting components a KUKU iiwa robotic arm and an OptiTrack camera setup were used in this system's implementation. The outcome of the experiments show that the robot could successfully finish the given task and precisely mimic the human movement displayed. The main advantage of this system is that it makes robot programming easier, making it possible for operators without programming experience to easily train robots. Although, minor trajectory errors may arise from variations in hum hand movements, requiring extra correction mechanisms.

1.11 Machine Vision Based Solder Joint Positioning

Yutong Yuan et al.[11] proposed system which is on machine vision for the positioning of solder joints in body in white welding inspection in the automotive industry.The main goal of the study is to precisely locate the solder joints, even in harsh environments where lighting, oil stains, and electrode wear are present. The authors categorized the issue under machine vision-based techniques for solder joint localization. Forget those old template-matching methods that can't handle noise or lighting changes—this study uses texture feature analysis and regional statistics to zero in on solder joint locations. The method starts by picking out regions of interest from the images, then switches them to grayscale and HSV modes. Next, texture filtering boosts edge features, making solder joints stand out. Using a circular sliding window, the algorithm hunts for regions packed with features, flagging the solder joint. They brought together cameras, lenses, proper lighting, and an industrial computer for image capture and processing. Tested in both simple and tough industrial setups, the method nailed solder joint positions and handled lighting interference well. Most joints were detected with minor errors, but you still need good

image quality and tight limits for it to work. Noisy environments remain a challenge. Overall, mixing texture features with statistics improves accuracy in solder joint positioning for industrial applications.

1.12 Adaptive Template Matching for BGA Solder Joint Detection

Here's a method for adaptively matching Ball Grid Array (BGA) solder joints on PCBs using X-ray images. The goal is to get more accurate detection, especially when joints are hidden behind chip packages. Because BGA joints don't show up under normal light, X-ray imaging gives a peek inside the solder structure. The process starts by isolating the solder joint area after the X-ray, then runs edge detection and thresholding to build an edge template. The template matching works because it uses edge direction vectors as prior knowledge. To make the system tougher, the algorithm searches globally using an image pyramid, helping it find joints at different scales. Final tweaks come through least optimization. Results show this method finds solder joints.

1.13 PCB Soldering Defect Detection Using Template-Based Image Processing

S. Ozturk and B. Akdemir [13] came up with a computer vision approach to inspect, and then spot soldering defects on printed circuit boards. From what it looks like this sits pretty firmly in the automated optical inspection (AOI) group, where image processing is used to notice issues. Their primary goal was kind of simple yet hard to miss: identify soldering problems , most notably short circuits that end up between solder points, and that can really disturb the circuit's function. These defects also have to be caught early because once they show up it becomes a real issue for the whole board. Manual checking is not good because it is slow and people can make mistakes. It does not work well for making a lot of electronics. So, we need a system that can automatically check for these problems. The method we suggest uses a way of looking at images along with a type of computer clustering called Fuzzy C-Means or FCM.First we change the picture of the printed circuit board or PCB from color to shades of Gray. Then the FCM algorithm groups the image into parts that show the background of the PCB, the paths of the circuit and the solder joints. This helps us to see the different parts of the PCB. We use this technique to make it easier to spot any soldering issues. Once the joints are identified, a joint template is created based on the PCB design to establish the relationship between adjacent solder joints. In the course of inspection, each solder joint is compared with the other solder joints near to it, as indicated by the template. This template-based inspection process reduces computation time and increases the inspection speed. The template also helps identify the solder joints that are deliberately connected, thus enabling the inspection system to identify only unwanted sort circuits. The results obtained from the experiments on a PCB with 50 solder joints indicated that the system can identify the soldering defects. The inspection process was repeated 85 times with 100% accuracy for the PCB model. The proposed method for inspection is effective and fast; however, it requires a template to be created manually and the PCB to be positioned correctly for inspection. The study proves that template-based computer vision can be highly effective for the inspection of solder defects on a PCB.

1.14 Deep Learning Based Solder Joint Defect Detection

Zhang et al. [14] introduced a system for detecting solder joint defects using deep learning techniques applied to X-ray images of industrial PCBs. The aim of the research is to minimize the workload of inspecting the solder joints manually. Their research falls under the deep learning techniques used in the analysis of the X-rays. In the research, the authors use different AI techniques, including CNN,3D CNN, and LSTM. In the research, the authors use the following steps: the authors pre-process the images. In the pre-processing, the authors use techniques that involve the extraction of the region of interest and stacking the slices. Different pre-processing techniques were used to handle the varying number of slices and the presence of noise. Four AI models were assessed to identify the most effective architecture for detecting defects. The experimental findings indicate that the deep learning models considerably enhanced detection accuracy compared to conventional inspection techniques. The new system also makes things easier for specialists. It filters out a lot of solder joints, which is about 73.29%. This means that experts can focus on the solder joints that are actually defective. The problem with using learning is that it needs a lot of labelled datasets and strong computers, which can make it hard to use in real life. A lot of work is needed to prepare the data. The computers need to be very powerful which might limit how deep learning approaches can be used.

1.15 SMT Solder Joint Quality Inspection Using Surface Reconstruction

Yu and Zhang[15] introduced a system using surface reconstruction techniques for inspecting 3D solder joints. By using 3D analysis instead of the traditional 2D image inspection their study looks for improvement of solder joint quality inspection. They classify their method for solder joint analysis under 3D image reconstruction techniques. Initially, the system applies image denoising through wavelet packet decomposition, Wiener filtering, and median filtering to eliminate noise from the image of solder joints. Following the denoising process the system rebuilds the surface of the solder joint using Shape-from-Shading technology. This method uses a lighting model that includes both diffuse and specular reflection parts to accurately rebuild the solder joint surface. The reconstructed 3D model of the solder joint is then examined. From this we can find out parameters like the shape, height and curvature of the solder joint. These parameters help in finding defects and checking the quality of the solder joint accurately than traditional 2D methods. The solder joint parameters give a picture of its quality. The 3D model of the solder joint is very helpful in this process. It provides information about the solder joint. The Shape-from-Shading technology plays a role here. It helps in creating a 3D model of the solder joint. Experimental findings indicate that the proposed system effectively extracts 3D quality information of solder joints and enhances inspection reliability. Nonetheless, the method involves intricate computations and relies significantly on precise illumination modelling, which may lead to increased processing time.

1.16 Vision-based soldering process parameters calculation for robotic soldering

Hao Ren, Xinyu, Wu, and Wanfeng Shang [16] focuses on Through Hole Technology (THT) assembly automation as it is highly required in aerospace and military to form more robust connections because of its high reliability rather than simple surface mount alternatives. The paper presents the current industrial need for building vision assisted fully autonomous frameworks rather than teaching playbacks, which require manual training and debugging. The method used here is mathematical modelling rather than the traditional trial and error method, as the mathematical

approach calculates the required solder volume precisely using high order curves and also approximately gives the longitudinal profile of a standard solder joint. Directly from the PCB, the pad radius, pin cross sectional area and through hole radius data are extracted using Hough transforms, HSV color space, asking and Canny edge detection. These data are then combined with a soldering database to provide consistency among different types of components like indicators and LEDs; it also prevents duplicate computations. During the performance analysis the system shows 99% of similarity when empirical data was optimized against existing literature data. The average computational time for vision is 18.99 ms and for calculation it requires 12.40 ms which means to solder a new joint the total average processing time is 31.39 ms which is very much faster and more efficient compared to the system requiring human intervention.

1.17 Experimental **Path Tracking of a Cartesian Robot Using Image Processing**

Allahyari and Fakhari [17] focuses on an essential framework that has high flexibility path tracking and is also cost effective using a 3 DOF cartesian robot. For simulation they have used MATLAB software which includes IP camera software, Machine vision toolbox, and stepper motor driven linear actuators as they provide a viable entry point for small scale soldering automation. To define nonlinear paths like sinusoidal, circular courses they have used passive optical sensing. It is based on image processing they have captured images to create a dataset which is exported to MATLAB to extract the coordinates of image, these coordinates are then sent to an Arduino UNO microcontroller using a USART protocol at 38,400 baud rates. For processing they have used an HSV color model where they set a threshold at >0.7 saturation level in order to obtain better robustness in industrial lighting conditions without the need of using an external light source. The system shows a high-level image processing precision with up to 1 mm of maximum image processing error and roughly 0.48 pixels at mean reprojection error but the mechanical tracking performance it depends on the path complexity, for linear paths it shows acceptable deviations while for circular path tracking it exhibits error up to 3mm. The major benefit of this approach is that it is accessible, reduces multi pixel pathways to single pixel trajectory but on the flip side the memory constraint of the Arduino UNO is major setback as thousands of paths are to be reduced to a maximum of 128 co-ordinate, due to the reduction in this data the discretization error are evident during high curvature paths as the robot does not travel in smooth true curve instead it follows points between short linear segments. In this paper the vision pipeline is very accurate for soldering trajectories but the hardware mechanical planning lacks accuracy in real time due to the memory limitation of the microcontroller.

1.18 Solderbot: Automated Soldering using Robot Arm

This paper presents various robotic arm applications and automated manufacturing techniques that includes deep learning-based robot controls, vision-based soldering, point cloud-based motion planning and visual programming for robots. In this paper they have used OpenCV for image processing, RoboDK for offline motion planning and Gazebo for 6 DOF robotic simulation. The reason they chose OpenCV instead of yolo or other deep learning methods is because calculation of soldering targets does not require the complex neural network computation as they are geometrically straightforward.

The advantage of this bot is that it is cost effective and does not use complex AI models or expensive hardware but it has significant limitations. The results are only validated through simulation while for real time hardware the marker-based recognition is not suitable for small metal contacts of common PCBs, the amount of solder to be dispensed is supposed to be calculated mathematically rather than by any closed loop feedback and this method does not generalize to densely populated boards. This method is only suitable for educational grade rather than industrial grade systems.

1.19 Optimization of the soldering process by the DMAIC methodology

This paper follows DMAIC (Define - Measure - Analyse - Improve - control) methodology. It focuses on automotive PCB soldering applications, mainly soldering of pins that connect a controller board and a wire harness plug. This process has three observable modes of failure i.e. solder alloy breakage of spool, operator correction time and part scrapping. The research shows that DMAIC effectively reduces the variability of soldering defects and also reduces its magnitude with a drop from 4000 - 16000 PPM range to 1000-2700 PPM which is below 3000 PPM target using PFMEA analysis, sustained control plan, and targeted corrective actions. The study specifically addresses seven highest RPN root causes which includes oxidised solder alloy, operator skill and soldering pin condition but it also has some limitations such as the comparative analysis against different approaches such as DOE based parameter optimization is not performed, it does not have any statistical, simulation or modelling data to validate the corrective actions before the implementation and mainly depends on human trainings and compliance with new procedures and thus the system stability is not guaranteed.

1.20 Optimization of Process Parameters in Lead-free Wave Soldering Based on Orthogonal Experiment and Analytic Hierarchy Process

Yin et al.[20] presents the research on the transition to lead free soldering driven by EU-RoHS directive, it was observed that though much research is done on lead free solder alloy formation using Sn- Zn, SnAgCu etc. To examine and figure out the four parameters, flux quantity, preheat temperature, track speed, and soldering temperature at three levels they have used the L9(3^4) orthogonal array as it is a structured and effective substitute for full factorial experiments. The analytical method to quantify and rank the influence of each parameter level the AHP (analytical hierarchy process) is introduced using a pairwise comparison matrix derived from observed PPM defect rates. The dataset consists of 100 PCBs per experimental group which consist of 446 solder joints each. The primary result of this paper gives the important process parameter that is track speed of 44.5% total AHP weight, followed by 26.6% of flux quantity, 16.1% soldering temperature, and 12.8% of preheat temperature with ideal settings of 40 mL/min flux, 110 degrees Celsius preheat, 1.6 m/min track speed and 265-degree Celsius soldering temperature. Another important finding is that if we reverse the pcb direction it reduces defects from 10,291 PPM to 4,574 PPM at the same parameter values.

1.21 Development of a vision system integrated with industrial robots for online weld

seam tracking Representing the stage of highly specialized industrial vision systems, this study conducted by Nguyen et al. uses triangulation technique to acquire 3D object data to develop a precise robotic tracking platform that

employs a CMOS camera and a line laser projector. The researchers integrated a discriminated convolution tracker refined by deep reinforcement ;earning (DRL) as it continuously outputs corrective pixel movement action based on visual states to handle complex demands like tracking the square groove butt joints. The system was trained on a custom dataset and was validated in controlled lab environments achieving an absolute mean welding error of 0.31mm.

1.22 RT-2: Vision-Language-Action Models Transfer Web Knowledge to Robotic Control

Marking a huge breakthrough toward generalist autoregressive policies, Brohan et al. RT-2 is a foundational Vision-Language-Action (VLA) model that directly combines internet-scale knowledge into robotic control systems. Rt-2 trains itself to produce robotic trajectories by representing physical action in the form of test token within standard language model vocabulary, which is achieved by employing massive pre-training on a set of visual question and image captioning datasets. Which enables Rt-2 for unprecedented emergent semantic generalization, this gives the robot a perspective about the properties of object or recognizes improvised tool zero short in real world performance hence allowing reasoning. One the other hand, the innovative ability of mapping visual and warble linguistic inputs to produce an action comes at a cost due to its massive parameter count of up to 55B making it difficult for local development and making it slow for high-frequency control application as it works on autoregressive generation mechanism.

1.23 OpenVLA: An Open-Source Vision-Language-Action Model

To modify and improve upon generalist autoregressive VLA architectures, Xiao et al. present OpenVLA is a 7B-parameter fully open-source foundation model. Built by combining advanced Vision Transformers DINOv2 and SigLIP with a LLaMA 2 backbone, the model was trained on the Open X-Embodiment dataset, encompassing over 970,000 robot episodes of robot morphologies. By capturing spatial-semantic features, OpenVLA exhibits a significant improvement in generalization capabilities, outperforming RT-2-X model by 16.5% in task success rates across unseen robot embodiments. Its open-source availability and support for parameter-efficient fine-tuning (PEFT) make it a highly versatile tool for researchers; nevertheless, because it relies on the same discrete token-by-token decoding as RT-2, it still faces limitations regarding inference latency in highly dynamic, real-time physical environments.

1.24 π_0 A Vision-Language-Action Flow Model for General Robot Control

To address the inference and skill challenges of autoregressive VLA models, Black et al. propose **π_0,** a versatile policy that moves towards continuous control methods. Rather than breaking down physical actions into text tokens, π_0 pairs a pre-trained Vision-Language Model backbone with an independent "action expert". This expert creates continuous actions through flow matching, which is a type of diffusion. The model was trained on a large cross-embodiment dataset with over 10,000 hours of robot manipulation data. It was tested on complex tasks that take time, like bussing tables and folding laundry in sequence. The flow matching design effectively connects high-level semantic thinking with fast (up to 50 Hz) continuous motor control. However, it still faces challenges due to the significant computing resources needed for pre-training and the lack of high-quality demonstration data for completely new robot designs.

1.25 Fine-Tuning Vision-Language-Action Models

Optimizing Speed and Success Focusing on the optimal deployment and adaptation of existing foundation models, Kim et al. developed the OpenVLA-Optimized Fine-Tuning (OFT) recipe to overcome the speed limitations of standard autoregressive generation. By modifying the base OpenVLA architecture to utilize parallel decoding, continuous action representations, and a simplified L1 regression learning objective, the model can predict entire chunks of future actions simultaneously in a single forward pass. Evaluated heavily on the LIBERO simulation benchmark and real-world ALOHA bimanual teleoperation setups, OpenVLA-OFT achieved a staggering 26x to 43x increase in action generation throughput, which directly correlated with boosting the average success rate from 76.5% to a state-of-the-art 97.1%. This methodology provides a vital strength by enabling high-speed, reliable fine-tuning for new environments, although fully capitalizing on these optimizations still requires starting from a heavily resource-intensive pretrained base model.

III. RESULTS

A. Literature Classification Overview

Based on a comprehensive review of 20 key articles represented in table 1, the dominant trend in modern automated soldering is the combination of visual imaging with neural networks, specifically YOLO based structures. All evaluated studies focus on Solder Joint Inspection as their final evaluation context while the component placement inspection and standalone solder paste are still understudied.

Table 1: Overview of Publication in the Literature Base

Paper Title/ Author(s)	Year	NDT Method/ Techniques	Key Finding
Gao et al. – Robotic manipulation of flexible PCBs	2020	Visual (YOLOv3)	Sub-pixel accuracy via Jacobian controller; 100% contact success
Milecki et al. – Robotized soldering station	2019	Visual (Classical)	Gerber + vision dual mode; 0.06 mm pad detection error
Lee et al. – Automated assembly line soldering	2018	Visual + AOI	Integrated offline programming with real-time visual inspection
Gao et al. – USB cable soldering system	2023	Visual (ResNet-50-FPN)	97.66% precision, 97.48% recall on wire detection
Li et al. – Vision-based flexible PCB manipulation	2018	Visual + NN adaptive	Lyapunov-stable deformation control for FPCBs
Fonseka & Jayasinghe – AOI for THT solder joints	2019	Visual (AOI)	>95% detection on 300 joints; k-means clustering
Zhang et al. – PCB defect via X-ray + CNN+LSTM	2022	X-ray (CNN+LSTM)	Reduced manual inspection workload by ~73%

Mendizabal-Arrieta et al. – Anomaly detection	2025	Visual (Deep Learning)	F1 score ~0.90; multi-view camera setup
Fridman et al. – ChangeChip unsupervised PCB	2021	Visual (PCA + k-means)	Precision 0.80, Recall 0.87; no labeled data needed
Hao Ren et al. – Vision-based parameter calc.	2021	Visual (Hough + Canny)	31.39 ms/joint; 99% empirical similarity
Allahyari & Fakhari – Cartesian robot path track	2023	Visual (HSV)	Max 1mm image error; circular path error up to 3mm
Ozturk & Akdemir – Template+FCM inspection	2017	Visual (FCM)	100% detection accuracy across 85 trials on 50-joint PCB
Xueli Hao et al. – Adaptive template BGA detect.	2018	X-ray (template)	Effective detection of hidden BGA joints
Zhang et al. – CNN/3DCNN /LSTM defect detection.	2022	X-ray (CNN/LSTM)	73% workload reduction; high accuracy on 3D X-ray
Yuan et al. – Solder joint positioning (BIW)	2024	Visual (texture + stats)	Accurate positioning under industrial lighting interference
Nguyen et al. – Weld seam tracking with DRL	2024	Visual (laser + CMOS)	0.31 mm mean welding error via deep reinforcement learning
RT-2 (Brohan et al.) – Vision-Language-Action	2023	Visual (VLA/LLM)	2-6x better generalization; 55B params, 1-3 Hz inference
OpenVLA (Xiao et al.) – Open-source VLA model	2024	Visual (VLA)	Beats RT-2-X by 16.5%; runs on consumer GPUs
Yin et al. – Lead-free wave soldering optimization	2010	Process (AHP/DOE)	Track speed dominant parameter; PCB reversal halves defects
Zasadzien – DMAIC soldering optimization	2016	Process (DMAIC)	Defects reduced from ~4000-16000 PPM to 1000-2700 PPM

B. Solder Joints Inspection

Among the 20 reviewed articles, post solder inspection is heavily documented pipeline intervention that has been reported the most.

i. Visual imaging and X-ray Imaging:

Visual imaging continues to be the most dominant inspection modality due to its ease of deployment and lower cost barrier. In this domain the conventional pipelines using FCM clustering achieve up to 100% accuracy on localized batches while the unsupervised ChangeChip models yield a precision of 0.87 without requiring a huge labelled corpus. The precise solder volume can be calculated in an efficient 31.39 ms per joint by mathematical feature extraction via Hough transforms and Canny filters. Specialized 3D X-ray systems using hybrid CNN-LSTM networks for subsurface inspection detect hidden internal defects and voids and reduces manual checking workloads by ~73%.

ii. Process and Trajectory Optimisation:

Simply reversing the PCB direction reduces wave soldering defects by more than half and statistical frameworks indicates that the track speed is the most crucial process variable having a 44.5% analytical weight. Additionally implementing DMAIC methodologies reduces industrial defects from 16000 PPM to a controlled range of 1000-2700 PPM range. SCARA and Cartesian hardware configurations are used for trajectory tracking as they offer reliable path execution, though along the sharp circular trajectories the data discretization error upto 3mm is displayed due to memory limitations of the microcontroller.

iii. Emerging Vision language Action Models:

Large networks such as 55 billion parameter RT-2 offer high semantic generalization by translating physical coordinates into standard language tokens though their slow inference rate of 1-3Hz makes them inappropriate for fast assembly lines. While specialized parallel decoding recipes (OFT) reduces latency by simultaneously predicting future action chunks to achieve a 97.1% success rate, OpenVLA increases baseline success rates by 16.5% and run on consumer grade hardware.

C. Robotic Architectures in Reviewed systems

Task specific kinematic configurations are essential for mechanical execution across automated assembly lines. High dexterity 6 DOF manipulators handle complex physical orientations while standard SCARA systems handle rigid planar assemblies with high repeatability. Dual arm cooperative systems are deployed for highly deformable substrates like flexible PCBs while a secondary end effector performs the actual soldering using an assistive vacuum clutch stabilizes the board. In accordance with the 3-axis gantry frame of the current project low cost cartesian coordinates offer an affordable substitute.

D. Distribution by NDT Method and ML Technique

The integrated distribution by NDT method and ML techniques based on classification is represented in table 2.

Table 2: Papers Classified by NDT method and ML techniques

NDT Method	**ML Technique**	**No. of Papers**	**Dominant Use Case**
Visual Imaging	Neural Network (CNN/YOLO)	14	Component placement, joint quality, pad detection

Visual Imaging	Conventional Supervised (SVM/DT)	3	Joint defect classification
Visual Imaging	Unsupervised (k-means/FCM)	2	Anomaly/defect detection without labels
X-ray Imaging	CNN + LSTM / 3D CNN	4	Hidden defect detection, void analysis
Visual (3D/Point Cloud)	Neural Network	1	Flexible PCB deformation control
Process Optimization	Statistical / DMAIC / AHP	2	Parameter tuning, defect rate reduction
Vision-Language-Action	LLM + Diffusion / Flow Matching	3	Generalist robot control, task adaptation

IV. DISCUSSION

A. Limitations of Existing Literature

The gap between mass production and experimental reality was revealed after the critical evaluation of literature. Due to limited sample sizes high performance metrices are frequently overstated with some studies validating complete correctness on fewer than 50 total joints. This lacks statistical data for actual industrial manufacture where high volume lines need continuous evaluation across thousands of samples to capture parts per million defect rates. Additionally existing training sets neglect factory floor factors like thermal distortion, mechanical vibrations and fluctuating ambient lighting as they are largely isolated within pristine laboratory environments with fixed camera perspectives. Most setups are extremely disjointed focusing on a single isolation module (such as standalone path planning or post hoc flaw classification) rather than a cohesive end to end pipeline. Further the optical systems remain to be highly susceptible to ambient conditions without specialized stationary scanning devices or shroud design ordinary camera networks become unreliable under changing workspace illumination.

B. Discussion of Intervention Points

The hardware setup and the total cost of handling industrial and manufacturing scrap are directly impacted by the choice of automation checkpoint. Although the pre solder paste inspection stops the components from being wasted on a badly printed board, the underlying algorithms are unable to meet strict throughput standards.

C. Discussion of ML Techniques

Neural networks clearly surpass traditional picture segmentation algorithms for automated computer vision applications. The implementation of YOLOv8 provides integrated multi class categorization that is ideal for high-speed verification loops, offers improved feature extraction for small scale objects and anchor free object detection. It's quick inference time less than 20 ms easily fits within the processing budgets needed for online factory operations. Traditional techniques like template comparison or Fuzzy C- means helpful backup option when the initial training data is severely limited.

D. Research Gaps & Future Directions

Gaps identified:

1. Lack of fully integrated robotic soldering systems

The majority of the studies focus on only one part of the process such as detection, trajectory planning or defect inspection and have different modules that operate independently due to which they lack effective communication with each other without human intervention. Only a few have an end system that integrates vision detection, robotic soldering and quality inspection in one complete automated framework.

2. Limited validation in real industrial environments

There is limited research about how these systems work in real conditions, high speed industrial production lines with continuous operation. The studies are tested only for short duration instead of continuous industrial operation, resulting in research to evaluate robotic soldering systems in real manufacturing environments with large production volumes and long-term operation while we already have sufficient evidence in laboratory setups and simulations.

3. Dependency on controlled lighting and imaging conditions

Change in illumination, occlusions, and reflection from metals significantly reduces the computer vision and image processing data accuracy, and hence the system still requires controlled and clean lighting conditions. Vision-based robotic soldering systems majorly depend on accurate image acquisition for detecting solder pads, wires, or joints. Good performance is achieved only when lighting conditions are carefully controlled and the captured images are clear, which is done by image processing and deep learning algorithms.

4. Datasets for Robotic soldering tasks are limited

There are very few generalized datasets on solder joints and solder pad detection although we have small custom datasets that are collected in specific environments. This limits the comparison, benchmarking, and optimization. As the datasets are not publicly available or standardized it becomes difficult for researchers to compare different algorithms fairly or benchmark their performance.

5. Lack of generalized robotic learning systems

Although the current trend follows RT 2 and OpenVLA, which help to build generalized robotic systems, they require high computational resources and have not been practically implemented for specific industrial tasks like PCB soldering. Models typically require very large datasets, powerful computing infrastructure, and extensive pre-training which makes them difficult to deploy in practical industrial settings. As a result, there is still a significant gap between general-purpose robotic learning systems and specialized industrial automation tasks which highlights the need for research on efficient and task specific adaptation of these models.

Future Work:

While the existing body of work has proposed a number of potential video vision-based soldering and inspection system, there remains substantial opportunity to enhance this technology, for example, further presentations based on this technology could focus on developing more robust vision system, allowing for reliable operations in various lighting conditions, while functioning in actual manufacturing settings.

Another major direction for further work will include the integration of real time performance monitoring and feedback control into robotics soldering system so that the robot can monitoring and feedback control into robotic soldering system so that the robot can dynamically adjust parameters such as soldering temperature, solder volume, and positional orientation, as necessary during the soldering operation. The implementation of this further would result in improved quality of solder joints as well as a reduction in solder joint defect rates.

It would also be highly advantageous if further systems were to combine deep learning methods with traditional images processing systems to achieve improved defect classification rates with lower processing costs. Another area of focus would be the development of improved methods of motion planning and optimizing the trajectory of individual robot motion, resulting in increased soldering speed and precision.

Having multiple sensors (e.g., 3D imaging, thermal imaging, and force sensing) will also allow for increased accuracy and fidelity of automated soldering systems in future.

V. CONCLUSION

In this literature review, automated inspection, vision-based robotic soldering, multi-research related automation and processes have been examined. As a result, based on all of the research studies reviewed, manufacturing automation and quality control can achieve higher levels of efficiency through the use of robotics, machine learning and computer vision when used together.

There are different methodologies such as defects detected by using machine vision-based inspection and by using deep learning-based inspection for the purposes of inspection.

The research findings provide valuable information regarding how robotics control systems and automated inspection algorithms combined with advanced imaging technology will improve the reliability of soldering results. However, there are also difficulties in obtaining sufficient training datasets needed to build deep learning models and the large amount of computing resources required.

In summary, the research conducted indicates that vision guided robotic soldering systems have a high potential to support manufacturing automation and to increase the level of quality control within present electromechanical manufacturing.

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2	en.wikipedia.org Internet Source
3	ijece.iaescore.com Internet Source
4	Yuan Gao, Zhi Chen, Jiahuang Lin, Xiang Li, Yun-Hui Liu. "Development of an automated system for the soldering of USB cables", Robotics and Computer-Integrated Manufacturing, 2023 Publication
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Word count:	7,824
Character count:	45,560
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Submission ID:	2993253550



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Vision-Assisted Robotic Soldering Process Optimization Using YOLOv8-Based PCB Component Detection and Solder Hole Detection

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Abstract*—Traditional Printed Circuit Board (PCB) assembly lines highly rely on conventional manual and semi-automated soldering, which causes high defect rates, joint inconsistencies, and inability to dynamically inspect defects and is highly constrained by rigid, pre-programmed coordinate paths. This study represents an integrated vision-assisted automated soldering framework that is driven by the need for localized production agility in flexible manufacturing high-mix environments. The complete system uses a two-stage vision verification procedure that utilizes two custom-made datasets: a front-side dataset of the PCB was trained using a fine-tuned and localized YOLOv8 neural network of the PCB for object detection to verify the right component count and the placement and orientation of the components, and a backside dataset of the PCB is used for solder hole detection and coordinate extraction required for robotic soldering process. A 3 DOF Cartesian robot is designed using SolidWorks and fabricated with mathematical evaluation for structural and kinematic limits. After target arrays are extracted, the mathematically optimal movement path for the robot is computed using an A* pathfinding optimization algorithm. This design shifts the industrial micro-assembly focus totally from mechanical rigid fixtures to highly flexible adaptive vision-driven software intelligence.

Index Terms—PCB, YOLOv8 model, Robotic Soldering, PCB Inspection, *A** Pathfinding, Automated Manufacturing, Object Detection, Vision-Assisted Robotics.

I. Introduction

The rapid development of the manufacturing of electronics has increased significantly, as has the demand for smart and adaptive automated soldering systems to handle compact and high-density printed circuit boards (PCBs). Traditional or current robotic soldering systems rely on the teach-playback method, which is fixed and is manually programmed for coordinate paths and, moreover, is not suitable for modern high-mix, low-volume manufacturing environments where there is a need for frequent altering of PCB layout. The accuracy of soldering and reliability of inspection in industries are highly influenced by the changes in illumination, reflective copper tracks, thermal glares, solder mask textures, and alignment of components. Due to these limiting factors, industries are shifting to vision-assisted manufacturing systems where image processing and artificial intelligence act as a primary control layer for inspection, coordinate extraction, defect detection, and motion planning for the robot. Therefore, to isolate solder pads, through-hole geometrics, and the components used in the PCB from noisy backgrounds, image processing operations such as grayscale conversion, Gaussian filtering, edge-gradient extraction, and adaptive threshold segmentation are implemented.

To overcome the limitations of traditional rigid automation systems, this research focuses on implementing a vision-assisted robotic soldering framework that is proposed and focuses on advanced image processing and intelligent perception pipelines. The proposed study consists of a system that is a dual-stage perception architecture, i.e., front side PCB verification and back-side solder hole localization. In the first step, to recognize electronic components and verify placement alignment prior to soldering, a specialized dataset of PCB images is utilized to train a localized YOLOv8 deep-learning model. In the second stage, by the means of mathematical image processing techniques, the coordinates of the holes of the components that need to be soldered are extracted from the Back PCB images for robotic alignment and solder positioning. The Back PCB images are also trained by the YOLOv8 model. Using a pathfinding optimization algorithm, A*, the coordinates are processed to generate efficient robot traversal paths, and the extracted coordinates are transformed into real-world Cartesian workspace coordinates. To physically perform

the generated trajectories, a supporting 3-DOF Cartesian robot platform is integrated, which shifts the operation from rigid hardware programming to adaptive software-driven manufacturing.

II. Literature Review

Fonseka and Jayasinghe [1] proposed an automated optical inspection system for detecting solder defects using image processing and machine vision methods. Using techniques like grayscale conversion, segmentation and feature extraction defects such as solder bridging, missing solder and excess solder are detected. This system improved the consistency of inspections and reduced dependency on manual quality control in manufacturing applications of PCBs. The study proposed by Zhang et al. focuses on a deep-learning-based framework for PCB defect detection by analyzing X-ray images. The system combines CNN and LSTM architectures that detect the hidden solder joint defects that are sometimes hard to be detected by the traditional optical inspection method. Experimental results demonstrated that the proposed method was more accurate in detection and had lower false-positive rates than conventional inspection systems [2].

A vision-assisted robotic soldering framework was presented by Ren et al. [3] that applies mathematical modeling techniques and image processing to automatically calculate the soldering parameters. Using methods such as Hough transformation, HSV masking and Canny edge recognition, accurate solder volume estimation and solder pad measurements were extracted. While achieving better similarity with actual solder data, the system maintained cheap computing time. For flexible PCBs, a robotic soldering system was employed using adaptive robotic control, and YOLOv3-based solder pad identification was proposed by Gao et al. The system had dual-arm architecture in which one arm of the robotic arrangement was employed to support the PCB while the other arm soldered. Pad localization accuracy was enhanced by deep learning and image processing approaches under different lighting situations. By achieving positioning errors of less than two pixels the study improved robotic accuracy and flexibility [4].

Milecki et al. [5] designed a robotic soldering station that combined machine vision and Gerber file parsing for solder pad identification and robotic path planning. For coordinate extraction and PCB inspection traditional image processing methods were utilized. The system increased automation efficiency and achieved an average positioning error of 0.06 mm using SCARA robots and industrial PLC hardware.

Allahyari and Fakhari constructed a 3 DOF Cartesian robotic system that was managed using trajectory extraction based on image processing. Using HSV thresholding and skeletonization algorithms the image pathways were transformed into robotic motion coordinates. The computed coordinates were sent to an Arduino controlled robot in order to achieve precise robotic route tracking and highlighted the significance of image processing in robotic automation systems.

Gao et al. proposed a system that used deep-learning-based wire recognition algorithms to create a fully automated robotic soldering system for USB cables. For detection of wire location in the crowded background the system used the ResNet-50-FPN vision framework for different lighting conditions. The system achieved a detection accuracy of 97% by decreasing the manual intervention and increasing automation efficiency. The study demonstrated how machine vision and deep learning may greatly increase the dependability of robotic soldering systems in challenging industrial environments [7].

An RT-2 Vision-Language-Action paradigm that uses transformer-based architectures to integrate robotic control with visual comprehension was introduced by Brohan et al. [8]. The framework allows robots to carry out generic tasks by directly converting visual and language inputs into robotic actions which enables the robot to perform generalized tasks through semantic reasoning.

Table I. Comparison of Vision and AI Techniques Used in Selected Literature

Ref	Vision / Image Processing	ML / AI	Limitations
[1]	Grayscale, segmentation, feature extraction	None	Sensitive to lighting & PCB variations
[2]	X-ray image analysis for defect detection	CNN + LSTM	Requires large datasets & high compute
[3]	Hough Transform, HSV masking, Canny edge	None	No real-time quality feedback loop
[4]	YOLOv3-based solder pad detection	YOLO v3 DL	Validated only in controlled lab setups
[5]	Machine vision pad recognition, coord. extraction	None	Performance affected by illumination changes

[6]	Deep vision-based wire detection	ResNet-50-FPN	Long cycle time; occasional sorting failures
[7]	PCB deformation tracking & manipulation	NN Adaptive Control	Highly dependent on vision & environment quality
[8]	HSV thresholding, skeletonization, coord. extraction	None	MCU memory limits trajectory accuracy

According to the reviewed literature table I shows, the majority of current automated robotic soldering systems primarily concentrate on distinct tasks such as solder joint identification, motion and path planning of the robot or defect detection rather than creating a comprehensive integrated framework. Many systems significantly rely on fixed camera surroundings and constant lighting which reduces the detection accuracy in industrial settings with shadows, reflections and fluctuating lighting.[9] Due to their vast dataset and high computing resource requirement, a number of deep learning-based techniques are challenging to apply in low-cost industrial applications. In contrast to real-time production circumstances the majority of research is justified in controlled environment setups or simulation setups. Also, in many robotic systems they lack adaptive motion planning and real-time feedback methods for fixing soldering faults during operation.

The proposed study focuses on developing an image-processing-based robotic soldering and inspection system that can accurately detect the solder points, i.e, the through holes on the PCB that needs to be soldered and guide robotic paths in real-world industrial settings in order to get beyond these restrictions. This system combines the robotic motion control, image processing and computer vision into one comprehensive framework. For solder joint identification, image processing techniques are used such as edge recognition, segmentation and object localization and the robotic motion planning will allow precise soldering tool positioning[10]. The system also aims to reduce dependency on expensive hardware and large datasets by using efficient image processing methods along with AI-based detection techniques which are lightweight. The project focuses on improving soldering accuracy, automation reliability and overall manufacturing efficiency through this approach. To summarize the literature analysis it emphasizes the growing importance of image processing and intelligent robotic automation in modern electronic manufacturing systems [11]. The evaluated results demonstrated that factors such as soldering accuracy, defect detection and production efficiency may all be greatly increased by integrating computer vision with robotic control. The proposed system seeks to expand upon these developments by creating a more efficient, cost-effective, and integrated vision-assisted automated robotic soldering framework appropriate for actual industrial applications[12].

III. METHODOLOGY

The system design isolates perception tasks into a dual-stage cognitive processing pipeline that separately handles the structural frontside confirmation, algorithmic edge filtering and schematic back side pin localization. Before physically deploying the system is verified through a simulation environment in CoppeliaSim that integrates motion control with multi view visual monitoring. This pipeline ensures adaptive and reliable execution without hardcoding the motion profiles.

A. Dual Stage Perception Architecture

The core intelligence of the system is structured as a sequential dual stage perception pipeline that distinguishes between the front side topological verification and back side geometric coordinate extraction. This decoupling ensures that only when both the component integrity and the target solder coordinates have been independently validated by computational models the physical tool actuation is initiated.

B. Front side Verification: Dataset Preparation and Augmentation

In order to replicate realistic deployment scenarios a custom dataset of 1500 high resolution top down images of a customised toy piano constructed PCB layout was collected under varying factory illumination conditions. At pixel level the bounding box annotation was performed manually using Roboflow platform classifying components into distinct classes namely resistors, push buttons, IC, electrolytic capacitors, ceramic capacitors and terminals and their number of annotation and images have been given in Table II. To reduce the neural network overfitting and improve generalization across changing lighting and camera views the dataset was enlarged using a structured augmentation pipeline. The augmentation transformations were Random flips in both the horizontal and vertical directions, Variation in randomized contrast, Shear adjustments within
±10° of the bounding box, Hue and Exposure offsets that are isolated. This increased the effective training set from 1500 to 4000 augmented frames.

TABLE II: Distribution of Images and Annotations Across Component Classes

Component Classes	Number of Images	Number of Annotations
Ceramic Capacitors	1646	1618
Electrolytic Capacitor	1646	1622
IC	1646	1627
Push Button	1646	11258

Resistor	1646	13040
Terminal	1646	1619

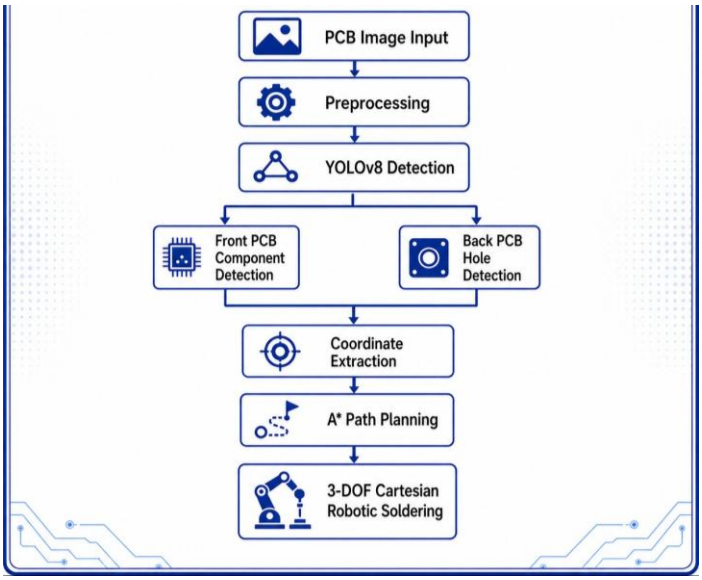


Fig. 1:Vision-Assisted Robotic Soldering Process Optimization Using YOLOv8

C. *Front side Verification: YOLOv8 Model Training, Optimization and Verification Logic*

A YOLOv8 deep convolution neural network was trained using the 4000 frame dataset. Its anchor free detection head combined with cross stage partial(CSP) bottleneck backbone enables efficient multi scale spatial feature extraction appropriate for crowded PCB. A stochastic gradient descent (SGD) optimizer with a batch size of 16 was used to run a training for 100 epochs. The trained weights were quantized using TensorRT, lowering floating point precision from FP32 to FP16 to enable real time deployment on hardware edge. This achieved a validated mAP50 of 94.2% across the augmented dataset by reducing the inference latency to sub millisecond scales with negligible loss in mean Average Precision. The trained model operates at inference time on Raspberry Pi 5 that is accelerated by Raspberry Pi AI HAT(26 TOPS). In order to reduce single frame noise a rolling window of the last 10 frames (implemented as a deque data structure) is maintained to establish a temporal stability. The maximum detection across this window is set by taking the stable component count for each class:

$$C_{stable}(k) = max_{\ i\in\{t-9,\ldots,t\ \}} C_i(k) \tag{1}$$

where, $C_i(k)$ is the detected count of components and k is the class in frame i.

The stable components are then compared against predefined BOM. If the stable count of any PCB is less than the value provided then the board is classified as BAD PCB and if it meets the expected values then the board is deemed as GOOD PCB and is approved for soldering stage. The front side model functions as a logical gatekeeper that prevents tool wear and material scrap by identifying the board with component misalignment, wrong counts or missing pieces and completely avoiding the soldering cycle when found.

D. *Backside Detection and Processing: Through hole localization via YOLOv8*

For the back side of PCB i.e., to detect the holes on PCB first the local image processing pipeline filters the raw camera frame and the cloud coordinate extraction runs using the provided Toy Piano PCB circuit design. The camera settings are specifically adjusted for back side imaging conditions with a fixed exposure period of 10,000 µs.

To ensure high recall across small, closely spaced through hole targets the model runs at an image size of 640 pixels with a confidence threshold of 0.09 and an IoU threshold of 0.30. After detection a size based filtering rule is used to suppress false positives where any bounding box with a height or width greater than 100 pixels is eliminated as true through holes yield compact detections while solder mask regions and board edges produce larger spurious boxes.

$$Valid\ Hole: w < 100px\ \wedge\ h < 100px \tag{2}$$

The centroid of each true bounding box is extracted as the holes pixel space coordinate:

$$P_i = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right), \quad \forall i \in \{1,2,\ldots,N\} \tag{3}$$

High frequency sensor noise and camera grain is suppressed using a localized spatial Gaussian kernel operator:

$$G(x,y) = \frac{1}{2\pi\sigma^2}\exp\left(-\frac{x^2+y^2}{2\sigma^2}\right) \tag{4}$$

Copper pad boundary gradients are extracted by computing partial spatial intensity derivatives across the image matrix:

$$G_x = \frac{\partial I}{\partial x}, \quad G_y = \frac{\partial I}{\partial y}$$
$$|\nabla I| = \sqrt{G^2{}_x + G^2{}_y} \tag{5}$$

The operator has real time visibility of detection completeness due to the live hole count displayed on the output frame prior to coordinate handoff to the path planner.

E. *Spatial Coordinate Mapping*

A linear scaling transform calibrated to the camera's fixed working weight is utilized to convert extracted hole centroid coordinates in pixel space into physical robot workspace coordinates:

$$X_{real} = x_{pixel} \cdot \left(\frac{W_{real}}{W_{image}}\right)$$
$$Y_{real} = y_{pixel} \cdot \left(\frac{H_{real}}{H_{image}}\right) \tag{6}$$

Where W_{real} and H_{real} are the actual dimensions of the camera's field of view at the calibrated Z axis operating height and W_{image} and H_{image} are the dimensions of image resolution 1280 x 720 pixels. The scaling constants remain consistent throughout all the PCB runs without requiring per board recalibration as the camera is fixed at the Z axis tool head and imaging always takes place at a fixed focus distance.

F. Path Optimization: A* Trajectory Planning

After the assembly of the physical coordinate set$\{X_{real,i}, Y_{real,i}\}$, an A* Path finding algorithm computes the optimal traversal sequence for the soldering tip reducing the total air travel distance between pad targets. At each node the algorithm evaluates a cost function:

$$f(n) = g(n) + h(n) \quad (7)$$

where g(n) is the precise cumulative travel distance from the start position and h(n) is a Euclidean heuristic estimating the remaining distance to the next target pad:

$$h(n) = \sqrt{X_{target} - X_n)^2 + Y_{target} - Y_n)^2} \quad (8)$$

An optimal trajectory matrix is produced by expanding nodes in ascending f(n) order and is dispatched directly to the motor drive control queue. A* routing contributes to validate 180.2 second cycle time for a 60 joint PCB by noticeably reducing total cycle air travel as compared to unoptimized sequential point to point traversal.

G. Pipeline Framework of Proposed Methodology

The summary of entire pipeline that connects every stage of the methodology is as follows:

- The front side of PCB is captured by the camera and YOLOv8 OBB uses a 10 frame temporal stability buffer to identify components across five classes.
- The BOM table is used to compare against stable counts, if any component is missing or misaligned the PCB is classified as BAD PCB and the soldering cycle is terminated.
- Once the board passes front side verification the PCB is flipped, the YOLOv8 model extracts pad centroids using size based false positive filtering while the camera records the solder face.
- The linear scaling transform is used to convert valid hole centroids to physical workspace coordinates.
- The optimized trajectory matrix is generated by a A* pathfinding algorithm which is further sent to the cartesian robot motor drivers for soldering execution.

IV. Results and Discussion

The proposed system of a vision-assisted robotic soldering system was evaluated for solder hole localization and PCB component identification using the YOLOv8 object detection model. The image processing analysis consisted of two distinct custom-made datasets namely the Front PCB and the Back PCB which were to evaluate the efficiency of the suggested framework under various PCB orientations. The obtained results demonstrated that the system can facilitate automated robotic soldering by accurately detecting and locating soldering locations while also validating the count of PCB components.

A. Front PCB Detection Performance

As presented in Table II. The YOLOv8 model shows a strong performance on the real-time detection results on the Front PCB dataset. The dataset consists of 28 images with 400 annotated instances of PCB components including 7 push buttons, 8 resistors, 1 IC, 1 terminal, 1 ceramic capacitor and 1 electrolytic capacitor corresponding to the application PCB of a piano toy which was also custom-made specifically for our application. With an overall precision of 100% and recall of 98.0% indicating highly accurate component detection and localization throughout the surface of the PCB. The mAP@50 score shows exceptional classification capabilities given in table III of 99.4% while the mAP@50-95 score of 79.0% validates the dependable detection performance even under more stringent IoU thresholds.

TABLE III: Performance of YOLOv8 in Front PCB component detection

Class	Images	Instances	P	R	mAP50	mAP50-95
all	1646	3879	0.95	0.97	0.92	0.81
Ceramic Capacitors	121	121	0.98	0.98	0.98	0.66
Electrolytic Capacitor	122	122	0.97	1.0	0.99	0.83
IC	122	122	0.98	0.99	0.99	0.84
Push Button	122	852	0.99	0.99	0.98	0.81
Resistor	121	969	0.90	0.85	0.93	0.52
Terminal	120	120	0.96	0.99	0.98	0.88

The confusion matrix shown in Fig. 2 demonstrates great classification accuracy with extremely few misclassifications among PCB component classes. For components like resistors, ICs, capacitors, push buttons and terminal strong true positive detections are noticed and are indicated by the majority of predictions being concentrated along the diagonal elements of the matrix. The model efficiently distinguishes between

visually identical components despite the thick PCB layout and various component sizes.

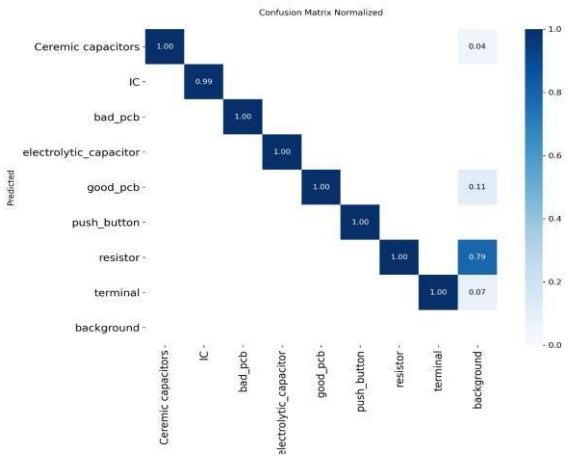


Fig. 2: Confusion matrix illustrating the classification performance of YOLOv8 on the Front PCB dataset.

There were some drawbacks too, like due to reflections, differences in lighting and similarity between microscopic electronic parts a few regions were mistakenly identified as components. However, the model successfully learned the visual properties of PCB components, as seen by the general lack of confusion between classes.

These results show how well the YOLOv8 model can identify and categorize PCB components for automated robotic soldering systems. The soldering efficiency is improved by the accurate component localization and also helps in the precise robotic arm positioning during the automated assembly process. The system also checks whether the PCB is assembled by identifying missing or faulty components before starting the soldering operation. Fig. 3 shows examples of real-time detection and categorization of the components on the Front PCB using the trained model. The model ability to correctly identify the PCB components can be validated by the generated bounding boxes under different viewing angles and lighting conditions.

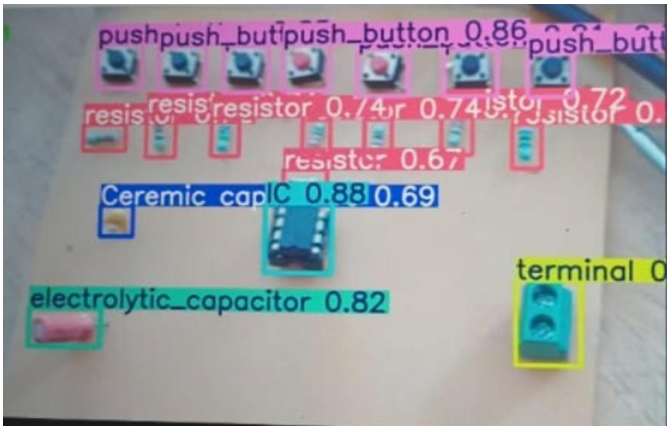


Fig. 3: Real-time Front PCB detection and component localization using YOLOv8.

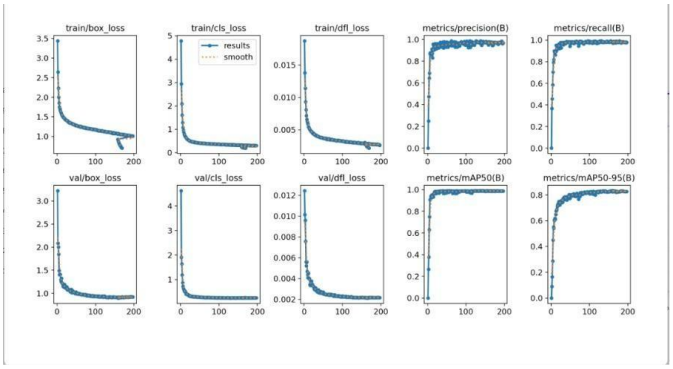


Fig. 4: Training performance metrics of the YOLOv8 model showing training and validation losses, precision, recall, mAP@50, and mAP@50-95 convergence over training epochs for Front PCB.

B. Back PCB Detection Performance

For the robotic solder positioning and alignment the Back PCB dataset was used to evaluate the effectiveness of YOLOv8. The detection task of the Back PCB is quite complex as in contrast to the Front PCB dataset the Back PCB images mostly consist of solder holes and conductive regions with few visual cues.

TABLE IV: Performance of YOLOv8 in Back PCB for solder hole detection

Class	Images	Instances	P	R	mAP 50	mAP 50-95
all	1152	6734	0.54	0.78	0.60	0.45

In terms of detection performance the Front PCB outperformed the Back PCB dataset as the model mAP@50 was 60.3% and its mAP@50-95 was 45.4% as given in table IV. The precision and recall of the model were 54.5% and 78.0% respectively the graphs of it are shown in fig 4. The primary cause of the decreased performance was the poor visible separation between solder holes and surrounding PCB backdrop areas.

The confusion matrix shown in Fig. 5 demonstrates the difficulties in detecting back PCB holes but the majority of solder holes were effectively identified by the trained model. The majority of the decrease in accuracy and mAP values was due to the missed holes, false positive detection from nearby PCB background regions was accounted for. The overlap between the background and holes classes shows how challenging it is to distinguish fine solder locations in different lighting and reflection conditions. The model was able to locate the necessary soldering spots even under difficult lighting situations with very little visual contrast.

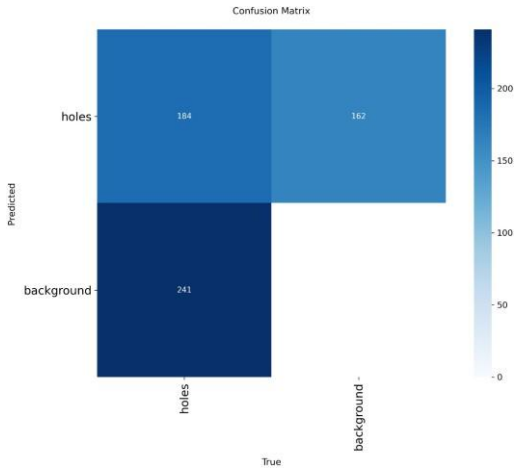


Fig. 5: Confusion matrix illustrating the detection performance of YOLOv8 on the Back PCB dataset.

Compared to the Front PCB dataset, the back PCB dataset has fewer training cases and distinguishing features which impacts the model's total learning capacity which can be proved by graphs in fig 7. Further improvement in the dataset like additional advancement in segmentation based models, controlled illumination, dataset size and preprocessing methods can improve the detection accuracy for industrial scale applications.

The example results are demonstrated in Fig. 6 of Back PCB for solder hole localization produced by the trained YOLOv8 model. Bounding boxes are used to identify the solder holes which helps the robotic system find the soldering points for automated procedures.

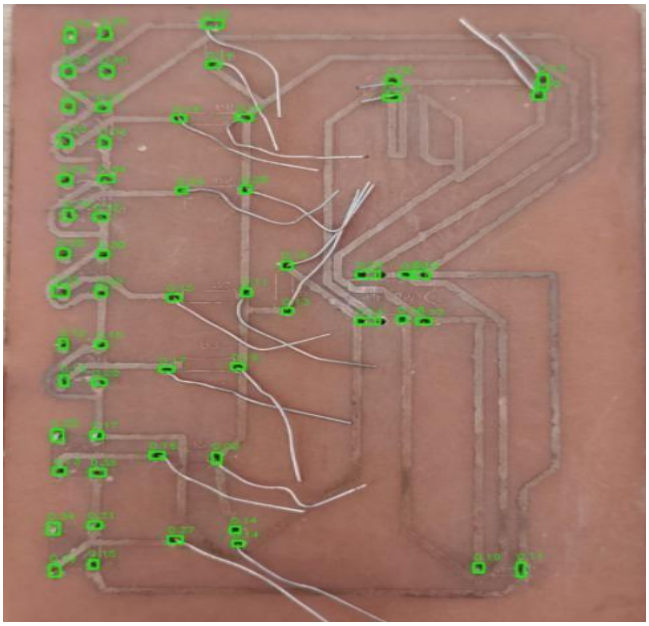


Fig. 6: Back PCB solder hole detection and localization using YOLOv8.

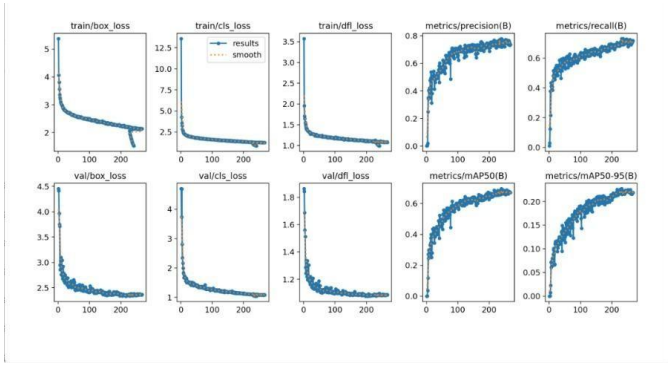


Fig. 7**:** Training performance metrics of the YOLOv8 model showing training and validation losses, precision, recall, mAP@50, and mAP@50-95 convergence over training epochs for Back PCB.

C. Simulation

A vision based automatic soldering system was developed and tested using the CoppeliaSim simulation environment in order to overcome the limitations of manual soldering such as inconsistent solder quality, positioning errors and reduced productivity. To carry out the automated soldering process of the PCB, a three-axis Cartesian robotic manipulator with X, Y, and Z axes was designed as shown in fig 8. A vision sensor was installed next to the soldering tip to continuously monitor the soldering environment and take real-time images of the PCB while it was in use. The PCB with several soldering holes was placed inside the robot's workspace and the manipulator moved the soldering tip in the direction of the intended soldering spots by following predetermined motion trajectories.

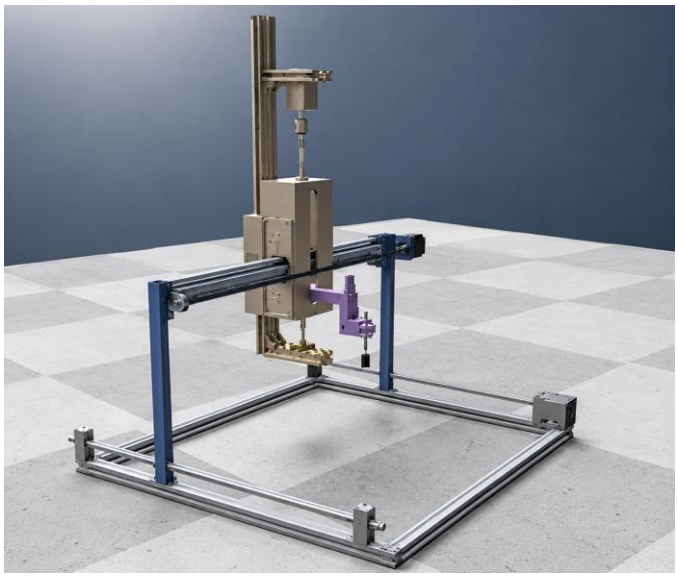


Fig. 8: CoppeliaSim simulation of the proposed vision-assisted robotic soldering system using a 3-DOF Cartesian manipulator.

Additional virtual cameras were integrated into the simulation environment to improve the visualization and monitoring of the

robotic operation. A top view camera was used to observe the overall PCB workspace and robot movement while a side view camera monitored the vertical motion of the soldering tool and tip positioning accuracy. To validate the appropriate workspace coverage and movement execution the vision sensor and virtual cameras continuously monitored the robot's interaction with the PCB during the simulation. The modeled simulation demonstrates that combining both robotic automation and machine vision is feasible for soldering applications. It also provides a framework for future research on intelligent trajectory planning, automatic pad detection and real-time soldering process monitoring.

V. Conclusion and Future Scope

This study presents a vision assisted robotic soldering framework that uses an adaptive dual stage YOLOv8 based perception pipeline instead of hardcoded inflexible motion profiles. The front side dataset of the customised toy piano PCB model achieved a mAP@50 of 99.4% and precision by 99% through the real time component verification against a preset BOM using a 10 frame temporal stability buffer and the back side of the PCB model was used to locate solder hole centroids using filtered bounding box detection, extracted coordinates were then converted into physical workspace positions through a calibrated linear scaling transform. The optimized tool traversal sequence was calculated and validated a cycle time of 180.2 seconds for a 60 joint PCB using an A* pathfinding algorithm. The entire pipeline was further evaluated using a three axis Cartesian Manipulator with coaxial and auxiliary camera monitoring in the CoppeliaSim simulation environment confirming the trajectory compliance prior to practical deployment. The results show that vision and deep learning may serve as the main control layer for PCB inspection, motion planning and coordinate extraction in standard manufacturing environments.

Future Work

- Physical Hardware Deployment: To verify the reliability, coordinate mapping and soldering quality on live PCBs the next phase is to fabricate the physical 3 DOF cartesian robot and deploy the entire dual stage image processing pipeline under actual factory floor lighting and thermal conditions.

- Real-time Thermal Feedback: Closed loop solder monitoring can be enabled by co-mounting a thermal imaging sensor with the soldering tip which will also help to prevent PCB delamination and cold joints by allowing the system to dynamically adjust dwell duration and temperature determined on live joint formation inputs.

- Multi board and Multi layout Generalization: Currently a single unique toy piano PCB is used to train and validate the current system. A variety of PCB designs and component configuration can be included to expand the dataset in proving the model generalization and enable the framework applicable to a wider range of standard production scenarios without the requirement of per board training.

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Vision-Assisted Robotic Soldering Process Optimization Using YOLOv8-Based PCB Component D Solder Hole Detection

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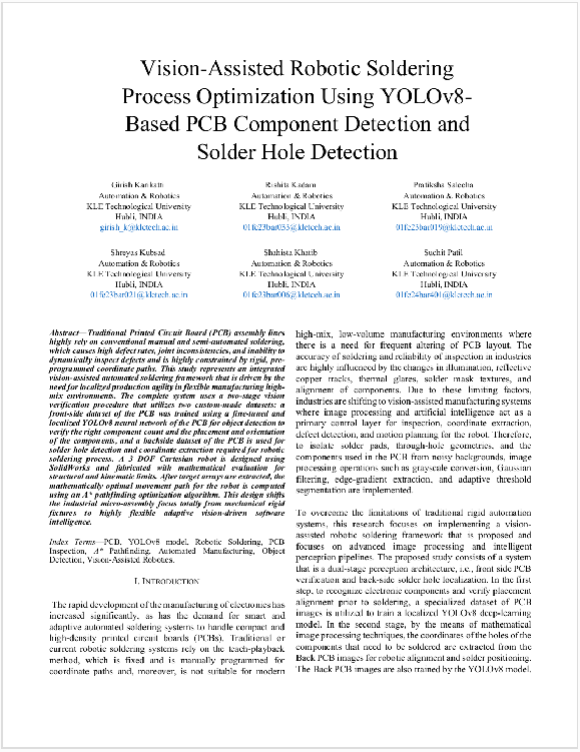
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Vision-Assisted Robotic Soldering Process Optimization Using YOLOv8-Based PCB Component Detection and Solder Hole Detection

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***Abstract*—Traditional Printed Circuit Board (PCB) assembly lines highly rely on conventional manual and semi-automated soldering, which causes high defect rates, joint inconsistencies, and inability to dynamically inspect defects and is highly constrained by rigid, pre-programmed coordinate paths. This study represents an integrated vision-assisted automated soldering framework that is driven by the need for localized production agility in flexible manufacturing high-mix environments. The complete system uses a two-stage vision verification procedure that utilizes two custom-made datasets: a front-side dataset of the PCB was trained using a fine-tuned and localized YOLOv8 neural network of the PCB for object detection to verify the right component count and the placement and orientation of the components, and a backside dataset of the PCB is used for solder hole detection and coordinate extraction required for robotic soldering process. A 3 DOF Cartesian robot is designed using SolidWorks and fabricated with mathematical evaluation for structural and kinematic limits. After target arrays are extracted, the mathematically optimal movement path for the robot is computed using an A* pathfinding optimization algorithm. This design shifts the industrial micro-assembly focus totally from mechanical rigid fixtures to highly flexible adaptive vision-driven software intelligence.**

***Index Terms*—PCB, YOLOv8 model, Robotic Soldering, PCB Inspection, A* Pathfinding, Automated Manufacturing, Object Detection, Vision-Assisted Robotics.**

I. Introduction

The rapid development of the manufacturing of electronics has increased significantly, as has the demand for smart and adaptive automated soldering systems to handle compact and high-density printed circuit boards (PCBs). Traditional or current robotic soldering systems rely on the teach-playback method, which is fixed and is manually programmed for coordinate paths and, moreover, is not suitable for modern high-mix, low-volume manufacturing environments where there is a need for frequent altering of PCB layout. The accuracy of soldering and reliability of inspection in industries are highly influenced by the changes in illumination, reflective copper tracks, thermal glares, solder mask textures, and alignment of components. Due to these limiting factors, industries are shifting to vision-assisted manufacturing systems where image processing and artificial intelligence act as a primary control layer for inspection, coordinate extraction, defect detection, and motion planning for the robot. Therefore, to isolate solder pads, through-hole geometries, and the components used in the PCB from noisy backgrounds, image processing operations such as grayscale conversion, Gaussian filtering, edge-gradient extraction, and adaptive threshold segmentation are implemented.

To overcome the limitations of traditional rigid automation systems, this research focuses on implementing a vision-assisted robotic soldering framework that is proposed and focuses on advanced image processing and intelligent perception pipelines. The proposed study consists of a system that is a dual-stage perception architecture, i.e., front side PCB verification and back-side solder hole localization. In the first step, to recognize electronic components and verify placement alignment prior to soldering, a specialized dataset of PCB images is utilized to train a localized YOLOv8 deep-learning model. In the second stage, by the means of mathematical image processing techniques, the coordinates of the holes of the components that need to be soldered are extracted from the Back PCB images for robotic alignment and solder positioning. The Back PCB images are also trained by the YOLOv8 model.

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